Forecasting Closing-Bar Variance for 0DTE SPX Options

Bullet-point draft of the full paper

James Chen Austin Pollok Chris Jones Mihai Cucuringu

Working draft, 15 October 2026 — not for citation

1 Introduction

- **Question.** At 15:30 ET, forecast the S&P 500's 15:30–16:00 realized variance RV_t.
  - A 0DTE SPXW straddle opened at 15:30 settles on exactly that variance.
  - Forecast and trade:

$$\hat{\sigma}_t^2 = \hat{y}_t^2 B_t, \qquad q_t = \operatorname{sign}(s_t), \qquad s_t = \hat{\sigma}_t^2 - (IV_t^{30})^2.$$

  - $y_t = \sqrt{RV_t / B_t}$; B_t: diurnal baseline; $(IV_t^{30})^2$: implied variance to 16:00.
  - $q_t = +1$ buys, $q_t = -1$ sells; P&L $q_t R_t$.
  - Sample: $N = 866$ expiration days, 3 January 2020 to 30 April 2024.
- **Why it matters.**
  - $s_t < 0$ on 72–83% of days: the value is in picking buy days.
  - 0DTE options: a large share of index option volume.
  - Payoff set by one known window.
- **Two axes, one model index:**

$$\hat{y}_t = f_{m,d,g}(x_t), \qquad x_t \in \mathbb{R}^p.$$

  - Design $d \in \{\text{HAR+cal}, \text{all}\}$: $p = 22$ (HAR ladder + calendar) vs $p = 628$ (8 exogenous buckets).
  - Fit $g \in \{\text{per bar}, \text{pooled}\}$: the 16:00 bar only vs all 48 half-hour bars.
  - Estimator m: linear, tree or neural network, crossed with (d, g).
- **Evaluation loss:**

$$\overline{\text{QLIKE}} = \frac{1}{N} \sum_{t=1}^{N} \text{QLIKE}(RV_t, \hat{\sigma}_t^2), \qquad \text{QLIKE}(RV, \hat{\sigma}^2) = \frac{RV}{\hat{\sigma}^2} - \log \frac{RV}{\hat{\sigma}^2} - 1.$$

  - Trade score: Sharpe ratio of $q_t R_t$.
- **Contributions.**
  - Value, availability and activity ladders: 47 exogenous series in one x_t.
  - Plain back-transform $\hat{y}_t^2 B_t$: no smearing, no recalibration.
  - Causal walk-forward test of tuning: refits, re-tuning, search budgets.
  - Native GPU MinGRU, regularised like the linear stack.
- **Axis 1, d: all vs HAR+cal** (pairs differ only in d).

  - $\Delta\overline{\text{QLIKE}} = \overline{\text{QLIKE}}_{\text{all}} - \overline{\text{QLIKE}}_{\text{HAR+cal}} < 0$ in 38 of 51; median -1.7%.
  - 2 individually significant; no significant reversals.
  - Δ Sharpe > 0 in 50 of 51; median $+0.54$; 4 significant.
  - All 20 forecasts significantly beating always-short ($q_t \equiv -1$) use $d = $ all.
- **Axis 2, g: per bar vs pooled.**
  - Per bar lowers $\overline{\text{QLIKE}}$ wherever both fits exist.
  - Significant on HAR+cal in every family; ridge -11.3%, DM -3.42.
  - Effect on the trade: not significant.
- **Accuracy is not the trade.**
  - Rank correlation of $\overline{\text{QLIKE}}$ and Sharpe across models: -0.18.
  - No single modelling step moves Sharpe significantly.
- **Roadmap:** data, features, methods, results (axis 1, axis 2), the trade, feature importance, conclusion.

2 Related work

- **Realized volatility and HAR:** $\mathbb{E}_\tau RV_{\tau+1} = \beta_0 + \sum_h \beta_h \overline{RV}_\tau^{(h)}$, $h \in \{1, 5, 22\}$ days.
  Corsi (2009), Andersen et al. (2003), Barndorff-Nielsen and Shephard (2004), Andersen et al. (2007), Corsi et al. (2008), Bollerslev et al. (2016), Bekierman and Manner (2018), Bennedsen et al. (2022), Hansen et al. (2012), Andersen et al. (2006).
  - Panel-pooled HAR, the anchor for $g =$ pooled: Bollerslev et al. (2018).
- **Intraday and diurnal patterns:** a U-shaped B_t, divided out in $y_t = \sqrt{RV_t / B_t}$.
  Andersen and Bollerslev (1997), Engle and Gallo (2006), Hizmeri and Bevilacqua (2025).
- **0DTE options and the variance risk premium:** $\mathbb{E}_t^{\mathbb{Q}}[RV_t] > \mathbb{E}_t^{\mathbb{P}}[RV_t]$ on average.
  Hizmeri and Bevilacqua (2025); pricing Black and Scholes (1973), Merton (1973).
  - *[to add: Bandi–Fusari–Renò 2023; Beckmeyer–Branger–Gayda 2023; Brogaard–Han–Won 2023; Dim–Eraker–Vilkov 2024; Vilkov, 0DTE trading rules; Almeida–Freire–Hizmeri 2025; Carr–Wu 2009; Bollerslev–Tauchen–Zhou 2009].*
- **Machine learning for volatility and returns:** nonlinear f on many inputs.
  Gu et al. (2020), Christensen et al. (2023); boosting Ke et al. (2017), Chen and Guestrin (2016); hyperparameter search Akiba et al. (2019).
  - *[to add: Breiman 2001; Bucci 2020; Rahimikia–Poon 2020; Zhang–Zhang–Cucuringu–Qian 2024; MinGRU (Feng et al. 2024); PufferLib].*
- **Forecast evaluation:** QLIKE is robust to proxy noise, $\arg\min_h \mathbb{E}_t \text{QLIKE}(RV_t, h) = \mathbb{E}_t RV_t$.
  Patton (2011) (robust losses), Diebold and Mariano (1995) with Harvey et al. (1997), Hansen et al. (2011) (MCS), Ehm et al. (2016) (elementary scores), Mincer and Zarnowitz (1969).
  - *[to add: Hansen 2005 (SPA); White 2000; Romano–Wolf 2005 (StepM); Newey–West 1987].*
- **Shrinkage with many weak signals:** $\hat{\beta} = \arg\min_\beta \|y - X\beta\|_2^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$; one λ per block in multi-penalty ridge.
  Giannone et al. (2021), Shen and Xiu (2025), Kozak et al. (2020), Hoerl and Kennard (1970), Tibshirani (1996), Zou and Hastie (2005), van de Wiel et al. (2021), van Wieringen (2023).
- **Interpretation:** additive attributions $\hat{y}_t = \phi_0 + \sum_j \phi_{t,j}$.
  Lundberg and Lee (2017), Lundberg et al. (2018), Aas et al. (2021).

- **Back-transform** (one sentence): smearing rescales $\hat{y}_t^2 B_t$; we do not.
  Duan (1983), Manning and Mullahy (2001).

3 Data

- **Sources.**
  - **E-mini S&P 500 futures (ES),** nearly 24 hours a day, 1998–2024. Bar t: 30 minutes.
  - $r_{t,i}$: the i-th 1-minute log return in bar t; $M_t \leq 30$ of them. Target:

$$RV_t = \sum_{i=1}^{M_t} r_{t,i}^2.$$

  - Other core statistics, same bar:

$$\sum_i r_{t,i}, \quad \sum_i |r_{t,i}|, \quad \sum_i r_{t,i}^3, \quad \sum_i r_{t,i}^4, \quad \sum_i r_{t,i}^2 \, \mathbf{1}\{r_{t,i} > 0\},$$

$$BV_t = \sum_{i \geq 2} |r_{t,i}| \, |r_{t,i-1}|, \qquad AC_t = \sum_{i \geq 2} r_{t,i} \, r_{t,i-1},$$

    plus volume and the observation count M_t.
  - **Constituent cross-section,** equal- and value-weighted, 1993–2024: six return moments, turnover, signed turnover and spreads. Signed turnover:

$$\mathrm{OFI}_t = \frac{\mathrm{buy}_t - \mathrm{sell}_t}{\mathrm{buy}_t + \mathrm{sell}_t} \in [-1, 1], \qquad \text{missing if } \mathrm{buy}_t + \mathrm{sell}_t = 0.$$

  - **StockTwits** attention and sentiment, from 2009 and 2010.
  - **Cboe** VIX (from 2006), VIX3M (from 2009) and VVIX (from 2012), all to February 2024.
  - Cboe option volume-demand series, 2011 to August 2023.
  - **FOMC** release times, 1993 to November 2023.
  - **SPXW expiration-day option chain,** every 30 minutes: bid, ask, mid, underlying, implied volatility, delta. 1,291 expirations, 2020–2025.
  - *[name the vendor of each file].*
- **Cleaning.**
  - Grid: 30 minutes from the first RV print. Bar kept iff RV printed and outside the weekend cut:

$$t \in \mathcal{T} \iff RV_t \text{ printed and } t \notin (\text{Fri } 20{:}00, \ \text{Sun } 18{:}30).$$

  - Holidays and early closes stay as futures bars; the trade sample drops early closes.
  - Exogenous series u: availability flag, taken before any fill,

$$a_t = \mathbf{1}\{u \text{ printed on bar } t\} \in \{0, 1\}.$$

  - Then u_t forward-filled, without limit. Every $a_t = 0$ row later gets a neutral median (Features, step 4).

– Diurnal baseline. $j(t)$: clock slot of t. S_t: the last 20 bars $s < t$ with $j(s) = j(t)$.

$$B_t = \frac{1}{|S_t|} \sum_{s \in S_t} RV_s, \qquad |S_t| \geq 5.$$

– $|S_t| < 5$: $B_t = 1$, bar not scored.
– Target and its winsorised version:

$$y_t = \sqrt{RV_t / B_t}, \qquad \tilde{y}_t = \mathrm{clip}\big(y_t,\ q_{0.01,t},\ q_{0.99,t}\big),$$

$$q_{p,t} = p\text{-quantile of } \{y_s : t - 240 \leq s \leq t - 1\}.$$

– Quantile window: 240 bars of all slots, lagged one bar.

- **Samples.**
  – Pooled panel: 300,317 half-hour bars from June 1998, with a 24,000-bar rolling window:

$$300{,}317 = 303{,}442 \text{ grid bars} - 3{,}125 \text{ burn-in bars}.$$

  – Per-bar 16:00 design: 3,469 sessions, July 2010 to April 2024, with a 2,000-session window.
  – 1,469 forecast rows from June 2018:

$$1{,}469 = 3{,}469 - 2{,}000.$$

  – Linear pre-tuning: 16:00 rows back to 2004; validation folds 2016–2019.
  – Test period: January 2020 to April 2024.

$$866 \text{ trade days} = 872 \text{ SPXW expirations} - 6 \text{ early closes}.$$

  – The 413 later expirations are held back.

4 Features

- **Eight exogenous buckets, 47 series:** moments (7), liquidity (14), equal-weighted market (6), value-weighted market (6), sentiment (3), implied volatility (3), volatility demand (4), FOMC (4).

$$47 = 7 + 14 + 6 + 6 + 3 + 3 + 4 + 4.$$

  – FOMC: $\mathbf{1}\{t \text{ is a release bar}\}$, $\mathbf{1}\{t \text{ on a release day}\}$, $1/(1 + n_t^+)$, $1/(1 + n_t^-)$.
  – $n_t^{\pm}$: bars to the next, and since the last, scheduled release.
- **Columns per series.**
  – Trailing-mean ladders of the adjusted value, shifted one bar:

$$\bar{u}_{t,k} = \frac{1}{k} \sum_{i=1}^{k} \tilde{u}_{t-i}, \qquad k \in \begin{cases} \{1, 5, 25, 125, 625, 3125\} & \text{per bar,} \\ \{2^0, 2^1, \ldots, 2^{11}\} \text{ (base 2, 12 rungs)} & \text{pooled.} \end{cases}$$

  – k counts bars of the full grid; the per-bar design keeps the 16:00 rows.
  – Same ladder on the availability flag, $\bar{a}_{t,k}$.
  – Activity ladder $\bar{\alpha}_{t,k}$ for 7 zero-inflated series.

Table 1: Data sources and spans.

Source	Content	Freq.	First	Last	Rows	Role
E-mini S&P 500 futures (ES); *[vendor]*	sums of 1-min log returns per bar: realized variance $\sum r^2$ (the target) and 9 other statistics (return moments, bipower variation, autocovariance, volume, return count)	30 min	1998-01-05	2024-04-30	303,444	target; features
Constituent cross-section, EW and VW; *[vendor]*	11 statistics each: 6 return moments, turnover (total, buyer-, seller-initiated), effective and quoted spread	30 min	1993-01-04	2024-04-30	213,578 each	features
StockTwits; *[vendor]*	attention; sentiment and sentiment count	30 min	2009-03-23 2010-06-01	2024-04-30	184,594 162,284	features
Cboe volatility indices; *[vendor]*	VIX VIX3M VVIX	30 min	2006-05-22 2009-08-12 2012-03-26	2024-02-12 2024-02-12 2024-02-12	85,625 50,150 38,508	features
Option volume demand; *[vendor]*	4 series: SPX or all options; opening and closing, or opening only	30 min	2011-01-03	2023-08-31	54,062	features
FOMC statement calendar; *[vendor]*	one bar per scheduled statement, 14:30 ET	event	1993-02-03	2023-11-01	246	features
SPXW option chain, expiration days; *[vendor]*	bid, ask, mid, underlying, implied volatility, delta, hours to expiry, early-close flag	30 min	2020-01-03	2025-12-31	8,198,778 quotes; 1,291 expirations	trade: entry, exit, implied variance

Notes: Bar files are keyed on the bar-end stamp (ET) and outer-merged on it (`src/data/loading.py`); Rows counts the 30-minute bars on which the series printed, before the grid and the weekend cut; First and Last are the first and last print. The chain is stamped in UTC and loaded on its own (`src/data/spxw.py`). The VIX family, the volume-demand series and the FOMC flag stop before the sample ends (2024-04-30) and are carried forward from their last print. Merged but never a feature: the other release flags (6), the SPXW slice and model-free implied variance, and the VIX representations. *[vendor]*: no file or document names the vendor (the chain is called OptionMetrics only in a code comment).

Table 2: Samples.

Sample	Start	End	Rows	Window	Role
Bar grid: bars on which RV printed, weekend cut	1998-01-05	2024-04-30	303,442 bars	–	construction
Pooled panel: after the 3,125-bar burn-in	1998-06-04	2024-04-30	300,317 bars	24,000 bars (trees, MinGRU); 96,000 (linear)	pooled designs
Pooled forecasts	2001-07-13	2024-04-30	276,317 bars	–	pooled scoring
Tree development span (pooled rows before 2003)	1998-06-04	2002-12-31	36,285 bars	24,000 bars, refit every 250	shipped tree settings
Per-bar 16:00 design	2010-07-12	2024-04-30	3,469 sessions	2,000 sessions	per-bar designs
Per-bar forecast rows	2018-06-25	2024-04-30	1,469 sessions	–	walk-forward
Linear pre-tune rows (16:00 rows of the last-hour design)	2004-04-13	2020-01-02	3,947 sessions	2,000 sessions	pre-tuning data
Pre-tune validation folds (4×250)	2016-01-06	2020-01-02	1,000 sessions	2,000-session fit, 25-session embargo	penalty choice
Test period: 16:00 rows	2020-01-03	2024-04-30	1,087 sessions	–	out of sample
Trade days: SPXW expirations (872) less early closes (6)	2020-01-03	2024-04-30	866 days	–	scoring, trading
Later SPXW expirations	2024-05-01	2025-12-31	419 (413 full)	–	held back

Notes: Dates are bar-end dates (ET). The burn-in is the longest HAR rung (3,125 rows) and is applied to each series as built: the half-hour grid (pooled), the one-bar 16:00 series (per bar) and the two-bar last-hour series (pre-tune), which is why the per-bar design starts in 2010 and the pre-tune borrows the last-hour rows. The pooled linear models' window is the per-bar 2,000 sessions, in bars. Full: expirations that are not early closes.

- **Backbone:** the HAR ladder of the adjusted target (6 columns) and 16 calendar indicators (weekday, opex, quad witching, month end, quarter end and others).

$$\bar{y}_{t,k} = \frac{1}{k}\sum_{i=1}^{k} \tilde{y}_{t-i}, \qquad 16 = \underbrace{5}_{\text{weekday}} + \underbrace{4}_{\text{clock}} + \underbrace{7}_{\text{expiry, month, quarter}}.$$

- **Transforms, in order:**
  1. diurnal adjustment, with a banded same-slot baseline (S_t as for B_t):

$$\tilde{u}_t = \frac{u_t}{b^{(u)}_{j(t)}}, \qquad b^{(u)}_{j(t)} = \begin{cases} \mathrm{clip}\big(\mathrm{mean}_{s\in S_t} u_s,\ 0.5\,\rho_t,\ 2\rho_t\big), & u \geq 0, \\ \max\big\{\mathrm{clip}\big(\mathrm{sd}_{s\in S_t} u_s,\ 0.5\,\rho_t,\ 2\rho_t\big),\ 0.1\,\bar{\sigma}^{(u)}\big\}, & u \text{ signed;} \end{cases}$$

     – ρ_t: median of the slot's last 250 unbanded baselines (needs 20).
     – $\bar{\sigma}^{(u)}$: full-sample median of the same-slot sd.
     – Unsigned baseline zero or undefined: $b = 1$.
     – Not adjusted: VIX family, StockTwits sentiment, FOMC.
  2. a variance-stabilising map by series type, $\tilde{u}_t \leftarrow g(\tilde{u}_t)$, first match by name:

$$g(v) = \begin{cases} \sqrt{v} & \text{squared returns, } RV, \text{ turnover, bipower, effective spread,} \\ \mathrm{sign}(v)\,|v|^{1/2} & \text{autocovariance,} \\ v^{1/3} & \text{cubed returns,} \\ \mathrm{sign}(v)\,|v|^{1/4} & \text{fourth-power returns,} \\ v & \text{other signed series, absolute returns,} \\ \log\max(v, 10^{-12}) & \text{otherwise.} \end{cases}$$

     – Signed, no name rule, divisor at its minimum on $\geq 50\%$ of rows (volatility demand):

$$\tilde{u}_t = \mathrm{asinh}(u_t/s_t), \qquad s_t = \max\big\{e_{t-1},\ 0.01\,\mathrm{med}_{s\leq t}\, e_{s-1}\big\}.$$

     – e_t: EWMA of $|u|$ over rows with $u \neq 0$, half-life 250 days.
  3. winsorisation at rolling 1/99% bounds, as for the target:

$$\tilde{u}_t \leftarrow \mathrm{clip}\big(\tilde{u}_t,\ q_{0.01,t},\ q_{0.99,t}\big), \qquad q_{p,t} = p\text{-quantile of } \{\tilde{u}_s : t-240 \leq s \leq t-1\};$$

  4. imputation with an indicator, median over the full sample:

$$\tilde{u}_t \leftarrow \begin{cases} \tilde{u}_t, & a_t = 1, \\ \mathrm{med}\{\tilde{u}_s : a_s = 1\}, & a_t = 0; \end{cases}$$

  5. activity flags, for the 7 series whose full-sample mode holds $\geq 20\%$ of rows:

$$\alpha_t = \mathbf{1}\{u_t \neq \mathrm{mode}(u)\};$$

  6. ladders $\bar{u}_{t,k}$, $\bar{a}_{t,k}$, $\bar{\alpha}_{t,k}$, $\bar{y}_{t,k}$;
  7. a trailing robust scale (median and IQR, with floors), value ladders only:

$$z_{t,k} = \frac{\bar{u}_{t,k} - \mathrm{med}_{[t-T_W,\,t)}\,\bar{u}_{\cdot,k}}{\max\big\{\mathrm{IQR}_{[t-T_W,\,t)},\ 0.01\,\max_{s\leq t}\mathrm{IQR}_{[s-T_W,\,s)},\ r_{t,k}\big\}}.$$

  - $r_{t,k}$: IQR of $\bar{u}_{s,k}$, $s < t$, over rows whose k-window is fully available.
  - Column whose mode holds $\geq 20\%$ of rows: $r_{t,k}$ also drops rows at the mode.
  - $r_{t,k} = 1$ until 8 such rows.
  - $\bar{a}$, $\bar{\alpha}$: unscaled. HAR, calendar: $r = 0$.
  - $t < T_W$: the first window's statistics.
  - Pooled only: median and IQR over $a_s = 1$ rows; fewer than 512 gives $z = 0$.
- **Filtering:** at every fit, on the training window $[t - T_W, t)$, keep

$$\mathcal{K}_t = \{j : \ x_{\cdot j} \text{ not constant}, \ x_{\cdot j} \neq x_{\cdot j'} \ \forall\, j' < j, \ j' \in \mathcal{K}_t\}.$$

  - Equality is bitwise; the first copy is kept.
  - The per-bar all-features design keeps a median of 369 of its 628 columns.
- **Designs:** HAR + calendar has 22 columns; all features has 628.

$$628 = \underbrace{22}_{\bar{y},\ \text{calendar}} + \underbrace{47 \times 6}_{\bar{u}} + \underbrace{47 \times 6}_{\bar{a}} + \underbrace{7 \times 6}_{\bar{\alpha}}, \qquad 22 = 6 + 16.$$

  - The pooled designs have 1,144 columns: 43 series, 12 rungs, 24 session-edge columns.

$$1{,}144 = \underbrace{12 + 16}_{\bar{y},\ \text{calendar}} + \underbrace{24}_{\bar{y} \times \text{open, close}} + 43 \times 12 + 43 \times 12 + 5 \times 12.$$

- **Causality:** every statistic at t uses rows before t: ladders, B_t, $q_{p,t}$, ρ_t, scaler, $\mathcal{K}_t$.
  - Full-sample elements, disclosed: the neutral median, the mode, $\bar{\sigma}^{(u)}$, the asinh routing.
  - The first window's scaler statistics touch warm-up rows only.

5 Methods

5.1 Problem formulation

- **Time-series regression,** run as a causal walk-forward backtest; every input known at 15:30.
- **Diurnal baseline** (t a bar here): trailing same-slot mean, strictly before t,

$$B_t = \frac{1}{|\mathcal{S}_t|} \sum_{j \in \mathcal{S}_t} RV_j, \qquad \mathcal{S}_t = \{\text{last 20 bars } j < t \text{ in the slot of } t\}, \qquad |\mathcal{S}_t| \geq 5.$$

  - Bars with $|\mathcal{S}_t| < 5$: not scored.
- **Target:** variance-stabilised, then winsorised on that scale:

$$y_t = \sqrt{RV_t / B_t}, \qquad \tilde{y}_t = \text{clip}\big(y_t;\ Q_{0.01}(y_{t-240}, \ldots, y_{t-1}),\ Q_{0.99}(y_{t-240}, \ldots, y_{t-1})\big).$$

  - Bounds: rolling quantiles of the previous 240 bars, all slots pooled.
  - Models fit $\tilde{y}_t$; from here on y_t denotes $\tilde{y}_t$.
  - Per bar: t is the session; its row is the 15:30–16:00 bar, stamped 16:00.
- **Back-transform:** square and rescale; no smearing, no recalibration:

$$\hat{\sigma}_t^2 = \hat{y}_t^{\,2}\, B_t.$$

  - Jensen bias, low, disclosed: $\mathbb{E}_t[y_t^2] - (\mathbb{E}_t y_t)^2 = \text{Var}_t(y_t) \geq 0$ ($\mathbb{E}_t$: given 15:30 information).

Table 3: Design composition: columns by block.

		Per bar (16:00)				Pooled
Block	Series	Value	Avail.	Active	Total	Total
HAR ladder of the target	1	6	–	–	6	12
HAR × open / close	–	–	–	–	–	24
Calendar	–	–	–	–	16	16
HAR + calendar					**22**	52
ES return moments	7	42	42	–	84	168
Liquidity	14	84	84	6	174	348
Market, equal-weighted	6	36	36	–	72	144
Market, value-weighted	6	36	36	–	72	144
Sentiment	3	18	18	–	36	72
Implied volatility	3	18	18	–	36	72
Volatility demand	4	24	24	24	72	144
FOMC calendar	4	24	24	12	60	–
Exogenous	47	282	282	42	606	1,092
All features					**628**	**1,144**

Notes: Value: trailing means $\bar{x}_{t,k}$ of the adjusted series, $k \in \{1, 5, 25, 125, 625, 3125\}$ bars of the one-bar series per bar, $k = 1, 2, 4, \ldots, 2048$ bars pooled, all lagged one bar. Avail.: the same ladder of the is-observed flag (taken before the forward fill). Active: the same ladder of an off-the-mode flag for the 7 mode-inflated series (`numobs`, `voldemand_spx_open_and_close`, `voldemand_spx_open_only`, `voldemand_all_open_and_close`, `voldemand_all_open_only`, `fomc_release`, `fomc_day`). Calendar: weekday dummies, hour, overnight / open / close session flags, option-expiration day and week, quad witching, rebalancing close, month and quarter end, days to expiration. Per bar, the twelve HAR × open / close columns are zero or copies of the HAR ladder at 16:00 and are dropped. The pooled panel of record predates the FOMC bucket (4 series) and so has no FOMC columns; with them it has 1,264. At each fit, columns constant on the training window or copies of an earlier column are dropped.

- **Scoring loss:** QLIKE, robust to noise in the realized proxy (Patton, 2011):

$$\mathrm{QLIKE}(RV, \hat{\sigma}^2) = \frac{RV}{\hat{\sigma}^2} - \log \frac{RV}{\hat{\sigma}^2} - 1, \qquad L_t = \mathrm{QLIKE}(y_t^2 B_t,\, \hat{\sigma}_t^2) = \mathrm{QLIKE}(y_t^2,\, \hat{y}_t^2).$$

  - B_t cancels: scale-invariant.
- **Fit objective:** squared error in fit space, on the fit block $\mathcal{F}_t$:

$$\hat{\theta}_t = \arg\min_{\theta} \sum_{i \in \mathcal{F}_t} \big(y_i - \hat{y}_i(\theta)\big)^2.$$

  - Exception: the QLIKE-trained network.
- **Windows:** training window $\mathcal{T}_t = \{t - T_W, \ldots, t-1\}$.
  - Per bar: the 16:00 bar only; $T_W = 2{,}000$ sessions; refit every session.
  - Pooled: all 48 bars; $T_W = 24{,}000$ bars; refit every bar (trees) or every 21 sessions (network).
- **Tuning splits are forward:** fit block, 25-row embargo, 125-row validation tail:

$$\mathcal{F}_t = \{t - T_W, \ldots, t - n_v - n_e - 1\}, \quad \mathcal{E}_t = \{t - n_v - n_e, \ldots, t - n_v - 1\}, \quad \mathcal{V}_t = \{t - n_v, \ldots, t - 1\},$$

  $n_e = 25$, $n_v = 125$. Row t never enters.

5.2 Models

- **HAR and HAR + calendar OLS** (Corsi, 2009):

$$\hat{y}_t = \beta_0 + \sum_{k \in \mathcal{K}} \beta_k\, \bar{y}_{t,k} + \gamma^\top c_t, \qquad \bar{y}_{t,k} = \frac{1}{k} \sum_{j=1}^{k} y_{\tau_t - j}, \qquad \mathcal{K} = \{1, 5, 25, 125, 625, 3125\}.$$

  - τ_t: the target bar; means run over half-hour bars, overnight included.
  - c_t: calendar dummies; HAR alone sets $\gamma = 0$.
  - $x_t \in \mathbb{R}^p$: these columns plus the same ladders of each exogenous series.
- **Ridge, lasso and elastic net,** in scikit-learn units, solved exactly online:

$$\hat{\beta}^{\mathrm{ridge}} = \arg\min_{\beta} \sum_{i \in \mathcal{T}_t} (y_i - x_i^\top \beta)^2 + \alpha \sum_j \mathrm{pf}_j\, \beta_j^2,$$

$$\hat{\beta}^{\mathrm{enet}} = \arg\min_{\beta} \frac{1}{2|\mathcal{T}_t|} \sum_{i \in \mathcal{T}_t} (y_i - x_i^\top \beta)^2 + \alpha\rho \sum_j \mathrm{pf}_j |\beta_j| + \frac{\alpha(1-\rho)}{2} \sum_j \mathrm{pf}_j\, \beta_j^2.$$

  - Lasso: $\rho = 1$. Elastic net: $\rho = 0.5$.
  - Ridge: rank-one add and drop on $(X^\top X + \alpha\, \mathrm{diag}(\mathrm{pf}))^{-1}$.
  - Lasso, elastic net: exact online homotopy (Garrigues and El Ghaoui, 2008), one row in, one out.
  - Window mask at every fit: drop window-constant columns and exact copies.
  - **Block-diagonal Tikhonov penalty:** backbone penalised 100 times more lightly than the exogenous block,

$$\mathrm{pf}_j = \begin{cases} 0.01 & j \in \text{HAR + calendar backbone}, \\ 1 & j \text{ exogenous}, \\ 0 & j = \text{intercept}. \end{cases}$$

  - HAR + calendar alone: $\mathrm{pf}_j = 1$, one penalty.
- **Per-bar trees:** LightGBM, XGBoost and random forest, on the same rows and windows as the linear models.
  - **Boosting** (LightGBM (Ke et al., 2017), XGBoost (Chen and Guestrin, 2016)), loss $\frac{1}{2}(y-F)^2$:

$$F_m = F_{m-1} + \eta f_m, \qquad f_m = \arg\min_f \sum_i \Big[g_i f(x_i) + \tfrac{1}{2} h_i f(x_i)^2 \Big] + \gamma J + \lambda_1 \|w\|_1 + \tfrac{1}{2}\lambda_2 \|w\|_2^2.$$

  - $g_i = F_{m-1}(x_i) - y_i$, $h_i = 1$; J leaves, leaf weights w.
  - Leaf weight at $\lambda_1 = 0$: $w_j^* = -G_j/(H_j + \lambda_2)$, $H_j = n_j$.
  - Leaf minimum: $n_j \geq$ min-child-samples (LightGBM); $H_j \geq$ min-child-weight (XGBoost).
  - **Random forest:** average of N trees, each on a bootstrap subsample,

$$\hat{y}_t = \frac{1}{N} \sum_{m=1}^{N} T_m(x_t);$$

    random feature subset at each split.
- **Pooled tree bank:** one frozen configuration per tree, refit every bar on a 1,144-column panel.
- **MinGRU network** (PufferLib 5.0, native CUDA): encoder, one MinGRU layer with highway gates, decoder; trained with Muon. On $\boldsymbol{x}_{t-L+1:t}$, $\tau = t-L+1, \ldots, t$:

$$\begin{aligned}
\xi_\tau &= W_e\,(x_\tau, 1), & [\,a_\tau;\, b_\tau;\, \pi_\tau\,] &= W_l\,\xi_\tau,\\
z_\tau &= \sigma(b_\tau), & \tilde{h}_\tau &= \phi(a_\tau),\\
h_\tau &= (1 - z_\tau) \odot h_{\tau-1} + z_\tau \odot \tilde{h}_\tau, & h_{t-L} &= 0,\\
u_\tau &= \omega_\tau \odot h_\tau + (1 - \omega_\tau) \odot \xi_\tau, & \omega_\tau &= \sigma(\pi_\tau),\\
o_t &= w_d^\top u_t = g(\boldsymbol{x}_{t-L+1:t}).
\end{aligned}$$

  - $\phi(a) = a + \frac{1}{2}$ for $a \geq 0$, $\sigma(a)$ otherwise; σ: logistic sigmoid.
  - **Softplus output,** so every forecast is positive:

$$\hat{y}_t = \mathrm{softplus}(c + s\,o_t), \qquad c = \mathrm{softplus}^{-1}(\mu_y), \qquad s = \sigma_y/\sigma(c).$$

    μ_y, σ_y: training-row mean and s.d. of y; $\hat{y}_t \approx \mu_y + \sigma_y o_t$ near $o_t = 0$.
  - **Linear shortcut:** a linear term beside the recurrent branch g,

$$\hat{y}_t = \mathrm{softplus}\big(c + s\,[\,v^\top (x_t, 1) + g(\boldsymbol{x}_{t-L+1:t})\,]\big).$$

    Zero-initialised decoder: $g \equiv 0$ at the start.
  - **Ridge init,** each window's closed form ($\mathcal{F}_t$ when tuning, the full window when refitting):

$$v_0 = \arg\min_v \frac{1}{|\mathcal{F}_t|} \sum_{i \in \mathcal{F}_t} \big(v^\top (x_i, 1) - \zeta_i\big)^2 + \frac{\lambda}{2} \sum_j \mathrm{pf}_j\, v_j^2, \qquad \zeta_i = \frac{y_i - \mu_y}{\sigma_y}, \qquad \lambda = \frac{\mathrm{wd}_{\mathrm{lin}}}{\sigma_y^2}.$$

  - **Muon with decoupled decay,** every matrix $W \in \mathbb{R}^{R \times C}$, at epoch rate η_e:

$$M \leftarrow 0.95\,M + G, \qquad W \leftarrow W(1 - \eta_e\,\mathrm{wd}) - \eta_e \sqrt{\max(1, R/C)}\;\mathrm{NS}_5(G + 0.95\,M).$$

    G: gradient, global norm clipped at 1.0. NS_5: five Newton–Schulz steps.

– **Exogenous decay:** encoder input columns, in proportion to the linear stack's penalty factors,

$$W_e[:, j] \leftarrow W_e[:, j]\,(1 - \eta_e \,\mathrm{wd}_{\mathrm{enc}}\,\mathrm{pf}_j), \qquad j \le p.$$

– **Shortcut penalty:** L2 under Adam, not decoupled; intercept unpenalised:

$$\nabla_v \leftarrow \nabla_v \mathcal{L} + \mathrm{wd}_{\mathrm{lin}}\,\mathrm{pf} \odot v = \nabla_v \Big(\mathcal{L} + \tfrac{1}{2}\,\mathrm{wd}_{\mathrm{lin}} \textstyle\sum_j \mathrm{pf}_j v_j^2 \Big).$$

Adam $(0.9, 0.999)$, own clip, own rate η_{lin} on the same cosine shape.

– **Cosine schedule** over $E_{\max} = 200$ epochs:

$$\eta_e = \tfrac{1}{2}\,\eta_0 \big(1 + \cos(\pi e / E_{\max})\big), \qquad e = 0, \ldots, E_{\max} - 1.$$

– **Training loss:** squared error, or QLIKE with a smooth floor ε:

$$\ell_t = r_t - \log r_t - 1, \qquad r_t = \frac{y_t^2}{\hat{y}_t^2 + \varepsilon^2}.$$

$|\partial \ell_t / \partial \hat{y}_t|$ capped at a training-window quantile.

– **Trained, early-stopped and selected on QLIKE.**

5.3 Tuned hyperparameters

- **Linear penalties,** chosen in pre-tuning, one grid value per decade:

$$\alpha^{\mathrm{ridge}} \in \{10^{-2}, \ldots, 10^{3}\}, \qquad \alpha^{\mathrm{lasso}}, \alpha^{\mathrm{enet}} \in \{10^{-6}, \ldots, 10^{-2}\}.$$

  – all features: ridge 10^3, lasso and elastic net 10^{-2} (top edge);
  – HAR + calendar: the lower edge of each grid.
- **Trees,** shipped settings from a 1998–2002 development search:
  – LightGBM and XGBoost: the top Optuna entry;
  – random forest: trial 255 of 400 (485 trees, depth 20, max-features 0.90, max-samples 0.41);
  – leaf minimums are rescaled to the 2,000-row window: $m_{\mathrm{leaf}} \leftarrow m_{\mathrm{leaf}} \cdot 2{,}000/24{,}000$.
- **Network grid** Θ:
  – per bar: $(L, H, \eta_0) \in \{5, 20\} \times \{16, 64\} \times \{0.003, 0.01, 0.03\}$;
  – crossed with $\mathrm{wd}_{\mathrm{enc}}$, $\mathrm{wd}_{\mathrm{lin}}$ and η_{lin};
  – pooled: $(H, L, \eta_0) \in \{64, 256\} \times \{20, 48\} \times \{0.001, 0.003\}$.

5.4 Tuning procedures

- **Linear:**
  – pre-tuning on 4 folds of 250 sessions (2016–2019), chosen on validation MSE:

$$\hat{\alpha} = \arg\min_{\alpha} \frac{1}{4} \sum_{f=1}^{4} \mathrm{MSE}_f^{\mathrm{val}}(\alpha);$$

  – a light re-tune at each block: neighbours $\mathcal{C} \subseteq \{\alpha_{k-1}, \alpha_{k+1}\}$ challenge the incumbent α_k;

  - the best challenger α' replaces it only on a significant Newey–West t:

$$t\big(\{e_i^2(\alpha') - e_i^2(\alpha_k)\}_i\big) < -z_{0.05/m}, \qquad m = |\mathcal{C}|, \qquad z_a = \Phi^{-1}(1-a);$$

  - $e_i(\alpha) = y_i - \hat{y}_i(\alpha)$, i over two 125-session validation folds.
- **Trees,** in rungs:
  - shipped settings refit every 10 sessions or every session;
  - random search every 250 sessions;
  - Optuna: 50 trials at tuning frequencies of 1, 5, 25 or 250 sessions, best of the first 10, 25 or 50.
- **Network:**
  - seed average over S seeds:

$$\hat{y}_t = \frac{1}{S}\sum_{s=1}^{S} \hat{y}_t^{(s)};$$

  - the grid is re-tuned at each of 5 blocks of 250 origins, on the validation tail:

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \frac{1}{|\mathcal{V}_t|}\sum_{i\in\mathcal{V}_t} \ell\big(y_i, \hat{y}_i(\theta)\big), \qquad \ell \in \{\text{squared error}, \text{QLIKE}\};$$

  - each seed's best epoch is kept and a cold refit is made at every origin: $e^{(s)} = \arg\min_e L_e^{\text{val}}$;
  - the forecast is the mean of 5 seeds (3 pooled).
- **Every selection is causal;** the test period is never used for a choice.

5.5 Statistical tests

- **Diebold–Mariano** (Diebold and Mariano, 1995) on daily QLIKE differences, HLN correction (Harvey et al., 1997):

$$d_t = L_t^A - L_t^B, \qquad \text{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}}, \qquad \text{DM}^* = \text{DM}\sqrt{\frac{n+1-2h+h(h-1)/n}{n}}, \qquad h = 1.$$

  Two-sided normal p; n scored days.
- **Newey–West variance,** Bartlett weights:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{k=1}^{\ell}\Big(1 - \frac{k}{\ell+1}\Big)\hat{\gamma}_k, \qquad \hat{\gamma}_k = \frac{1}{n}\sum_{t=k+1}^{n}(d_t - \bar{d})(d_{t-k} - \bar{d}).$$

  - DM: $\ell = \max\big(h-1, \lfloor 4(n/100)^{2/9}\rfloor\big)$, Newey–West lag 6.
- **HAC t on daily P&L differences:** $t = \bar{d}/\sqrt{\hat{\Omega}/n}$, no HLN; $\ell = \lfloor 1.5\, n^{1/3}\rfloor$, Newey–West lag 14.
- **Model confidence set, SPA and StepM** for the feature comparison (Hansen et al., 2011).
- **Point estimates only;** no bootstrap intervals in the main tables.

Table 4: Hyperparameters per model (i): linear models and the shipped tree settings.

Model	Hyperparameter	Grid or search space	Chosen	Chosen by
Linear, per bar				
OLS	none	–	–	minimum-norm least squares
Ridge	α	$\{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$ (6)	HAR + cal. 10^{-2}; all 10^{3}	pre-tune (mean validation MSE, 4 folds), then the margin retune
Lasso	α	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	HAR + cal. 10^{-6}; all 10^{-2}	as ridge
Elastic net	α	$\{10^{-6}, 10^{-5}, \ldots, 10^{-2}\}$ (5)	HAR + cal. 10^{-6}; all 10^{-2}	as ridge
	ρ (L1 share)	fixed	0.5	set a priori
Block penalty	R_{BB}, backbone / exogenous	fixed	0.01	set a priori
Trees, per bar: shipped settings from the 1998–2002 pooled development search				
LightGBM	`num_leaves`	[15, 255]	20	top entry of the menu
	`learning_rate`	[0.01, 0.3]	0.013	`lgbm_opt_00`
	`n_estimators`	[50, 500]	392	
	`min_child_samples`	[20, 500]	$98 \to 8$	
	`feature_fraction`	[0.5, 1]	0.70	
	`bagging_fraction`	[0.5, 1]	0.56	
	`lambda_l1`	$[10^{-8}, 10]$	10^{-8}	
	`lambda_l2`	$[10^{-8}, 10]$	0.016	
XGBoost	`max_depth`	[3, 10]	5	top entry of the menu
	`learning_rate`	[0.01, 0.3]	0.0123	`xgb_opt_00`
	`n_estimators`	[50, 500]	367	
	`min_child_weight`	[1, 100]	$50.2 \to 4.18$	
	`subsample`	[0.5, 1]	0.66	
	`colsample_bytree`	[0.5, 1]	0.65	
	`reg_alpha`	$[10^{-8}, 10]$	0.0215	
	`reg_lambda`	$[10^{-8}, 10]$	1.2×10^{-5}	
	`gamma`	$[10^{-8}, 1]$	1.5×10^{-6}	
Random forest	`n_estimators`	[50, 500]	485	top entry of the menu
	`max_depth`	[2, 32]	20	`rf_opt_00`
	`max_features`	[0.02, 1]	0.90	trial 255 of 400
	`max_samples`	[0.05, 1]	0.41	
	`min_samples_leaf`	[1, 1000]	$12 \to 1$	

Notes: Linear: sklearn units, one grid value per decade (`specs/causal_tune_linear.py`); picks are the pre-tuned penalties on the HAR + calendar (HAR + cal.) and all-features designs. The margin retune at the 4 block starts then switched (HAR + cal. and all): ridge 1 and 0; lasso 1 and 0; elastic net 2 and 0. R_{BB} multiplies the penalty on the HAR + calendar columns; the intercept is unpenalised. Trees: the search box of the development menu (LightGBM and XGBoost: `experiments/tree_menu_dev.py`; forest: its 400-trial study), searched by Optuna on the pooled 24,000-bar window and ranked by development-span MSE; $a \to b$: a leaf minimum tuned on 24,000 rows, scaled to the 2,000-row per-bar window. Fixed: LightGBM `bagging_freq` = 1; XGBoost `tree_method` = hist; forest `bootstrap` = True. Seed 42.

Table 5: Hyperparameters per model (ii): searched in the walk.

Model	Hyperparameter	Search space or grid	Chosen	Chosen by
Trees, per bar, re-tuned in the walk: Optuna TPE, 50 trials, trial 0 = shipped				
LightGBM	`num_leaves`	[2, 128] (log)	per tuning point	validation MSE
	`min_child_samples`	[1, 256] (log)		(`tp25_q`: QLIKE)
	`feature_fraction`	[0.05, 1]		
	`bagging_fraction`	[0.3, 1]		
	`lambda_l2`	$[0.001, 10^4]$ (log)		
	`learning_rate`	[0.005, 0.1] (log)		
XGBoost	`max_depth`	[1, 8]	per tuning point	as LightGBM
	`min_child_weight`	[0.1, 256] (log)		
	`subsample`	[0.15, 1]		
	`colsample_bytree`	[0.05, 1]		
	`reg_lambda`	$[10^{-5}, 10^4]$ (log)		
	`learning_rate`	[0.005, 0.1] (log)		
Random forest	`max_features`	[0.02, 1]	per tuning point	as LightGBM
	`min_samples_leaf`	[1, 256] (log)		
	`max_depth`	$\{2, 4, 8, 16, 32, \text{none}\}$		
Random search	the same axes	32 fixed candidates from 700 / 2,800 / 112 configurations	per tuning point	validation MSE
MinGRU network				
Per bar, base (12 configurations)	L, sequence length	$\{5, 20\}$	20 (6/10)	lowest validation MSE
	H, width	$\{16, 64\}$	16 (7/10)	of the 5-seed mean
	learning rate	$\{0.003, 0.01, 0.03\}$	0.03 (10/10)	
Per bar, QLIKE (216 configurations)	L, sequence length	$\{5, 20\}$	20 (8/10)	lowest validation QLIKE
	H, width	$\{16, 64\}$	64 (7/10)	of the 5-seed mean
	learning rate	$\{0.003, 0.01, 0.03\}$	0.03 (9/10)	
	w_{enc}, encoder decay	$\{0, 3, 15\}$	0 (4/10)	
	w_{lin}, shortcut L2	$\{0.02, 0.2, 2\}$	2 (5/10)	
	shortcut learning rate	$\{0.0003, 0.001\}$	0.0003 (5/10)	
Pooled (48 configurations)	L, sequence length	$\{20, 48\}$	48 (6/10)	lowest validation MSE
	H, width	$\{64, 256\}$	64 (8/10)	of the 3-seed mean
	learning rate	$\{0.001, 0.003\}$	0.003 (9/10)	
	weight decay	$\{0, 0.01, 0.03, 0.1, 0.3, 1\}$	0.3 (6/10)	

Notes: Re-tuned trees: log: a log-uniform axis; early stopping with patience $\lceil 1/\text{lr} \rceil$ and a cap of 4× the shipped shrinkage budget; best of the first 10, 25, 50 trials recorded; the XGBoost `reg_lambda` floor is 10^{-5} so that the shipped value is inside; other parameters stay shipped. Random search: the shipped configuration and 31 drawn once (seed 20260929) from the LightGBM / XGBoost / forest product grids. MinGRU: in parentheses, the grid size per design; base, the softplus network trained on squared error; QLIKE, the network with the linear shortcut started at the window's ridge, trained, early-stopped and selected on QLIKE; pooled, the 48-bar network. w_{enc}: decay on the encoder's input columns, scaled by the linear penalty factors; w_{lin}: the shortcut's L2 penalty. Fixed: 1 layer, batch 128, at most 200 epochs (pooled 100), patience 20 (pooled 10), Muon momentum 0.95, gradient clip 1.0, cosine schedule, softplus output. Chosen: the modal pick over the 5 blocks × 2 designs, with its count.

Table 6: Tuning protocol per model family.

Family	Window	Refit	Tuning	Validation tail / embargo	Criterion	Seeds
Per bar (16:00 design)						
OLS	2,000 sessions	every session	none	–	–	none
Ridge, lasso, elastic net	2,000 sessions	every session	pre-tune once (4 folds × 250 sessions); margin retune every 250 sessions	pre-tune: 250 / 25 sessions; retune: 2 × 125 / 25	validation MSE; a switch needs HAC $t < -z_{0.05/m}$	none
Trees, shipped	2,000 sessions	every 10 or 1 sessions	none (frozen)	–	–	42
Trees, random search	2,000 sessions	every 10 sessions	every 250 sessions	125 / 25 sessions	validation MSE	42; grid 20260929
Trees, Optuna	2,000 sessions	every session	every 1, 5, 25, 250 sessions	125 / 25 sessions	validation MSE (`tp25_q`: QLIKE)	42; TPE 20260929 + row
MinGRU	2,000 sessions	every session, cold	every 250 sessions (5 blocks)	125 / 25 sessions	validation MSE or QLIKE of the seed mean; epochs frozen per block	5 (20261012–16)
Pooled (all 48 bars)						
Ridge, lasso, elastic net	96,000 bars	every bar	pre-tune once (4 folds × 12,000 bars); margin retune every 250 bars	pre-tune: 12,000 / 1,200 bars; retune: 125 / 25	as per bar	none
Tree bank (20 experts)	24,000 bars	every bar	configurations frozen; expert chosen every 250 bars	125 / 25 bars	tail QLIKE: the best expert, or exponential weights	42
MinGRU	24,000 bars	every 21 sessions	5 blocks, as per bar	2,400 / 48 bars	validation MSE of the seed mean	3 (20261012–14)

Notes: Every split is forward: fit block, embargo, then the validation tail, all inside the trailing window; the test period is never used for a choice. Validation tail / embargo in sessions or bars; the linear pre-tune folds tile 2016-01-06 to 2020-01-02. Seeds: none, the fit is deterministic. m: the number of challengers (the two grid neighbours). Fit, tuning and selection use squared error on the fit scale ($\sqrt{RV/B}$) unless QLIKE is named. Trees, Optuna: a fresh TPE study at each tuning point; the configuration in force at a row is the best of the latest point. MinGRU: each seed is refitted for its own best epoch, and the forecast is the seed mean. The tree bank's experts refit at every bar (part of the bank every 10th bar).

6 Results

6.1 Scoring convention

- Forecast $\hat{\sigma}_t^2 = \hat{y}_t^2 B_t$ (plain back-transform); target RV_t, 15:30–16:00; $N = 866$ trade days.
- Daily loss and mean QLIKE:

$$L_t = \mathrm{QLIKE}(RV_t, \hat{\sigma}_t^2) = \frac{RV_t}{\hat{\sigma}_t^2} - \log \frac{RV_t}{\hat{\sigma}_t^2} - 1, \qquad \overline{\mathrm{QLIKE}} = \frac{1}{N}\sum_{t=1}^{N} L_t.$$

- Reference: per-bar OLS on HAR + calendar; QLIKE 0.0946, Sharpe 1.37.
- Change vs reference, with $d_t = L_t^{(a)} - L_t^{(\mathrm{ref})}$:

$$\Delta\% = 100\,\frac{\bar{d}}{\bar{L}^{(\mathrm{ref})}}, \qquad \bar{d} = \frac{1}{N}\sum_{t=1}^{N} d_t.$$

- Diebold–Mariano, HLN factor at $h = 1$, two-sided normal p:

$$\mathrm{DM} = \sqrt{\frac{N-1}{N}}\,\frac{\bar{d}}{\sqrt{\hat{\Omega}/N}}, \qquad p = 2\{1 - \Phi(|\mathrm{DM}|)\}.$$

- Newey–West long-run variance, Bartlett weights:

$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{\ell} w_j\,\hat{\gamma}_j, \qquad w_j = 1 - \frac{j}{\ell+1}, \qquad \hat{\gamma}_j = \frac{1}{N-j}\sum_{t=j+1}^{N} (d_t - \bar{d})(d_{t-j} - \bar{d}).$$

- Lag $\ell = \max(h-1, \lfloor 4(N/100)^{2/9} \rfloor) = 6$. DM < 0 favours a.
- Variance-space errors:

$$\mathrm{MSE} = \frac{1}{N}\sum_{t=1}^{N} (RV_t - \hat{\sigma}_t^2)^2, \qquad \mathrm{MAE} = \frac{1}{N}\sum_{t=1}^{N} |RV_t - \hat{\sigma}_t^2|.$$

- Out-of-sample R^2, benchmark the full-sample mean:

$$R^2_{\mathrm{OOS}} = 1 - \frac{\sum_t (RV_t - \hat{\sigma}_t^2)^2}{\sum_t (RV_t - \overline{RV})^2}, \qquad \overline{RV} = \frac{1}{N}\sum_{t=1}^{N} RV_t.$$

- Calibration: $\overline{RV}/\overline{\hat{\sigma}^2}$ and $\frac{1}{N}\sum_t 1\{RV_t > \hat{\sigma}_t^2\}$.

6.2 Axis 1: HAR + calendar vs all features

- **Paired change,** same family, $d_t = L_t^{(\mathrm{all})} - L_t^{(\mathrm{base})}$:

$$\Delta\% = 100\Big(\frac{\bar{L}^{(\mathrm{all})}}{\bar{L}^{(\mathrm{base})}} - 1\Big) = 100\,\frac{\bar{d}}{\bar{L}^{(\mathrm{base})}}, \qquad \text{DM on } d_t,\ \ell = 6.$$

- $\Delta\% < 0$ **in most families;** largest, and only significant, gains in the pooled trees.
- **51 pairs:** $\bar{d} < 0$ in 38, 2 with $p < 0.05$; $\Delta\mathrm{Sharpe} > 0$ in 50, 4 significant.
- **Squared error disagrees:** $\Delta\mathrm{MSE} > 0$ in all 51 pairs, none significantly.
  - Five days in March 2020 carry 83–91% of $\sum_t (RV_t - \hat{\sigma}_t^2)^2$.
  - Excluding February–April 2020: all-features lasso beats the OLS on MSE.
- **Network** (linear shortcut, QLIKE training): $\Delta\% = -3.8$, not significant; ties elastic net and ridge on all features.

Family	QLIKE, HAR + cal.	QLIKE, all	Change	DM
Per-bar ridge	0.0946	0.0958	+1.2%	+0.26
Per-bar lasso	0.0946	0.0914	−3.4%	−1.22
Per-bar elastic net	0.0946	0.0942	−0.4%	−0.10
Per-bar LightGBM	0.0986	0.0939	−4.7%	−1.43
Per-bar XGBoost	0.0976	0.0935	−4.2%	−1.41
Per-bar random forest	0.0968	0.0966	−0.2%	−0.05
Per-bar MinGRU (QLIKE-tuned)	0.0985	0.0948	−3.8%	$p = 0.34$
Pooled ridge / lasso / elastic net	0.107	0.102 / 0.104 / 0.102	−4.3/ − 1.7/ − 4.7%	−1.29/ − 0.50/ − 1.42
Pooled LightGBM / XGBoost / RF	0.121–0.124	0.106–0.107	−12 to −14%	about −4.5

Table 7: All features vs HAR + calendar within each family, on 866 days (paired DM; negative favours all features).

6.3 Axis 2: per bar vs pooled

- **Pooled minus per-bar,** $\Delta\% = 100(\bar{L}^{(\text{pool})}/\bar{L}^{(\text{bar})} - 1) > 0$ in every family:
  - linear: +12–13% on HAR + calendar (DM about 3.4), +7–14% on all features;
  - trees: +24–27% on HAR + calendar, +11–14% on all features.
- ΔSharpe**, pooled vs per-bar:** none significant.
- **Pooled network:** *[pending: the pooled ridge-init, QLIKE-tuned walk].*

6.4 Model families

- **Leading group, all on all features,** within noise of the OLS ($\overline{\text{QLIKE}} = 0.0946$):
  - lasso 0.0914, XGBoost 0.0935, LightGBM 0.0939, elastic net 0.0942, MinGRU 0.0948;
  - none significantly better than the OLS.
- **Trees vs per-bar ridge, same inputs:** $\bar{d} > 0$ in 46 of 48, 18 significantly.
- **Pooled trail per-bar;** weakest: pooled trees on HAR + calendar (0.121–0.124).

Table 8: Axis 1: HAR + calendar against all features, within each family. QLIKE of the two input sets, the change (all features minus HAR + calendar), its Diebold–Mariano test and the change in the trade's Sharpe ratio.

	QLIKE		All vs HAR + cal.			Sharpe change	
Model	HAR + cal.	All	Δ (%)	DM	p	ΔSR	t
Per-bar (one model per bar)							
Ridge	0.0946	0.0958	+1.2	+0.26	0.799	+0.05	+0.09
Lasso	0.0946	0.0914	−3.4	−1.22	0.223	+0.50	+1.47
Elastic net	0.0946	0.0942	−0.4	−0.10	0.920	+0.55	+1.34
LightGBM	0.0986	0.0939	−4.7	−1.43	0.153	+0.54	+1.11
XGBoost	0.0976	0.0935	−4.2	−1.41	0.159	+0.16	+0.30
Random forest	0.0968	0.0966	−0.2	−0.05	0.959	+0.18	+0.34
MinGRU	0.0985	0.0948	−3.8	−0.96	0.336	+0.49	+0.95
Pooled (one model for all 48 bars)							
Ridge	0.1067	0.1021	−4.3	−1.29	0.198	+0.53	+1.12
Lasso	0.1062	0.1044	−1.7	−0.50	0.614	+0.51	+1.19
Elastic net	0.1065	0.1016	−4.7	−1.42	0.156	+0.61	+1.21
LightGBM	0.1236	0.1060	−14.3	**−4.57**	**<0.001**	+0.51	+1.02
XGBoost	0.1213	0.1063	−12.4	**−4.46**	**<0.001**	+0.30	+0.87
Random forest†	0.1229	0.1073	−12.7	**−4.57**	**<0.001**	+0.30	+0.70
MinGRU	0.1151	0.1121	−2.7	−0.64	0.524	−0.45	−1.11

Notes. QLIKE is the mean over the 866 trade days (2020-01-03 to 2024-04-30, early closes excluded) of the daily loss $L_t = RV_t/\hat{\sigma}_t^2 - \log(RV_t/\hat{\sigma}_t^2) - 1$, with the plain forecast $\hat{\sigma}_t^2 = \hat{y}_t^2 B_t$ (no recalibration) and RV_t the winsorised 15:30–16:00 realised variance. $\Delta = 100(\overline{L}^{\text{all}}/\overline{L}^{\text{HAR+cal.}} - 1)$. DM is the Diebold–Mariano statistic of the daily QLIKE difference (all features minus HAR + calendar, same model), with a Bartlett Newey–West variance at lag 6 and the Harvey–Leybourne–Newbold correction; p is two-sided (normal). Negative Δ and DM favour all features. ΔSR is the change in the annualised Sharpe ratio of the midpoint straddle trade (sign of forecast minus implied variance); t is the Newey–West t (lag 14) of the daily return difference. Bold: $p < 0.05$ (DM, or the t against the normal). Per-bar linear models are pre-tuned; the per-bar trees use fixed settings refit every session; the MinGRU is the ridge-initialised network trained and selected on QLIKE (seed mean: 5 per-bar, 3 pooled). The pooled MinGRU is refit monthly, each refit on its trailing 24,000 bars; the pooled trees are refit at every bar. The pooled LightGBM and XGBoost on all features are the earlier paper's pooled trees. †Interim: the pooled forest's 16:00 forecasts on all features; its full every-bar walk is still running. Across all 51 input-set pairs of the model set, all features lower QLIKE in 38 (2 with $p < 0.05$) and raise variance-space MSE in 51 (0 with $p < 0.05$).

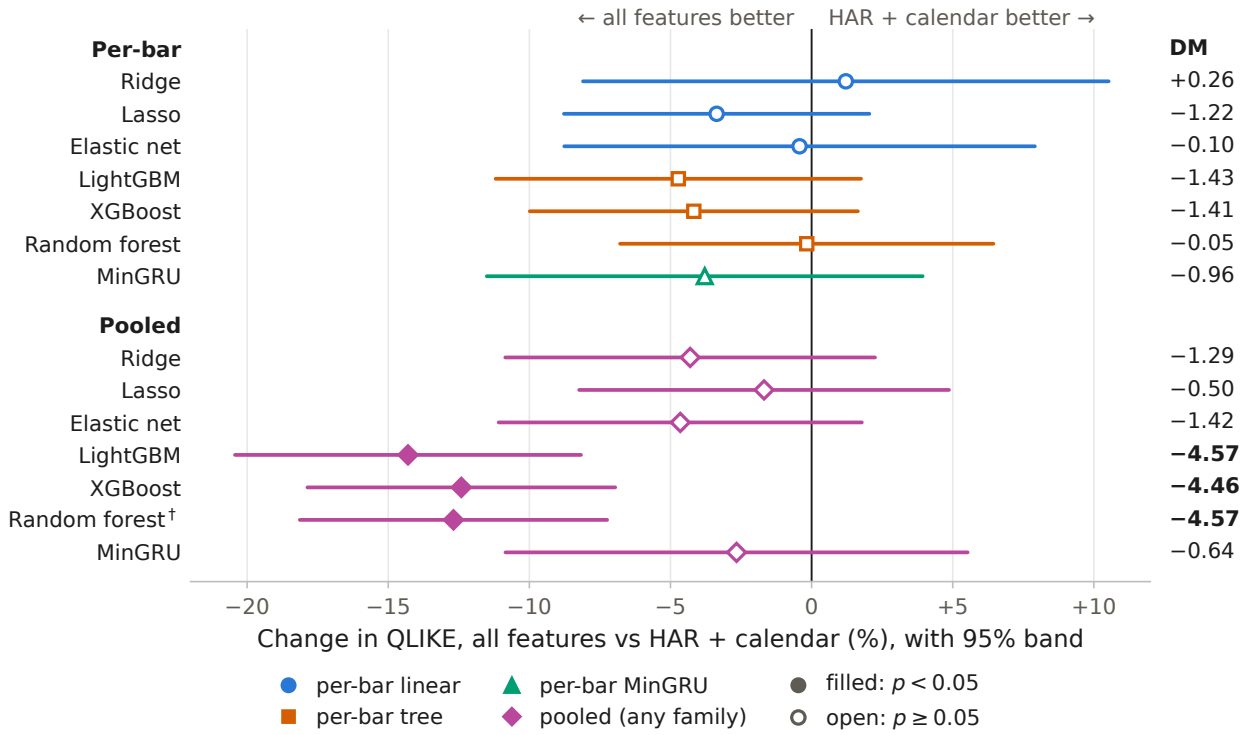


Figure 1: All features against HAR + calendar, within each family: the change in mean QLIKE (all features minus HAR + calendar, in % of the HAR + calendar value) with its 95% band, $\Delta \pm 1.96\,|\Delta/\mathrm{DM}|$ (the interval the Diebold–Mariano test inverts), and the DM statistic on the right (bold: $p < 0.05$). Left of zero, all features forecast better. Colour and marker give the family for per-bar fits; every pooled fit is a diamond. Filled markers have $p < 0.05$. †Interim (the pooled forest's 16:00 forecasts on all features). Numbers in Table 8.

Table 9: Axis 2: pooled against per-bar, by family and input set. QLIKE change (pooled minus per-bar), its Diebold–Mariano test and the t of the trade's return difference.

	HAR + calendar				All features			
Model	Δ (%)	DM	p	t	Δ (%)	DM	p	t
Linear								
Ridge	+12.8	**+3.42**	**<0.001**	−1.05	+6.6	+1.56	0.118	+0.07
Lasso	+12.2	**+3.42**	**<0.001**	−1.69	+14.2	**+3.19**	**0.001**	−1.66
Elastic net	+12.6	**+3.50**	**<0.001**	−1.12	+7.8	**+2.00**	**0.046**	−0.88
Trees								
LightGBM	+25.4	**+4.37**	**<0.001**	+0.36	+12.8	**+2.74**	**0.006**	+0.15
XGBoost	+24.4	**+4.50**	**<0.001**	+0.01	+13.7	**+2.98**	**0.003**	+0.30
Random forest†	+27.0	**+4.75**	**<0.001**	+0.95	+11.1	**+2.71**	**0.007**	+0.85
Neural network								
MinGRU	+16.8	**+3.29**	**<0.001**	−0.62	+18.2	**+4.00**	**<0.001**	**−2.30**

Notes. QLIKE is the mean over the 866 trade days (2020-01-03 to 2024-04-30, early closes excluded) of the daily loss $L_t = RV_t/\hat{\sigma}_t^2 - \log(RV_t/\hat{\sigma}_t^2) - 1$, with the plain forecast $\hat{\sigma}_t^2 = \hat{y}_t^2 B_t$ (no recalibration) and RV_t the winsorised 15:30–16:00 realised variance. $\Delta = 100(\overline{L}^{\text{pooled}}/\overline{L}^{\text{per-bar}} - 1)$, the same estimator and input set. DM is the Diebold–Mariano statistic of the daily QLIKE difference (pooled minus per-bar), with a Bartlett Newey–West variance at lag 6 and the Harvey–Leybourne–Newbold correction; p is two-sided (normal). Positive Δ and DM favour the per-bar fit. t is the Newey–West t (lag 14) of the daily difference of the midpoint straddle returns (pooled minus per-bar). Bold: $p < 0.05$. Per-bar trees: fixed settings refit every session; pooled trees: the bank's walk refit at every bar (24,000-bar window); linear models are pre-tuned in both fits. The pooled MinGRU is refit monthly (each refit on its trailing 24,000 bars), not at every bar like the pooled trees; both MinGRU fits are ridge-initialised and trained and selected on QLIKE. †The pooled forest on all features is interim (16:00 forecasts; the full walk is still running).

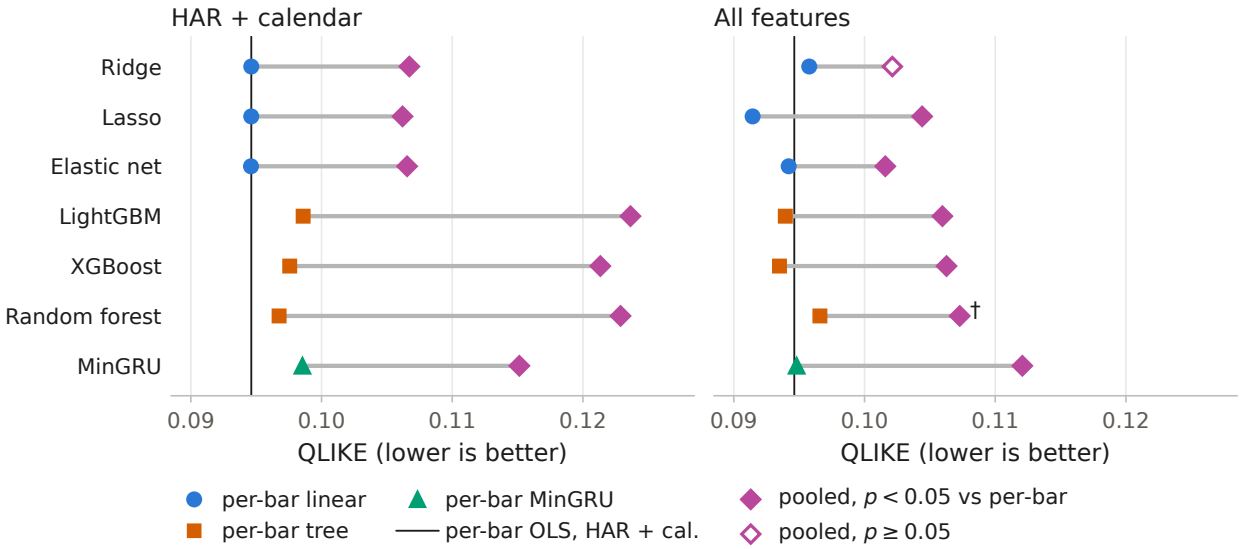


Figure 2: Per-bar against pooled fits. Mean QLIKE of each family's per-bar fit (family colour and marker) and its pooled fit (diamond), on HAR + calendar (left) and all features (right), on the same 866 days; the vertical line is the per-bar OLS on HAR + calendar (QLIKE 0.0946). A filled diamond is a pooled fit significantly worse than its per-bar twin (Diebold–Mariano $p < 0.05$). Per-bar trees use fixed settings refit every session. †Interim. Numbers in Table 9.

Table 10: The headline forecasts ranked by QLIKE, against the per-bar OLS on HAR + calendar. Rank is the position among all 120 forecasts of Table 11.

					vs per-bar OLS		
Rank	Model	Fit	Inputs	QLIKE	Δ (%)	DM	p
1	Lasso	per-bar	All	0.0914	−3.4	−1.22	0.222
2	XGBoost, fixed, daily refit	per-bar	All	0.0935	−1.2	−0.39	0.697
3	LightGBM, fixed, daily refit	per-bar	All	0.0939	−0.7	−0.21	0.831
4	Elastic net	per-bar	All	0.0942	−0.5	−0.11	0.915
8	**OLS** (reference)	per-bar	HAR + cal.	**0.0946**			
9	MinGRU (ridge-init, QLIKE-tuned)	per-bar	All	0.0948	+0.2	+0.05	0.962
11	Ridge	per-bar	All	0.0958	+1.2	+0.25	0.799
14	Random forest, fixed, daily refit	per-bar	All	0.0966	+2.1	+0.61	0.543
16	Random forest, fixed, daily refit	per-bar	HAR + cal.	0.0968	+2.2	+1.09	0.276
21	XGBoost, fixed, daily refit	per-bar	HAR + cal.	0.0976	+3.1	+1.43	0.154
37	MinGRU (ridge-init, QLIKE-tuned)	per-bar	HAR + cal.	0.0985	+4.1	+1.88	0.060
38	LightGBM, fixed, daily refit	per-bar	HAR + cal.	0.0986	+4.2	+1.92	0.054
77	Elastic net	pooled	All	0.1016	+7.4	+1.50	0.134
81	Ridge	pooled	All	0.1021	+7.9	+1.70	0.090
93	Lasso	pooled	All	0.1044	+10.3	**+1.96**	**0.0497**
101	LightGBM	pooled	All	0.1060	+12.0	**+2.28**	**0.022**
102	Lasso	pooled	HAR + cal.	0.1062	+12.2	**+3.41**	**<0.001**
104	XGBoost	pooled	All	0.1063	+12.3	**+2.43**	**0.015**
105	Elastic net	pooled	HAR + cal.	0.1065	+12.6	**+3.49**	**<0.001**
107	Ridge	pooled	HAR + cal.	0.1067	+12.8	**+3.42**	**<0.001**
108	Block-diagonal ridge, earlier panel	pooled	All	0.1068	+12.8	**+2.38**	**0.017**
111	Random forest†	pooled	All	0.1073	+13.4	**+2.53**	**0.011**
115	MinGRU (ridge-init, QLIKE-tuned)	pooled	All	0.1121	+18.4	**+4.53**	**<0.001**
116	MinGRU (ridge-init, QLIKE-tuned)	pooled	HAR + cal.	0.1151	+21.7	**+4.88**	**<0.001**
117	XGBoost	pooled	HAR + cal.	0.1213	+28.2	**+5.03**	**<0.001**
118	Random forest	pooled	HAR + cal.	0.1229	+29.9	**+5.04**	**<0.001**
119	OLS, earlier panel	pooled	HAR + cal.	0.1232	+30.2	**+4.80**	**<0.001**
120	LightGBM	pooled	HAR + cal.	0.1236	+30.7	**+5.02**	**<0.001**

Notes. QLIKE is the mean over the 866 trade days (2020-01-03 to 2024-04-30, early closes excluded) of the daily loss $L_t = RV_t/\hat{\sigma}_t^2 - \log(RV_t/\hat{\sigma}_t^2) - 1$, with the plain forecast $\hat{\sigma}_t^2 = \hat{y}_t^2 B_t$ (no recalibration) and RV_t the winsorised 15:30–16:00 realised variance. $\Delta = 100(\overline{L}/\overline{L}^{\,\mathrm{OLS}} - 1)$. DM is the Diebold–Mariano statistic of the daily QLIKE difference (model minus the per-bar OLS), with a Bartlett Newey–West variance at lag 6 and the Harvey–Leybourne–Newbold correction; p is two-sided (normal). Negative Δ and DM favour the model. Bold: $p < 0.05$. Rows: the per-bar OLS, the canonical rung of each family on each input set (per-bar trees with fixed settings refit every session) and the earlier paper's pooled OLS and block-diagonal ridge. The per-bar ridge, lasso and elastic net on HAR + calendar are within 0.03 % of the OLS and are not shown. †Interim (see Table 8). The pooled MinGRU is refit monthly, the pooled trees at every bar. Of the 111 model-set forecasts other than the OLS, 7 have lower QLIKE (0 with $p < 0.05$) and 48 are significantly worse.

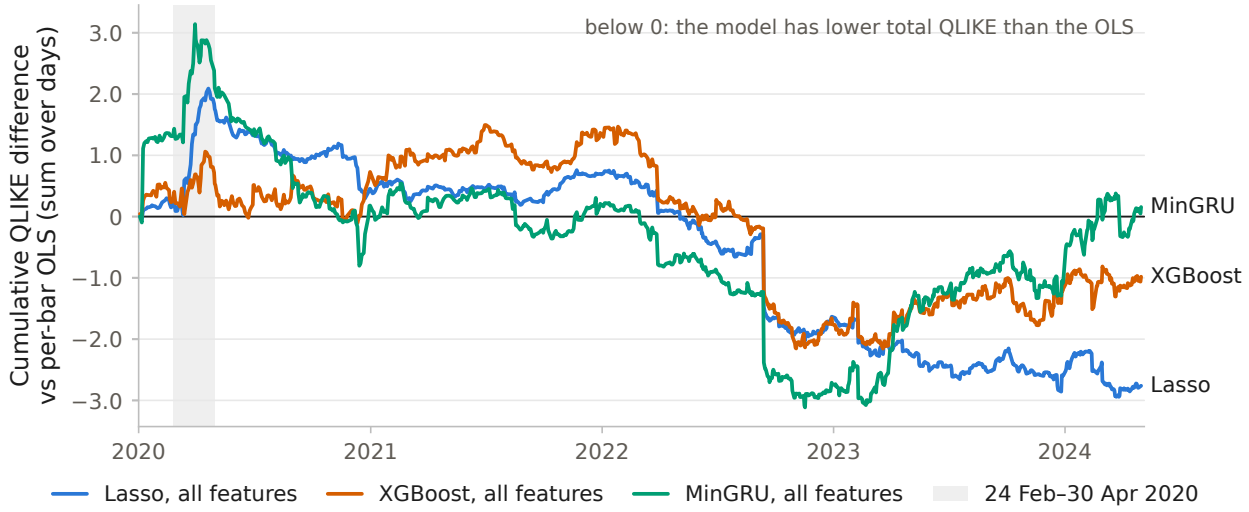


Figure 3: Cumulative daily QLIKE difference against the per-bar OLS on HAR + calendar, $\sum_{s\le t}(L_s^{\text{model}} - L_s^{\text{OLS}})$, for the best per-bar forecast of each family, all on all features: the lasso, XGBoost (fixed settings, refit every session) and the MinGRU (ridge-initialised, trained and selected on QLIKE). Below zero, the model has accumulated less QLIKE than the OLS; the end value divided by 866 is the difference in mean QLIKE. Over the shaded window, 24 February to 30 April 2020, the sums change by +1.60 (lasso), +0.30 (XGBoost) and +0.78 (MinGRU); over the full sample they end at −2.76, −0.98 and +0.16.

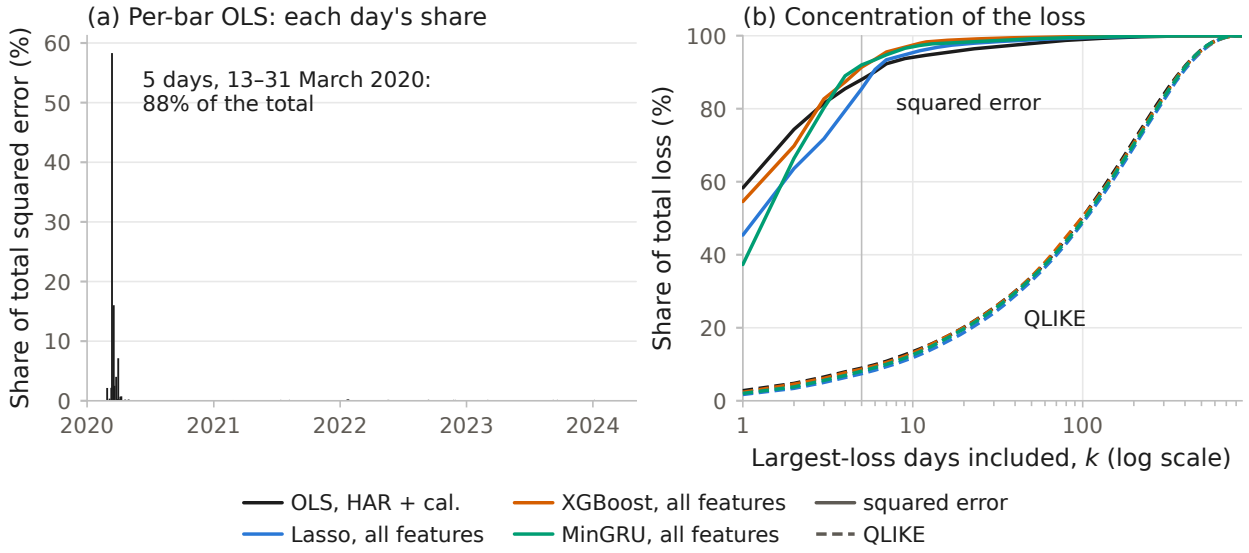


Figure 4: Concentration of the loss. (a) Each day's share of the per-bar OLS's total squared error, $\sum_t (RV_t - \hat{\sigma}_t^2)^2$: its five largest days, all in March 2020, carry 88% of it. (b) Share of the total loss carried by the k largest-loss days, for squared error (solid) and QLIKE (dashed), for the OLS and the three forecasts of Figure 3. For each of the four, the five largest squared errors fall in March 2020 and carry 85–92% of the total squared error; the five largest QLIKE days carry 7–9% of the total QLIKE. Across the 51 input-set pairs, all features raise the squared error in 51 (0 significantly) while lowering QLIKE in 38.

Table 11: Every forecast ranked by QLIKE, against the per-bar OLS on HAR + calendar (120 forecasts: the 112-forecast model set, the per-bar and pooled MinGRU and the pooled trees outside the master table).

Rank	Model	Fit	Inputs	QLIKE	Δ (%)	DM	p
1	Lasso	per-bar	All	0.0914	−3.4	−1.22	0.222
2	XGBoost, fixed, daily refit	per-bar	All	0.0935	−1.2	−0.39	0.697
3	LightGBM, fixed, daily refit	per-bar	All	0.0939	−0.7	−0.21	0.831
4	Elastic net	per-bar	All	0.0942	−0.5	−0.11	0.915
5	Elastic net	per-bar	HAR + cal.	0.0946	0.0	−1.19	0.235
6	Lasso	per-bar	HAR + cal.	0.0946	0.0	−1.38	0.169
7	Ridge	per-bar	HAR + cal.	0.0946	0.0	−0.73	0.467
8	**OLS** (reference)	per-bar	HAR + cal.	**0.0946**			
9	MinGRU (ridge-init, QLIKE-tuned)*	per-bar	All	0.0948	+0.2	+0.05	0.962
10	LightGBM, rand. search (QLIKE), daily refit	per-bar	All	0.0956	+1.0	+0.29	0.773
11	Ridge	per-bar	All	0.0958	+1.2	+0.25	0.799
12	LightGBM, fixed, refit 10	per-bar	All	0.0962	+1.7	+0.47	0.635
13	Random forest, rand. search (QLIKE), refit 10	per-bar	HAR + cal.	0.0965	+2.0	+0.95	0.342
14	Random forest, fixed, daily refit	per-bar	All	0.0966	+2.1	+0.61	0.543
15	LightGBM, rand. search (SE), daily refit	per-bar	All	0.0966	+2.1	+0.54	0.589
16	Random forest, fixed, daily refit	per-bar	HAR + cal.	0.0968	+2.2	+1.09	0.276
17	XGBoost, fixed, refit 10	per-bar	All	0.0970	+2.5	+0.65	0.518
18	Random forest, rand. search (QLIKE), daily refit	per-bar	HAR + cal.	0.0971	+2.6	+1.25	0.211
19	XGBoost, fixed, refit 10	per-bar	HAR + cal.	0.0974	+3.0	+1.41	0.159
20	LightGBM, rand. search (QLIKE), refit 10	per-bar	All	0.0975	+3.0	+0.72	0.470
21	XGBoost, fixed, daily refit	per-bar	HAR + cal.	0.0976	+3.1	+1.43	0.154
22	XGBoost, Optuna (SE), tune 1, top 25	per-bar	All	0.0976	+3.2	+0.78	0.436
23	Random forest, rand. search (QLIKE), daily refit	per-bar	All	0.0976	+3.2	+0.93	0.352
24	Random forest, Optuna (SE), tune 25, top 10	per-bar	All	0.0979	+3.5	+0.96	0.336
25	Random forest, Optuna (SE), tune 1, top 25	per-bar	All	0.0979	+3.5	+1.00	0.317
26	Random forest, Optuna (SE), tune 1, top 10	per-bar	All	0.0981	+3.7	+1.04	0.299
27	Random forest, fixed, refit 10	per-bar	All	0.0982	+3.7	+1.09	0.277
28	LightGBM, rand. search (SE), refit 10	per-bar	All	0.0982	+3.8	+0.85	0.396
29	Random forest, fixed, refit 10	per-bar	HAR + cal.	0.0982	+3.8	+1.79	0.073
30	XGBoost, Optuna (SE), tune 5, top 50	per-bar	All	0.0982	+3.8	+0.99	0.321
31	LightGBM, Optuna (SE), tune 250, top 50	per-bar	All	0.0983	+3.8	+0.85	0.395
32	Random forest, Optuna (QLIKE), tune 25, top 50	per-bar	HAR + cal.	0.0983	+3.9	+1.77	0.077
33	XGBoost, rand. search (SE), daily refit	per-bar	All	0.0984	+4.0	+0.94	0.347
34	Random forest, Optuna (SE), tune 1, top 50	per-bar	All	0.0985	+4.1	+1.12	0.261
35	Random forest, Optuna (SE), tune 5, top 50	per-bar	All	0.0985	+4.1	+1.08	0.279
36	XGBoost, rand. search (QLIKE), daily refit	per-bar	All	0.0985	+4.1	+0.96	0.338
37	MinGRU (ridge-init, QLIKE-tuned)*	per-bar	HAR + cal.	0.0985	+4.1	+1.88	0.060
38	LightGBM, fixed, daily refit	per-bar	HAR + cal.	0.0986	+4.2	+1.92	0.054
39	Random forest, Optuna (SE), tune 1, top 25	per-bar	HAR + cal.	0.0988	+4.4	**+2.03**	**0.043**
40	Random forest, Optuna (SE), tune 1, top 10	per-bar	HAR + cal.	0.0989	+4.5	**+2.03**	**0.042**
41	Random forest, Optuna (SE), tune 5, top 50	per-bar	HAR + cal.	0.0990	+4.6	**+2.00**	**0.045**
42	Random forest, Optuna (SE), tune 1, top 50	per-bar	HAR + cal.	0.0990	+4.6	+1.94	0.053
43	Random forest, Optuna (SE), tune 25, top 25	per-bar	HAR + cal.	0.0990	+4.7	**+2.06**	**0.040**
44	XGBoost, Optuna (SE), tune 250, top 50	per-bar	All	0.0991	+4.7	+0.96	0.336
45	Random forest, Optuna (SE), tune 25, top 50	per-bar	HAR + cal.	0.0992	+4.9	**+2.20**	**0.028**
46	XGBoost, Optuna (SE), tune 1, top 50	per-bar	All	0.0993	+4.9	+1.28	0.201
47	LightGBM, fixed, refit 10	per-bar	HAR + cal.	0.0993	+5.0	**+2.20**	**0.028**
48	LightGBM, Optuna (SE), tune 5, top 50	per-bar	All	0.0994	+5.1	+1.26	0.208
49	Random forest, Optuna (SE), tune 25, top 10	per-bar	HAR + cal.	0.0995	+5.2	**+2.15**	**0.032**
50	Random forest, Optuna (SE), tune 25, top 25	per-bar	All	0.0996	+5.2	+1.47	0.141
51	Random forest, Optuna (QLIKE), tune 25, top 50	per-bar	All	0.0997	+5.3	+1.39	0.165
52	XGBoost, Optuna (SE), tune 1, top 10	per-bar	HAR + cal.	0.0999	+5.5	**+2.21**	**0.027**
53	XGBoost, Optuna (SE), tune 5, top 50	per-bar	HAR + cal.	0.0999	+5.6	**+2.20**	**0.028**

continued on the next page

Table 11, continued

Rank	Model	Fit	Inputs	QLIKE	Δ (%)	DM	p
54	Random forest, Optuna (SE), tune 250, top 50	per-bar	All	0.1001	+5.8	+1.58	0.113
55	LightGBM, rand. search (QLIKE), daily refit	per-bar	HAR + cal.	0.1002	+5.9	**+2.64**	**0.008**
56	Random forest, rand. search (QLIKE), refit 10	per-bar	All	0.1003	+6.0	+1.62	0.105
57	LightGBM, Optuna (SE), tune 25, top 25	per-bar	All	0.1003	+6.0	+1.32	0.187
58	XGBoost, Optuna (SE), tune 25, top 10	per-bar	HAR + cal.	0.1003	+6.0	**+2.48**	**0.013**
59	LightGBM, rand. search (QLIKE), refit 10	per-bar	HAR + cal.	0.1004	+6.1	**+2.73**	**0.006**
60	LightGBM, Optuna (SE), tune 1, top 25	per-bar	All	0.1004	+6.1	+1.53	0.125
61	XGBoost, rand. search (QLIKE), refit 10	per-bar	All	0.1006	+6.3	+1.37	0.170
62	LightGBM, Optuna (SE), tune 1, top 10	per-bar	All	0.1006	+6.4	+1.47	0.143
63	XGBoost, rand. search (SE), refit 10	per-bar	All	0.1007	+6.4	+1.39	0.164
64	XGBoost, Optuna (SE), tune 25, top 50	per-bar	HAR + cal.	0.1007	+6.4	**+2.37**	**0.018**
65	XGBoost, Optuna (SE), tune 1, top 10	per-bar	All	0.1007	+6.4	+1.63	0.103
66	LightGBM, Optuna (QLIKE), tune 25, top 50	per-bar	HAR + cal.	0.1007	+6.5	**+2.64**	**0.008**
67	XGBoost, Optuna (SE), tune 25, top 25	per-bar	All	0.1008	+6.6	+1.43	0.152
68	XGBoost, Optuna (SE), tune 25, top 25	per-bar	HAR + cal.	0.1009	+6.6	**+2.40**	**0.016**
69	Random forest, Optuna (SE), tune 25, top 50	per-bar	All	0.1009	+6.7	+1.77	0.077
70	LightGBM, Optuna (SE), tune 1, top 50	per-bar	All	0.1010	+6.7	+1.64	0.101
71	XGBoost, Optuna (SE), tune 1, top 25	per-bar	HAR + cal.	0.1011	+6.8	**+2.30**	**0.022**
72	XGBoost, Optuna (QLIKE), tune 25, top 50	per-bar	HAR + cal.	0.1011	+6.9	+1.93	0.053
73	XGBoost, Optuna (SE), tune 1, top 50	per-bar	HAR + cal.	0.1011	+6.9	**+2.39**	**0.017**
74	XGBoost, Optuna (SE), tune 25, top 50	per-bar	All	0.1015	+7.3	+1.61	0.108
75	LightGBM, Optuna (SE), tune 25, top 25	per-bar	HAR + cal.	0.1016	+7.3	**+2.91**	**0.004**
76	XGBoost, Optuna (QLIKE), tune 25, top 50	per-bar	All	0.1016	+7.4	+1.53	0.125
77	Elastic net	pooled	All	0.1016	+7.4	+1.50	0.134
78	LightGBM, Optuna (SE), tune 5, top 50	per-bar	HAR + cal.	0.1016	+7.4	**+2.98**	**0.003**
79	Random forest, rand. search (SE), daily refit	per-bar	All	0.1020	+7.7	+1.67	0.094
80	LightGBM, Optuna (QLIKE), tune 25, top 50	per-bar	All	0.1020	+7.8	+1.54	0.124
81	Ridge	pooled	All	0.1021	+7.9	+1.70	0.090
82	LightGBM, Optuna (SE), tune 1, top 10	per-bar	HAR + cal.	0.1025	+8.3	**+3.45**	**<0.001**
83	LightGBM, Optuna (SE), tune 25, top 50	per-bar	HAR + cal.	0.1026	+8.5	**+3.00**	**0.003**
84	LightGBM, Optuna (SE), tune 1, top 50	per-bar	HAR + cal.	0.1028	+8.6	**+3.03**	**0.002**
85	XGBoost, rand. search (QLIKE), daily refit	per-bar	HAR + cal.	0.1028	+8.6	+1.90	0.057
86	LightGBM, Optuna (SE), tune 1, top 25	per-bar	HAR + cal.	0.1029	+8.7	**+2.91**	**0.004**
87	XGBoost, rand. search (QLIKE), refit 10	per-bar	HAR + cal.	0.1030	+8.8	+1.94	0.053
88	LightGBM, Optuna (SE), tune 25, top 10	per-bar	All	0.1031	+9.0	+1.76	0.078
89	LightGBM, Optuna (SE), tune 25, top 10	per-bar	HAR + cal.	0.1032	+9.1	**+3.11**	**0.002**
90	Random forest, Optuna (SE), tune 250, top 50	per-bar	HAR + cal.	0.1036	+9.5	**+2.18**	**0.029**
91	LightGBM, Optuna (SE), tune 25, top 50	per-bar	All	0.1042	+10.1	**+2.02**	**0.043**
92	LightGBM, rand. search (SE), refit 10	per-bar	HAR + cal.	0.1043	+10.2	**+2.55**	**0.011**
93	Lasso	pooled	All	0.1044	+10.3	**+1.96**	**0.0497**
94	LightGBM, rand. search (SE), daily refit	per-bar	HAR + cal.	0.1045	+10.4	**+2.60**	**0.009**
95	XGBoost, Optuna (SE), tune 25, top 10	per-bar	All	0.1053	+11.2	**+2.15**	**0.032**
96	Lasso (fixed penalty), earlier panel	pooled	All	0.1054	+11.4	**+2.13**	**0.033**
97	Block-diag. ridge, no FOMC, earlier panel	pooled	All	0.1057	+11.7	**+2.21**	**0.027**
98	Random forest, rand. search (SE), refit 10	per-bar	HAR + cal.	0.1057	+11.7	**+2.44**	**0.015**
99	Random forest, rand. search (SE), daily refit	per-bar	HAR + cal.	0.1058	+11.8	**+2.49**	**0.013**
100	Random forest, rand. search (SE), refit 10	per-bar	All	0.1059	+11.9	+1.88	0.060
101	LightGBM	pooled	All	0.1060	+12.0	**+2.28**	**0.022**
102	Lasso	pooled	HAR + cal.	0.1062	+12.2	**+3.41**	**<0.001**
103	XGBoost, Optuna (SE), tune 250, top 50	per-bar	HAR + cal.	0.1062	+12.3	**+3.02**	**0.003**
104	XGBoost	pooled	All	0.1063	+12.3	**+2.43**	**0.015**
105	Elastic net	pooled	HAR + cal.	0.1065	+12.6	**+3.49**	**<0.001**
106	XGBoost, rand. search (SE), daily refit	per-bar	HAR + cal.	0.1066	+12.7	**+2.63**	**0.009**
107	Ridge	pooled	HAR + cal.	0.1067	+12.8	**+3.42**	**<0.001**
108	Block-diagonal ridge, earlier panel	pooled	All	0.1068	+12.8	**+2.38**	**0.017**

continued on the next page

Table 11, continued

Rank	Model	Fit	Inputs	QLIKE	Δ (%)	DM	p
109	LightGBM, Optuna (SE), tune 250, top 50	per-bar	HAR + cal.	0.1068	+12.9	**+2.95**	**0.003**
110	XGBoost, rand. search (SE), refit 10	per-bar	HAR + cal.	0.1069	+12.9	**+2.72**	**0.007**
111	Random forest†*	pooled	All	0.1073	+13.4	**+2.53**	**0.011**
112	Lasso (fixed penalty), earlier panel	pooled	All	0.1076	+13.7	**+2.53**	**0.011**
113	Lasso (causally tuned), earlier panel	pooled	All	0.1090	+15.2	**+2.64**	**0.008**
114	Elastic net (causally tuned), earlier panel	pooled	All	0.1110	+17.3	**+3.06**	**0.002**
115	MinGRU (ridge-init, QLIKE-tuned)*	pooled	All	0.1121	+18.4	**+4.53**	**<0.001**
116	MinGRU (ridge-init, QLIKE-tuned)*	pooled	HAR + cal.	0.1151	+21.7	**+4.88**	**<0.001**
117	XGBoost*	pooled	HAR + cal.	0.1213	+28.2	**+5.03**	**<0.001**
118	Random forest*	pooled	HAR + cal.	0.1229	+29.9	**+5.04**	**<0.001**
119	OLS, earlier panel	pooled	HAR + cal.	0.1232	+30.2	**+4.80**	**<0.001**
120	LightGBM*	pooled	HAR + cal.	0.1236	+30.7	**+5.02**	**<0.001**

Notes. Conventions as in Table 10: plain $\hat{y}_t^2 B_t$ forecasts on 866 days; DM against the per-bar OLS (lag 6, HLN), negative favours the model; bold $p < 0.05$. *Not in the master table's model set (results/close_master_table/paper_model_set.csv). †Interim. Per-bar tree rungs: fixed settings refit every 10 sessions or every session (daily refit); random search on squared error (SE) or QLIKE, re-run every 250 sessions; Optuna on SE or QLIKE re-run every 1, 5, 25 or 250 sessions (tune), keeping the best of the first 10, 25 or 50 trials (top). Earlier panel: the pooled 48-bar models of the earlier draft. The pooled MinGRU is refit monthly, the pooled trees at every bar.

7 Economic application: the closing trade

7.1 The trade

- **Legs:** SPXW 0DTE, 15:30 spot S; nearest out-of-the-money call and put,

$$K_C = \min\{K \geq S\}, \qquad K_P = \max\{K \leq S\},$$

  over live strikes. Straddle iff $K_C = K_P = S$; else a one-strike strangle.
- **Entry premium (mid)** and package quotes, from leg bids b and asks a at 15:30:

$$P_t = \tfrac{1}{2}(b_t^C + a_t^C) + \tfrac{1}{2}(b_t^P + a_t^P), \qquad P_t^b = b_t^C + b_t^P, \qquad P_t^a = a_t^C + a_t^P.$$

- **Settlement** at the official close S_T, cash, no hedge:

$$X_t = (S_T - K_C)^+ + (K_P - S_T)^+, \qquad R_t = \frac{X_t}{P_t} - 1 \;\geq\; -1.$$

- **Implied variance:** IV_t^C, IV_t^P are the legs' quoted hourly vols at 15:30;

$$IV_t^{30} = \frac{IV_t^C + IV_t^P}{2\sqrt{2}}.$$

- **Signal:**

$$s_t = \hat{\sigma}_t^2 - \left(IV_t^{30}\right)^2 = \hat{y}_t^2 B_t - \left(IV_t^{30}\right)^2.$$

- **Rules,** position q_t, daily return $r_t = q_t R_t$:
  - always short (benchmark): $q_t = -1$;
  - sign: $q_t = \text{sign}(s_t)$, with $q_t = -1$ at $s_t = 0$;
  - robust clip: $q_t = \text{clip}(s_t/m_t, -1, 1)$, $m_t = \text{median}_{u<t}\,|s_u|$;
  - full long, scaled short: $q_t = 1$ if $s_t > 0$, else $q_t = \max(s_t/m_t, -1)$;

  - rank of $|s|$:

$$q_t = \operatorname{sign}(s_t) \cdot \frac{1}{t-1} \sum_{u<t} 1\{|s_u| \le |s_t|\}.$$

- **Sizing sample:** q_t uses $u < t$ only; scored on $t > 252$ (614 days).
- **Fills:** mid, $r_t = q_t R_t$; crossed, spread paid at entry only:

$$r_t^x = \begin{cases} |q_t|\,(X_t/P_t^a - 1), & q_t > 0, \\ |q_t|\,(1 - X_t/P_t^b), & q_t < 0. \end{cases}$$

- Crossing costs a median 2.9% of premium. No other costs.

7.2 Economic measures

- **Sharpe,** per trade day, mid and crossed, over n days:

$$\text{Sharpe} = \sqrt{252}\,\frac{\bar r}{\text{sd}(r)}, \qquad \text{sd}(r)^2 = \frac{1}{n-1}\sum_t (r_t - \bar r)^2.$$

- **Mean return, Newey–West** t**:** $t = \bar r / \sqrt{\hat\Omega_r / n}$, $\hat\Omega_r$ as in Section 6, lag $\ell = \lfloor 1.5\, n^{1/3} \rfloor$, 14 at $n = N$.
- **Hit rate** $\frac{1}{n}\sum_t 1\{r_t > 0\}$; buys $\#\{t : q_t > 0\}$.
- **Gain over always short** (or any b), daily difference $\Delta_t = r_t^{(a)} - r_t^{(b)}$:

$$\Delta\text{Sharpe} = \text{Sharpe}(r^{(a)}) - \text{Sharpe}(r^{(b)}), \qquad t_{\text{HAC}} = \frac{\bar\Delta}{\sqrt{\hat\Omega_\Delta / n}}.$$

- **Elementary score** at threshold $(IV_t^{30})^2$, i.e. $c = 1$ on the scaled values (Ehm et al., 2016):

$$\text{ES}_t = |\kappa_t - 1|\; 1\{\min(\hat\kappa_t, \kappa_t) \le 1 < \max(\hat\kappa_t, \kappa_t)\}, \qquad \hat\kappa_t = \frac{\hat\sigma_t^2}{(IV_t^{30})^2}, \quad \kappa_t = \frac{RV_t}{(IV_t^{30})^2}.$$

- **Nonzero** only when $\hat\sigma_t^2$ and RV_t sit on opposite sides of the threshold.
- **Compounding,** $f = 0.03$ of wealth per trade (3%):

$$W_T = \prod_{t=1}^{T} (1 + f\, q_t R_t), \qquad g = 252 \cdot \frac{1}{T}\sum_{t=1}^{T} \log(1 + f\, r_t).$$

- Crossed: r_t^x in place of $q_t R_t$. Drawdown $\max_t (1 - W_t / \max_{u \le t} W_u)$.

7.3 What the trade shows

- **Always short:** Sharpe 0.20 mid, −0.27 crossed; $\bar r$ indistinguishable from zero; loses in 3 of 5 years.
- $\Delta\text{Sharpe} > 0$ **vs always short for every forecast** (point estimate):
  - 20 of 112 significant (t_{HAC});
  - all 20 read the exogenous inputs.
- **Best Sharpe:**
  - per-bar elastic net on all features: 1.92 mid, 1.43 crossed;

  - per-bar lasso: 1.88;
  - tuned LightGBM: 1.81;
  - MinGRU on all features: 1.53 / 1.05.
- **All features vs HAR + calendar:** median ΔSharpe = +0.54 across pairs (elastic net +0.55, lasso +0.50).
- **Sizing,** median Sharpe across the model set: sign 0.92; robust clip and rank 1.34.
  - $\text{Sharpe}^{\text{clip}} > \text{Sharpe}^{\text{sign}}$ for 98 of 112 forecasts.
- **A tail trade:**
  - without its 10 best days, $\bar{\Delta}$ vs always short < 0 in 102 of 112 forecasts;
  - pin days supply most of the return.
- **Accuracy is a weak guide,** Spearman over forecasts:

$$\rho_S(\overline{\text{QLIKE}}, \text{Sharpe}) = -0.18, \qquad \rho_S(-\overline{\text{ES}}, \text{Sharpe}) = +0.50.$$

- **Robustness** (appendix):
  - date shift, $q_t = \text{sign}(\hat{y}^2_{t,k} B_t - (IV_t^{30})^2)$, row k bars off: swings of ±1 Sharpe are noise;
  - sample splits around May 2022;
  - compounding: median W_T 4.4 at mid, 1.8 crossed; drawdowns up to 64–78%.

8 Feature importance

- **Models:** the per-bar 16:00 models of record, all features ($p = 628$).
  - Lasso and ridge, pre-tuned: $\alpha = 10^{-2}$ and 10^3, backbone at $\text{pf}_j = 0.01$, re-solved every session; the margin retune never fired.
  - Not the old study's linear arm (one penalty, re-chosen every 250 sessions): that arm is not in the paper.
  - Trees on their top development-span entries, window-masked: LightGBM lgbm_opt_00, XGBoost xgb_opt_00, random forest trial 255.
- **Notation.**
  - Refit $k = 1, \ldots, K$, $K = 147$, one every 10 sessions: fit on the 2,000 sessions W_k before its anchor; score its causal tail T_k (≤ 10 sessions); 1,469 scored sessions, 25 June 2018 to 30 April 2024.
  - Input $x_t \in \mathbb{R}^p$; forecast $\hat{y}_t = \hat{f}_k(x_t)$, $t \in T_k$, in the fit space.
  - Unit $A \subseteq \{1, \ldots, p\}$: all columns of one series (ladders, flags). 49 units: 47 exogenous series in 8 buckets, the HAR ladder, the calendar block.
  - Linear: $\beta^{(k)} \in \mathbb{R}^p$, the coefficients that forecast T_k's first session.
  - Trees: SHAP values $\phi_{t,j}$ and interaction values $\phi_{t,jl}$,

$$\hat{y}_t = \phi_0 + \sum_{j=1}^{p} \phi_{t,j}, \qquad \phi_{t,j} = \sum_{l=1}^{p} \phi_{t,jl}, \qquad \phi_{t,A} = \sum_{j \in A} \phi_{t,j}.$$

  - Path-dependent TreeSHAP (Lundberg et al. 2020): node covers stand in for the background.
  - Number triples below: LightGBM, XGBoost, random forest.

Table 12: The closing trade under $q_t = \text{sign}(s_t)$: always short and the key forecasts, 866 trade days.

	Mid return		Sharpe ratio				vs always short	
Forecast	Mean (%)	t	Mid	Crossed	Hit (%)	Buys (%)	ΔSharpe	t
Always short ($q_t = -1$)	1.4	0.43	0.20	−0.27	61.8	0.0	–	–
Per-bar linear (16:00 bar)								
OLS, HAR + calendar	**9.7**	**2.83**	1.37	0.89	56.7	27.9	+1.17	1.85
Elastic net, all features	**13.5**	**3.66**	1.92	1.43	57.7	28.3	**+1.72**	**2.56**
Lasso, all features	**13.2**	**3.81**	1.88	1.40	57.6	29.6	**+1.67**	**2.54**
Ridge, all features	**10.0**	**2.86**	1.42	0.93	57.5	27.1	**+1.21**	**1.98**
Per-bar trees, all features								
LightGBM[a]	**12.8**	**3.73**	1.81	1.33	56.9	30.5	**+1.61**	**2.33**
XGBoost[a]	**12.5**	**3.82**	1.77	1.27	58.7	27.8	**+1.56**	**2.56**
Random forest[a]	**11.2**	**3.38**	1.58	1.11	56.8	30.4	**+1.38**	**2.07**
Per-bar neural net								
MinGRU, all features[b]	**10.8**	**3.09**	1.53	1.05	56.2	32.8	**+1.33**	**2.02**
Pooled linear (all 48 bars)								
Elastic net, all features	**10.6**	**2.81**	1.50	1.01	57.3	27.4	+1.29	1.90
Lasso, all features	**8.5**	**2.45**	1.20	0.71	58.0	23.7	+1.00	1.66
Ridge, all features	**10.2**	**2.78**	1.44	0.96	57.5	27.1	+1.24	1.93

Notes: One 0DTE SPXW straddle (a one-strike strangle when spot is between strikes) is bought ($q_t = +1$) or sold ($q_t = -1$) at 15:30 ET and held to the close; $r_t = q_t R_t$, with R_t the return on the premium; 3 January 2020 to 30 April 2024. Mid: the straddle is bought or sold at the 15:30 midpoint; crossed: a buy pays the ask and a sale receives the bid, the spread paid at entry only (settlement is cash at intrinsic). Mean: the mean mid return, in % of the premium. Sharpe ratio $= \sqrt{252}\,\bar{r}/\text{sd}(r)$, annualised per trade day (the trade runs about 200 days a year). Hit: share of days with $r_t > 0$ at mid; Buys: share of days with $q_t = +1$. ΔSharpe: mid Sharpe ratio minus always short's (0.20). Every t is Newey–West (Bartlett kernel, lag $\lfloor 1.5\, n^{1/3} \rfloor = 14$ at $n = 866$); the t of a ΔSharpe is that of the mean daily return difference, so it tests the mean difference, not the Sharpe difference. Bold: $|t| > 1.96$ (5 %, two-sided), on the estimate and its t.
[a] Of each estimator's 15 configurations on all features in the model set, the one with the highest mid Sharpe ratio, so selected on the evaluation sample. LightGBM: random search, refit every 10 sessions, MSE-selected; XGBoost: random search, refit daily, MSE-selected; random forest: Optuna, retuned every 25 sessions, best of 10 trials, MSE-selected.
[b] Ridge-initialised linear shortcut plus a MinGRU branch, trained and selected on QLIKE; the mean of 5 seeds.

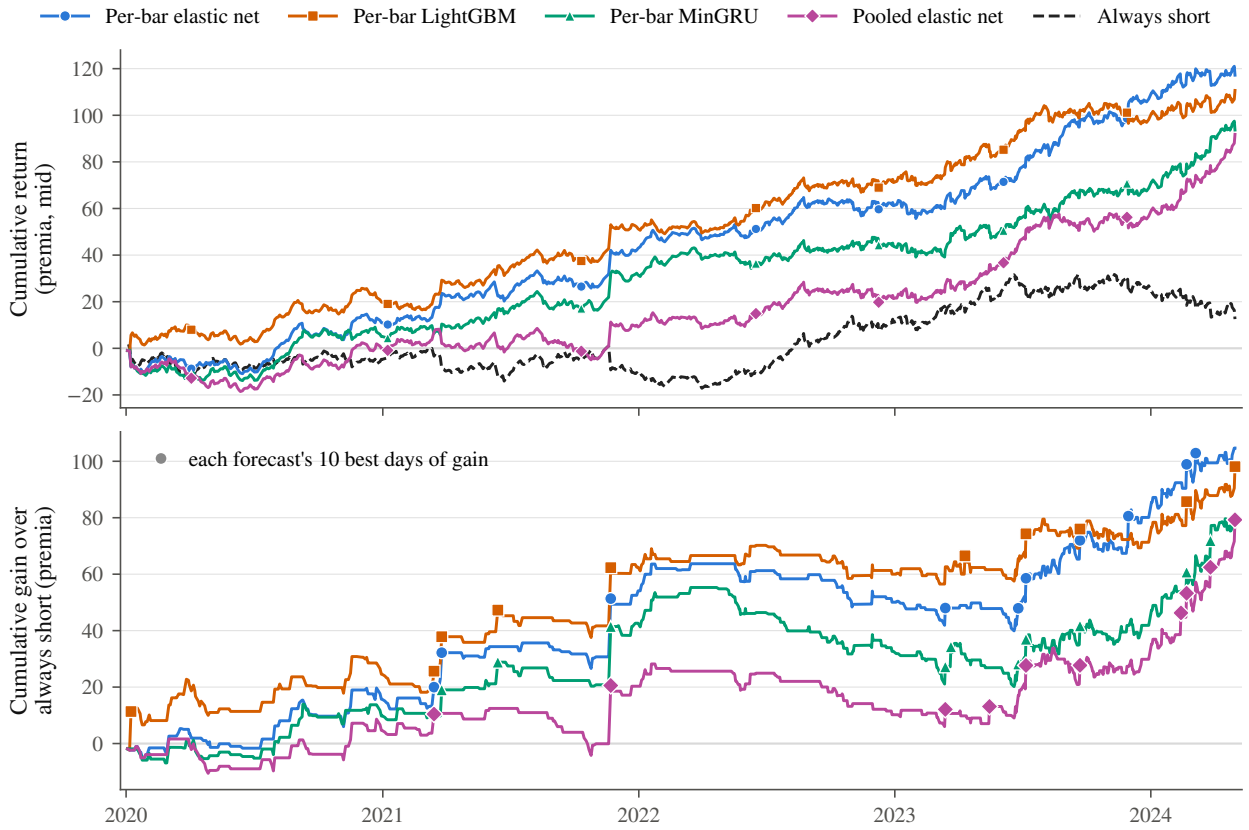


Figure 5: The closing trade over time, $q_t = \text{sign}(s_t)$ at mid, for one forecast per family (all features) and always short. Top: cumulative return $\sum_{u \le t} r_u$, in units of the straddle premium. Bottom: cumulative gain over always short, $\sum_{u \le t} \left(r_u - r_u^{\text{short}}\right)$; the markers are each forecast's 10 days of largest gain, each a day on which it bought and $R_t \ge 2.99$. The gain over the 866 days is per-bar elastic net +104.7, per-bar LightGBM +98.1, per-bar MinGRU +81.1, pooled elastic net +79.3; without each forecast's 10 best days it is +15.6, +2.4, −6.2, −6.3. For 102 of the 112 forecasts of the model set the gain is negative without their 10 best days.

Table 13: Adding the exogenous inputs: the Sharpe ratio from HAR + calendar to all features, within each family ($q_t = \text{sign}(s_t)$, 866 trade days).

	Sharpe ratio, mid		All features minus HAR + calendar			
	HAR + cal.	All	ΔSharpe mid	t	ΔSharpe crossed	t
Per-bar linear (16:00 bar)						
Elastic net	1.37	1.92	+0.55	1.34	+0.54	1.35
Lasso	1.37	1.88	+0.50	1.47	+0.51	1.48
Ridge	1.37	1.42	+0.05	0.09	+0.04	0.10
Per-bar neural net						
MinGRU	1.04	1.53	+0.49	0.95	+0.49	0.95
Pooled linear (all 48 bars)						
Elastic net	0.89	1.50	+0.61	1.21	+0.60	1.20
Lasso	0.69	1.20	+0.51	1.19	+0.50	1.19
Ridge	0.91	1.44	+0.53	1.12	+0.52	1.12
Per-bar trees: medians over configurations; the t columns count $t > 1.96$						
LightGBM (15)	0.73	1.33	+0.60	0/15	+0.60	0/15
XGBoost (15)	0.80	1.48	+0.55	3/15	+0.54	3/15
Random forest (15)	0.64	1.07	+0.47	1/15	+0.47	1/15
All trees (45)	0.70	1.24	+0.55	4/45	+0.54	4/45
Model set, all pairs (51)	0.73	1.27	+0.54	4/51	+0.53	4/51

Notes: Each pair is one model fitted on HAR + calendar (`baseline`) and on all features, with nothing else changed; the change is all features minus HAR + calendar on the same days. The per-bar linear models are pre-tuned; on HAR + calendar all three take the OLS's position on every day, so they share its Sharpe ratio. The tree rows summarise 45 pairs (every tuning configuration of the model set, 15 per estimator): the Sharpe and ΔSharpe columns are medians, and each t column gives the number of pairs with $t > 1.96$. The model-set row covers the 51 pairs of the 112-forecast model set (the MinGRU is not in it): ΔSharpe mid > 0 in 50 of 51, none significantly negative. Mid: the straddle is bought or sold at the 15:30 midpoint; crossed: a buy pays the ask and a sale receives the bid, the spread paid at entry only (settlement is cash at intrinsic). Sharpe ratio $= \sqrt{252}\,\bar{r}/\text{sd}(r)$, annualised per trade day (the trade runs about 200 days a year). Every t is Newey–West (Bartlett kernel, lag $\lfloor 1.5\,n^{1/3} \rfloor = 14$ at $n = 866$); the t of a ΔSharpe is that of the mean daily return difference, so it tests the mean difference, not the Sharpe difference. Bold: $|t| > 1.96$ (5 %, two-sided), on the estimate and its t.

Table 14: Per-bar against pooled linear fits: the trade and accuracy ($q_t = \text{sign}(s_t)$, 866 trade days).

	Sharpe ratio, mid		Per-bar minus pooled					
	Pooled	Per-bar	ΔSharpe mid	t	ΔSharpe crossed	t	ΔQLIKE (%)	DM
All features								
Elastic net	1.50	1.92	+0.42	0.88	+0.42	0.88	**−7.3**	**−2.00**
Lasso	1.20	1.88	+0.68	1.66	+0.68	1.67	**−12.4**	**−3.19**
Ridge	1.44	1.42	−0.03	−0.07	−0.02	−0.06	−6.2	−1.56
HAR + calendar								
Elastic net	0.89	1.37	+0.49	1.12	+0.48	1.12	**−11.2**	**−3.50**
Lasso	0.69	1.37	+0.68	1.69	+0.68	1.69	**−10.9**	**−3.42**
Ridge	0.91	1.37	+0.46	1.05	+0.46	1.04	**−11.3**	**−3.42**

Notes: Pooled: one model fitted on all 48 half-hour bars of the day, read at the 16:00 bar; per-bar: fitted on the 16:00 bar alone; both pre-tuned, on the same inputs and days. ΔQLIKE: per-bar mean QLIKE over pooled, minus 1, in %; DM: Diebold–Mariano statistic of the daily QLIKE difference (Harvey–Leybourne–Newbold corrected, automatic Newey–West lag), negative when the per-bar fit is more accurate. Mid: the straddle is bought or sold at the 15:30 midpoint; crossed: a buy pays the ask and a sale receives the bid, the spread paid at entry only (settlement is cash at intrinsic). Sharpe ratio $= \sqrt{252}\,\bar{r}/\text{sd}(r)$, annualised per trade day (the trade runs about 200 days a year). Every t is Newey–West (Bartlett kernel, lag $\lfloor 1.5\,n^{1/3} \rfloor = 14$ at $n = 866$); the t of a ΔSharpe is that of the mean daily return difference, so it tests the mean difference, not the Sharpe difference. Bold: $|t| > 1.96$ (5 %, two-sided), on the estimate and its t. Bold DM: $|\text{DM}| > 1.96$.

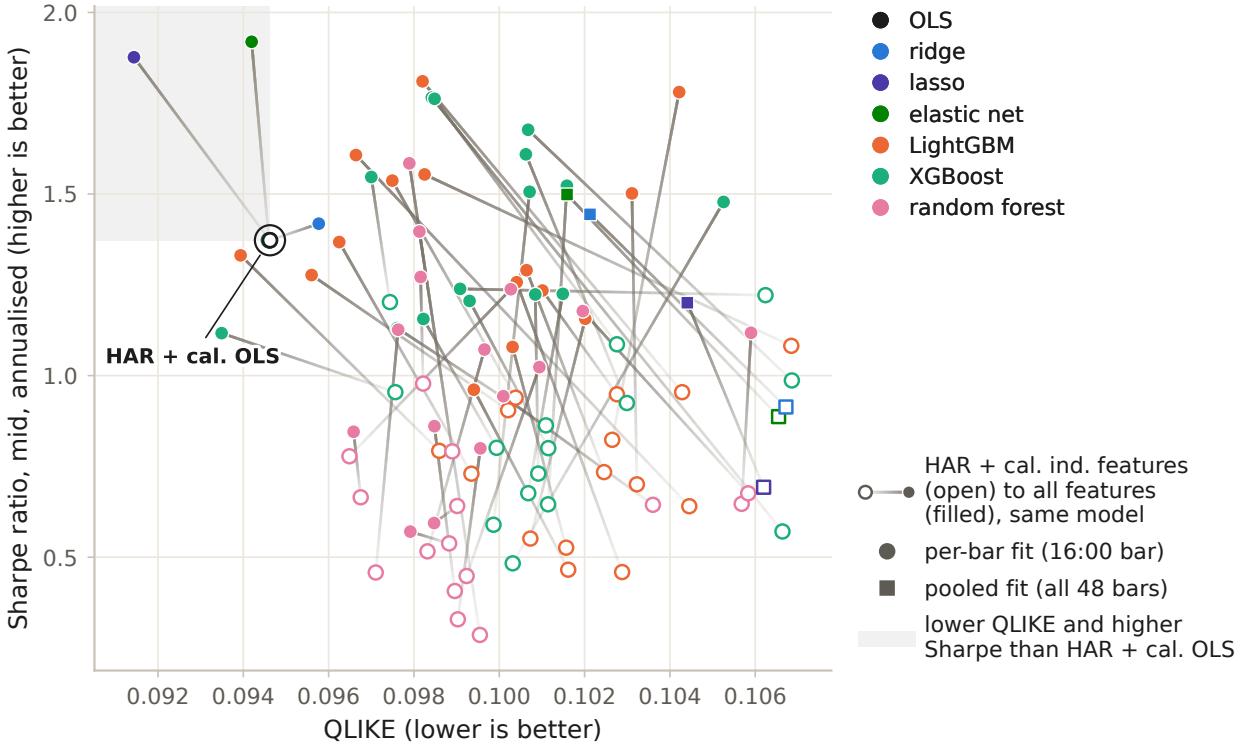


Figure 6: Accuracy against the trade: QLIKE and the sign(s) Sharpe ratio (mid) of the 51 models of the model set fitted on both HAR + calendar (open markers) and all features (filled), 866 trade days; each pair is joined from its HAR + calendar end. Circles are per-bar fits and squares pooled fits; the colour marks the estimator. Shaded: lower QLIKE and higher Sharpe ratio than the HAR + calendar OLS (QLIKE 0.0946, Sharpe 1.37). All features raise the Sharpe ratio for 50 of the 51 models (4 significantly, by the HAC t of the daily return difference) and lower QLIKE for 38 (2 significantly, by Diebold–Mariano), at 5 %.

Table 15: Sizing rules: Sharpe ratio on days 253–866 (614 days).

	Mid				Crossed			
Forecast	sign(s)	Clip	Long/scaled	Rank $\lvert s\rvert$	sign(s)	Clip	Long/scaled	Rank $\lvert s\rvert$
Always short ($q_t = -1$)	0.34 under every rule				-0.03 under every rule			
Per-bar linear (16:00 bar)								
OLS, HAR + calendar	1.64	1.55	1.69	1.67	1.27	1.20	1.38	1.31
Elastic net, all features	1.93	1.82	1.84	1.90	1.56	1.46	1.52	1.54
Lasso, all features	1.76	1.75	1.74	1.85	1.40	1.39	1.42	1.49
Ridge, all features	1.51	1.70	1.56	1.75	1.14	1.35	1.24	1.39
Per-bar trees, all features (as in Table 12)								
LightGBM	1.57	1.78	1.64	1.66	1.20	1.44	1.32	1.31
XGBoost	1.34	1.63	1.58	1.67	0.96	1.27	1.21	1.30
Random forest	1.20	1.28	1.28	1.26	0.84	0.93	0.95	0.90
Per-bar neural net								
MinGRU, all features	1.59	1.74	1.57	1.79	1.22	1.39	1.24	1.44
Pooled linear (all 48 bars)								
Elastic net, all features	1.91	1.75	1.72	1.69	1.54	1.38	1.40	1.32
Lasso, all features	1.56	1.69	1.65	1.59	1.19	1.33	1.32	1.22
Ridge, all features	1.73	1.79	1.68	1.73	1.36	1.43	1.35	1.35
Model set (112): median	0.92	1.34	1.12	1.34	0.56	0.99	0.79	0.98
Model set: above sign(s)	–	98	100	98	–	98	106	100

Notes: Position q_t in straddles, from the gap $s_t = \hat{\sigma}_t^2 - (IV_t^{30})^2$ and the earlier days only: sign(s), $q_t = \pm 1$; Clip (robust clip), $q_t = \text{clip}(s_t/m_t, -1, 1)$ with m_t the median $|s_u|$ over $u < t$; Long/scaled (full long, scaled short), $q_t = 1$ if $s_t > 0$, else $\max(s_t/m_t, -1)$; Rank $|s|$, $q_t = \text{sign}(s_t)$ times the share of earlier days with $|s_u| \leq |s_t|$. The first 252 days only build the scale, so every rule, sign(s) included, is scored on the last 614 of the 866 days. Crossed: the spread is paid on $|q_t|$ straddles. Model-set rows: the median over the 112 forecasts of the model set (the MinGRU is not in it), and the number of them whose Sharpe ratio is above their own sign(s) Sharpe ratio. Point estimates; no test. Sharpe ratio $= \sqrt{252}\,\bar{r}/\text{sd}(r)$, annualised per trade day (the trade runs about 200 days a year).

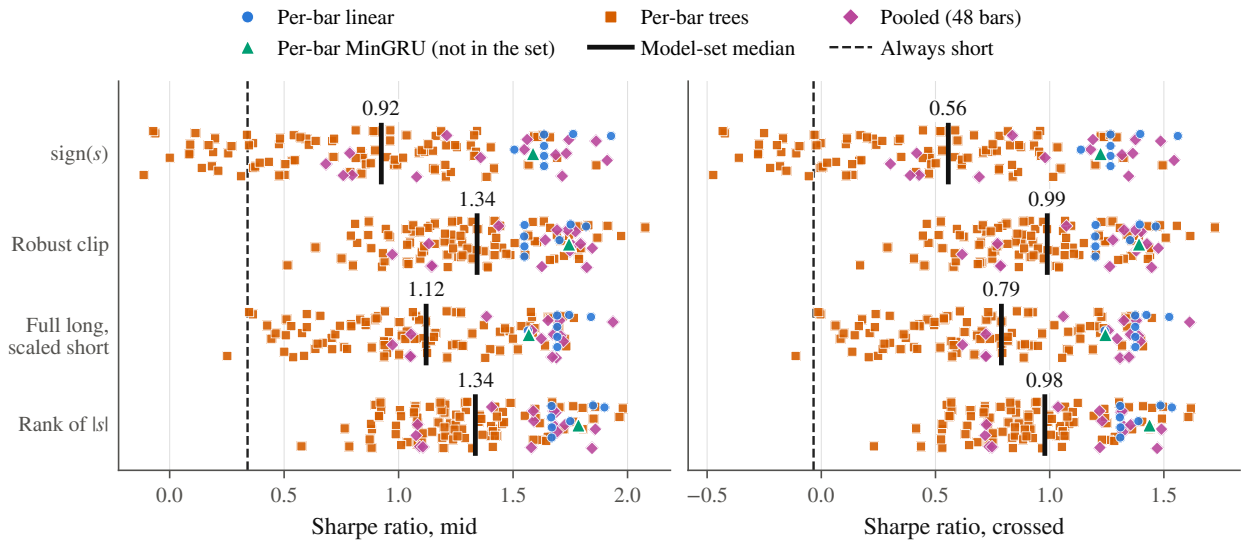


Figure 7: Sharpe ratio by sizing rule over the 112 forecasts of the model set, mid (left) and crossed (right), on days 253–866. One marker per forecast, coloured by family and jittered vertically; the bar and its label mark the model-set median (mid: 0.92, 1.34, 1.12, 1.34 for sign(s), robust clip, full long / scaled short and rank of $|s|$). The per-bar MinGRU on all features (triangle) is drawn but not in the median. Dashed: always short (0.34 mid, −0.03 crossed). The robust clip is above sign(s) for 98 of the 112 forecasts at mid (median gain +0.45). Rules and scoring as in Table 15.

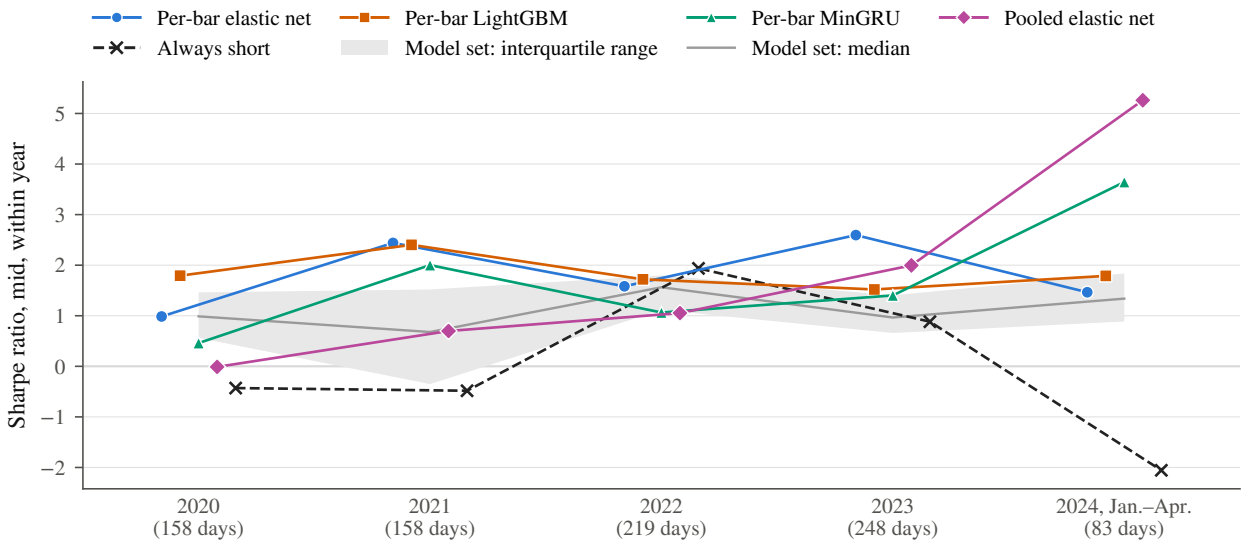


Figure 8: Sharpe ratio by calendar year at mid, $q_t = \text{sign}(s_t)$, for the forecasts of Figure 5 and always short (2020 to 2024: −0.43, −0.48, 1.94, 0.88, −2.06); 2024 runs from January to April. Each year's ratio is annualised per trade day, as in the tables. Grey: the interquartile range and median of the 112 forecasts of the model set (median 0.99, 0.68, 1.56, 0.97, 1.34).

- **Lasso, aggregate:** the selection count,

$$N_A = \sum_{k=1}^{K} \mathbf{1}\{\exists j \in A : \ \beta_j^{(k)} \neq 0\}, \qquad N_j = \sum_{k=1}^{K} \mathbf{1}\{\beta_j^{(k)} \neq 0\}.$$

  - Per refit: 30 nonzero coefficients (median; 27–35), of which 14 exogenous.
  - Columns: 60 of 628 ever nonzero, 29 in at least half of the refits.
  - Exogenous series: 27 of 47 ever selected; 11 in at least half of the refits; 3 in every refit (`sumret`, `sumabsret_vwstock`, `sumautocov`).
  - Backbone: selected in every refit (penalty factor 0.01); 88 % of $\sum_j |\beta_j|$ (median refit).
  - Noise bar: the continuous probe is selected in 2.0 % of its fits; 54 columns, in 23 exogenous series, are selected more often.
- **Lasso, time-varying:** magnitude and on/off pattern per unit and refit,

$$b_{k,A} = \sum_{j \in A} |\beta_j^{(k)}|, \qquad \mathbf{1}\{b_{k,A} > 0\}$$

  (Figure 14, left). Rank stability: Kendall's $W = 0.74$ (definition below).
- **Ridge, aggregate:** the share of the coefficient mass,

$$S_A = \frac{\sum_k \sum_{j \in A} |\beta_j^{(k)}|}{\sum_k \sum_{j=1}^{p} |\beta_j^{(k)}|}.$$

  - Scale check: only the value ladders are robust-scaled.
    - Value ladders (282 columns): window sd 0.84 (median; 5–95 %: 0.03–1.76), IQR 0.94.
    - Availability / activity ladders (324): sd 0.022; calendar dummies 0.39; HAR ladder 1.07.
    - So $|\beta_j|$ is per unit of different scales. Check: $\sum_k \sum_{j \in A} |\beta_j^{(k)}| \, \mathrm{sd}_{k,j}$, $\mathrm{sd}_{k,j}$ over W_k; Spearman with S_A over the units 0.85.
    - Leading units on that scale: HAR ladder 30.6 %, Calendar 9.3 %, `sumret` 4.5 %.
  - Leading units: Calendar 31.0 %, HAR ladder 22.8 %, `sumret` 3.6 %; backbone 54 % (median refit).
  - All 49 units have $N_A = K$: no count measure.
  - Noise bar: 107 columns, in 38 exogenous series, have $|\beta_j|$ above the continuous probe's (robust-scaled) in most probe fits.
- **Ridge, time-varying:** $s_{k,A} = b_{k,A} / \sum_B b_{k,B}$ (Figure 14, right); $W = 0.85$.
- **Trees, per-refit ranking:** unit importance on the refit's own held-out rows, then its rank,

$$I_{k,A} = \frac{1}{|T_k|} \sum_{t \in T_k} |\phi_{t,A}|, \qquad r_{k,A} = 1 + \#\{B : I_{k,B} > I_{k,A}\} \quad \text{(ties averaged)}.$$

- **Trees, global ranking:** aggregate the K rankings,

$$\bar{r}_A = \frac{1}{K} \sum_{k=1}^{K} r_{k,A}, \qquad B_A = \sum_{k=1}^{K} \max(0, \ 11 - r_{k,A}).$$

  - B_A: truncated Borda count (a full Borda count is affine in $\bar{r}_A$).

– Kendall's W over the K rankings of the n units, $R_A = \sum_k r_{k,A}$:

$$W = \frac{12 \sum_A (R_A - \bar{R})^2}{K^2(n^3 - n) - K \sum_k \sum_g (\tau_{kg}^3 - \tau_{kg})},$$

τ_{kg} the size of tie group g in refit k; $W = 1$: the same ranking at every refit.
– LightGBM: HAR ladder ($\bar{r} = 1.0$), `sumret` (2.9), `sumabsret` (4.8); $W = 0.57$.
– XGBoost: HAR ladder (1.0), `sumret` (3.0), `sumabsret` (3.9); $W = 0.60$.
– Random forest: HAR ladder (1.0), `sumabsret` (2.6), `sumret` (4.0); $W = 0.62$.
– Mean rank vs Borda, Spearman over units: 0.90, 0.87, 0.92.
– Aggregate of ranks vs rank of the aggregate ($N^{-1} \sum_t |\phi_{t,A}|$ over all scored sessions): 0.96, 0.97, 0.98.
– Column level ($n = 628$): $W = 0.91$, 0.90, 0.93.
– Agreement of $\bar{r}_A$ across models (Spearman): trees 0.83–0.96; tree vs linear 0.17–0.55; lasso vs ridge 0.47.

- **Interactions, global:** the mean absolute SHAP interaction over the scored sessions,

$$\Phi_{t,AB} = \sum_{j \in A} \sum_{l \in B} (\phi_{t,jl} + \phi_{t,lj}), \qquad \bar{I}_{AB} = \frac{1}{N} \sum_t |\Phi_{t,AB}|, \quad A \neq B,$$

$N = 1{,}469$; columns: $|\phi_{t,jl} + \phi_{t,lj}|$, $j \neq l$.
– $\phi_{t,jl} = \frac{1}{2}(\phi_{t,l} \mid j \text{ on} - \phi_{t,l} \mid j \text{ off})$, $j \neq l$; $\phi_{t,jj} = \phi_{t,j} - \sum_{l \neq j} \phi_{t,jl}$.
– Top pair: LightGBM HAR ladder × `sumabsret` (12.6); XGBoost HAR ladder × `sumabsret` (15.2); random forest HAR ladder × `sumabsret` (23.8); $\times 10^{-3}$, fit space.
– Scale: largest main effect $N^{-1} \sum_t |\phi_{t,A}|$ 195, 192, 224 ($\times 10^{-3}$); all series-pair $\bar{I}_{AB}$ summed, as % of all main effects summed: 90, 74, 72.
– Within a unit ($\sum_{j \neq l \in A} \phi_{t,jl}$): largest HAR ladder (18.7), HAR ladder (19.1), HAR ladder (34.3); $\times 10^{-3}$.
– Noise floor: the continuous probe's largest $\bar{I}$ with any unit, median over probe fits: 0.49, 0.41, 0.50 ($\times 10^{-3}$); series pairs above it: 116, 115, 52 of 1,176.

- **Interactions, cross-check:** Friedman–Popescu H^2 on each model's top 10 series pairs,

$$H^2_{AB} = \frac{\sum_i [\mathrm{PD}_{AB}(x_i) - \mathrm{PD}_A(x_i) - \mathrm{PD}_B(x_i)]^2}{\sum_i \mathrm{PD}_{AB}(x_i)^2}, \qquad \mathrm{PD}_S(x) = \frac{1}{n} \sum_{b=1}^{n} \hat{f}(x_S, x_{b,-S}),$$

each PD centred over the x_i.
– Points: the 100 held-out sessions from the anchor, at 15 refits (every tenth); sums pooled over refits.
– The denominator holds the HAR ladder's own PD whenever it is in the pair. Second normalisation, by the smaller main PD:

$$H^2_{\min,AB} = \frac{\sum_i [\mathrm{PD}_{AB}(x_i) - \mathrm{PD}_A(x_i) - \mathrm{PD}_B(x_i)]^2}{\min\{\sum_i \mathrm{PD}_A(x_i)^2,\ \sum_i \mathrm{PD}_B(x_i)^2\}}.$$

– Top pair, H^2 (%): 0.33, 0.37, 0.85; over the top 10: 0.02–0.33, 0.02–0.68, 0.02–0.85.
– Top pair, $H^2_{\min}$: 0.19, 0.39, 1.50; over the top 10: 0.05–0.54, 0.08–0.80, 0.17–5.46.
– Top column pair, H^2 (%) and $H^2_{\min}$: LightGBM 0.36, 0.02; XGBoost 0.39, 0.02; random forest 1.16, 0.06.

- **Interactions, time-varying:** $\Phi_{t,AB}$ on each refit's rows, ranked per refit, aggregated as the main effects,

$$I_{k,AB} = \frac{1}{|T_k|} \sum_{t \in T_k} |\Phi_{t,AB}| \;\rightarrow\; r_{k,AB} \;\rightarrow\; \bar{r}_{AB},\; W.$$

  - W over each model's top 50 pairs: 0.45, 0.47, 0.57.
  - Mean rank vs $\bar{I}_{AB}$ (Spearman, all pairs): 0.99, 0.99, 1.00.
- **Known issues, reconciled:**
  - MDI's cardinality bias: MDI not used; SHAP and coefficients only.
  - Noise probe (every third refit, 49 fits; continuous, 20-level and binary noise columns):
    - Lasso: the probes are selected in 1, 0, 0 of 49 fits (robust-scaled, penalty 1).
    - Ridge: 107 columns have $|\beta_j|$ above the continuous probe's in most probe fits.
    - Trees: the continuous probe's median rank among 631 columns by mean $|\phi_j|$: 83, 110, 97; exogenous series with a column above it in most probe fits: 27, 35, 29.
  - Correlated-column dilution: a unit's $\phi_{t,A}$ and $b_{k,A}$ sum over its columns, so credit split among its lags stays inside the unit.
  - Old linear arm (re-tuned, one penalty): median 23 nonzero coefficients per refit; 18 exogenous series selected in half the refits; Spearman of N_A with the pre-tuned lasso 0.70.
  - Refits: LightGBM reproduces the stored refits bit for bit; XGBoost and the forest differ with the library version (forecast gaps up to 0.17 and 0.11). Main effects: the stored refits'; interactions: the local ones (Spearman of $\bar{r}_A$, local vs stored: 1.000, 0.993, 0.996).
- **Compute** (one laptop, 11 processes):
  - Tree refits and SHAP interaction values: 133 minutes. Per refit: forest fit 212 s, its SHAP and interaction values 182 s; LightGBM 0.9 s, XGBoost 0.4 s.
  - Linear coefficient paths and probes: 7 minutes; H^2: 7 minutes.

Table 16: Draft, measures under review. The `har_ma_*` ladder's importance in the trees, first pass to the current design (all features), old → new.

Model	MDI share (%)	TreeSHAP share (%)	Perm. P2 ΔQLIKE (%)	`_x_close` share (%)	MDI rank
LightGBM	57 → 52	47 → 44	+237 → +200	22	1 → 1
XGBoost	69 → 54	55 → 47	+224 → +181	22	1 → 1
Random forest	62 → 59	58 → 61	+290 → +271	54	1 → 1

Notes: first pass: the old design (with the 12 session-edge interactions), trees unmasked; new: those columns dropped and every tree fit window-masked. MDI and TreeSHAP: the ladder's share of the model's total. Perm. P2 ΔQLIKE: the loss increase when the ladder's rows are replaced by rows drawn from the training window, in % of the model's tail loss. `_x_close` share: the part of the old ladder's MDI that sat on its 6 `_x_close` copies. MDI rank among the series. XGBoost and the random forest also changed settings between the passes, so their before/after mixes that change with the design and the mask.

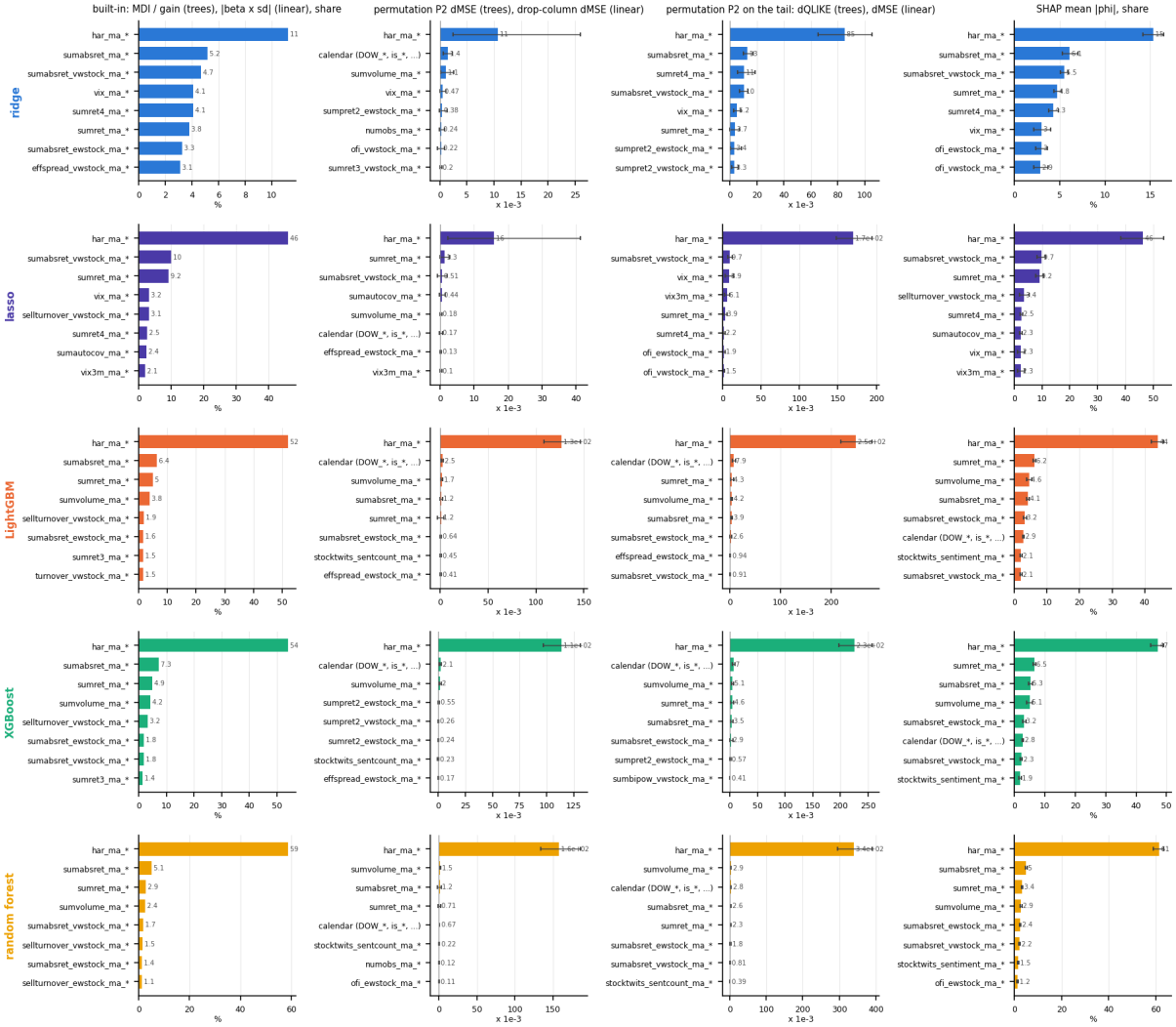


Figure 9: Draft, measures under review. The leading series by each measure, per model (rows), all features (628 columns in 49 series), averaged over the 147 refits. Columns: the built-in measure (MDI / gain share for the trees, $|\beta \cdot \mathrm{sd}|$ share for the linear models); permutation P2, ΔMSE in the fit space (trees), or drop-column ΔMSE (linear); permutation P2 on the held-out tail, ΔQLIKE (trees) or ΔMSE (linear); SHAP mean $|\phi|$ share. Whiskers: 95 % circular-block bootstrap intervals over refits (block 6), which the paper's point-estimate convention would drop.

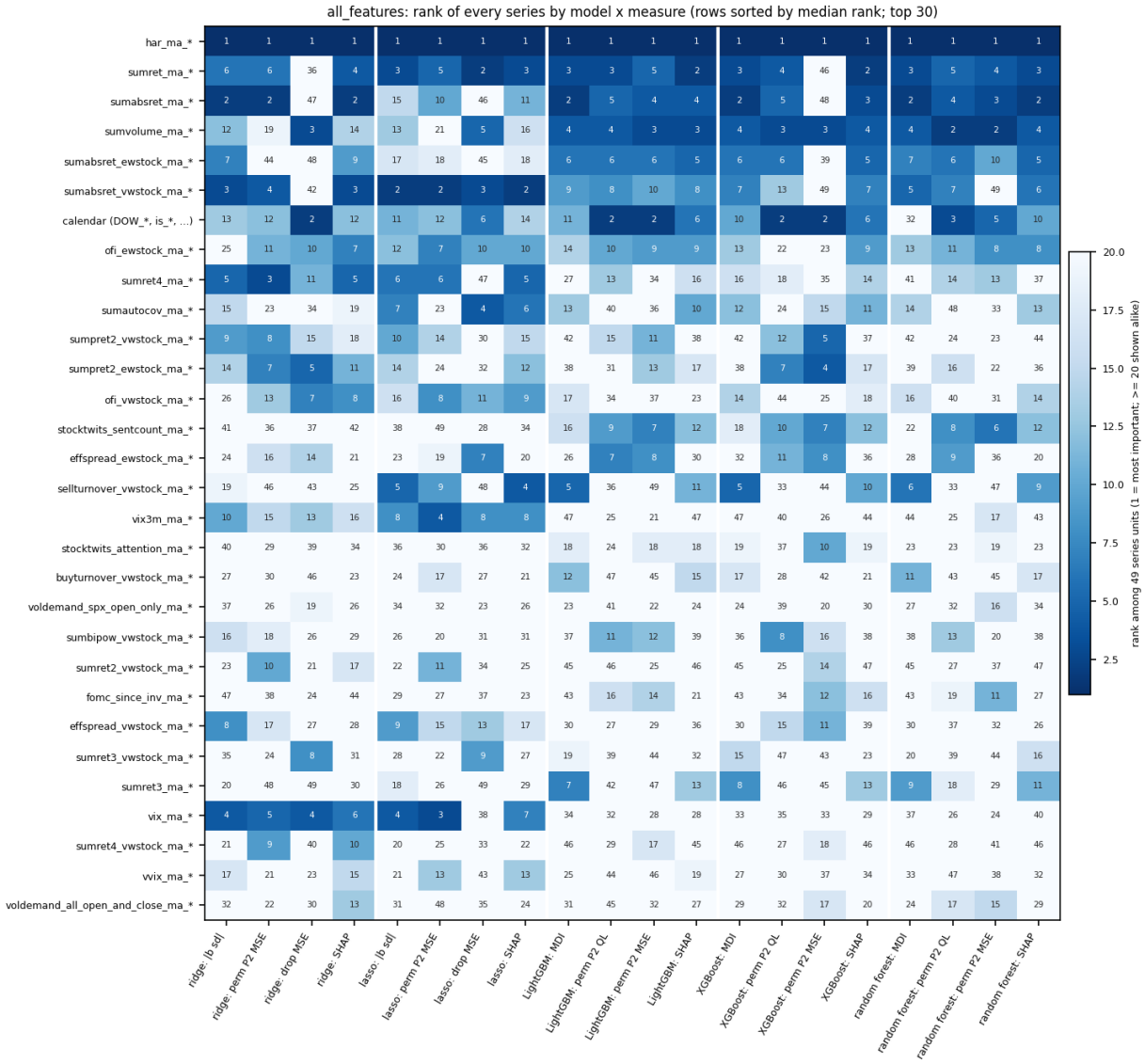


Figure 10: Draft, measures under review. The rank of each series among the 49 by model × measure, all features (1 = most important; the colour scale saturates, so the low ranks share a shade); rows sorted by median rank, the leading rows shown. Measures: $|\beta \cdot \mathrm{sd}|$ and drop-column ΔMSE (linear), MDI / gain (trees), permutation P2 on QLIKE (QL) or MSE, and SHAP.

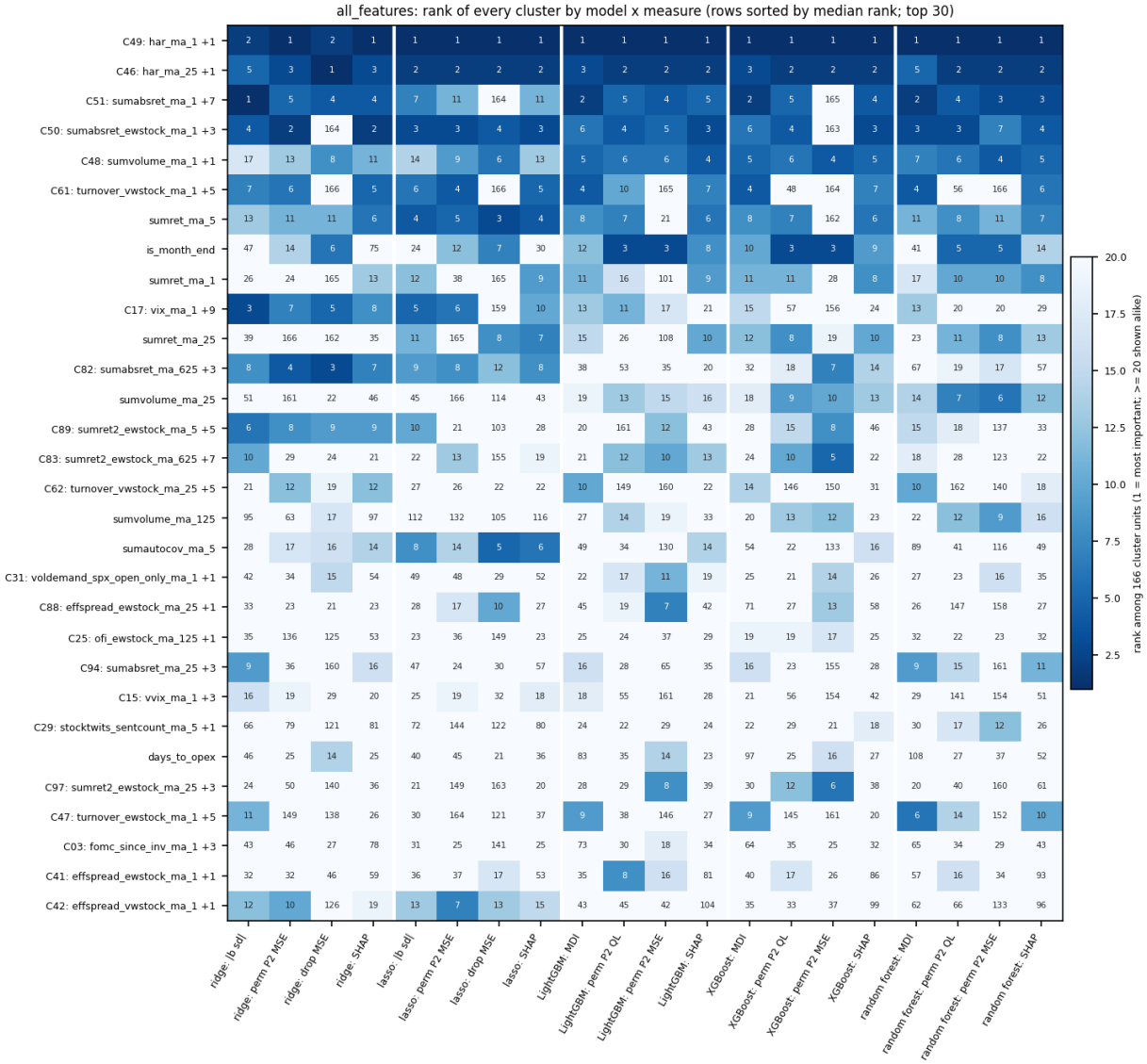


Figure 11: Draft, measures under review. The same at the cluster level: 166 clusters of columns, every pair in a cluster correlated at $|\rho| \geq 0.8$ (complete linkage). A cluster is named by its first member and the number of further members ($+k$); single columns keep their name.

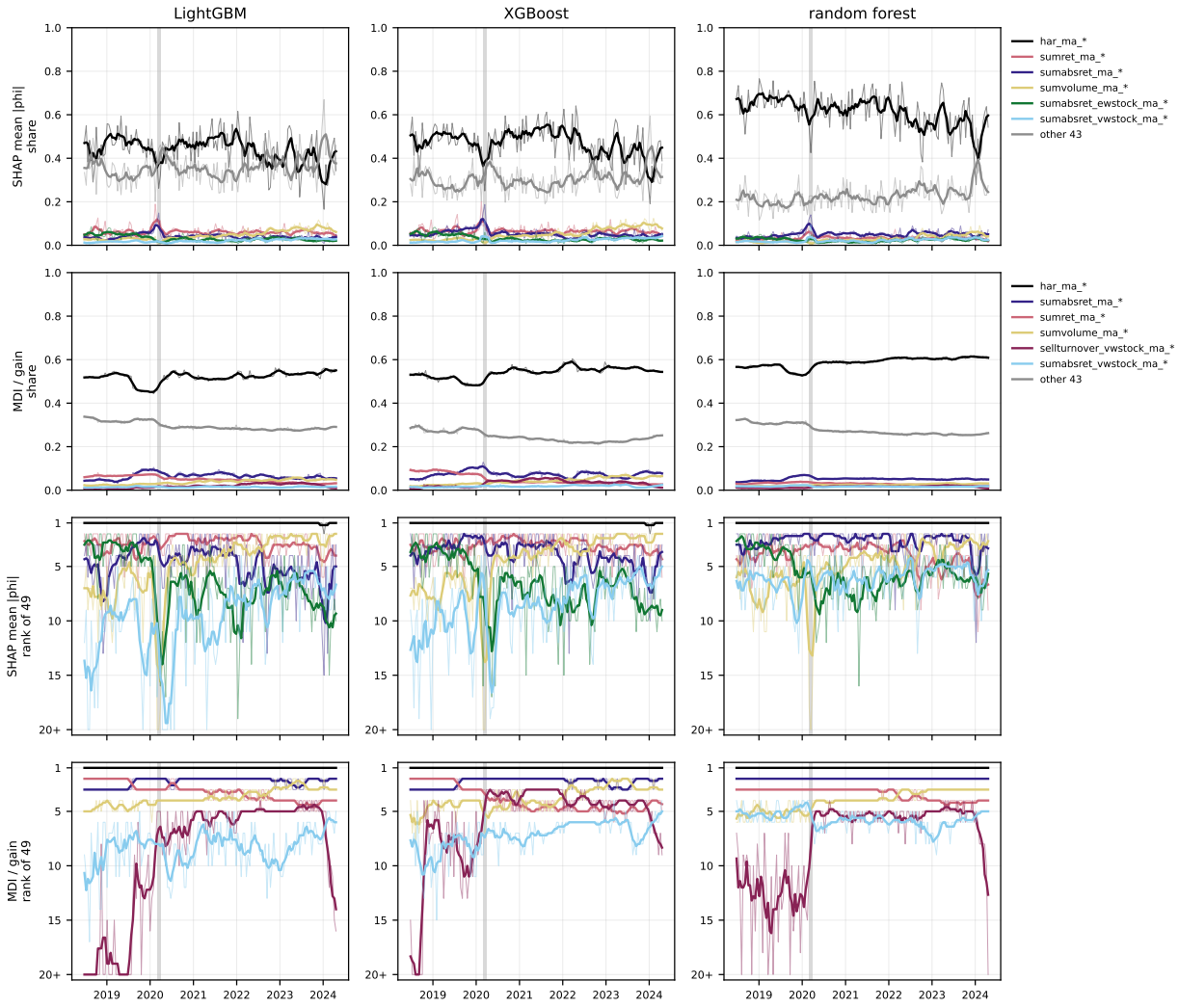


Figure 12: Draft, measures under review. Importance over time, the three trees, all features, series level. Rows 1–2: each series' share of the refit's total SHAP mean $|\phi|$ (on the held-out tail) and MDI / gain; rows 3–4: the same series' rank among the 49 (ranks beyond 20 drawn at 20). Lines: `har_ma_*` and the next 5 series by overall share, the rest summed as 'other'. Thin: one point per refit (every 10 sessions, 147 refits), at its tail's start date; thick: a 5-refit centred rolling mean. Grey band: the 2020 crash tails. Shares are split credit: correlated inputs share it, and MDI and SHAP favour many-valued columns.

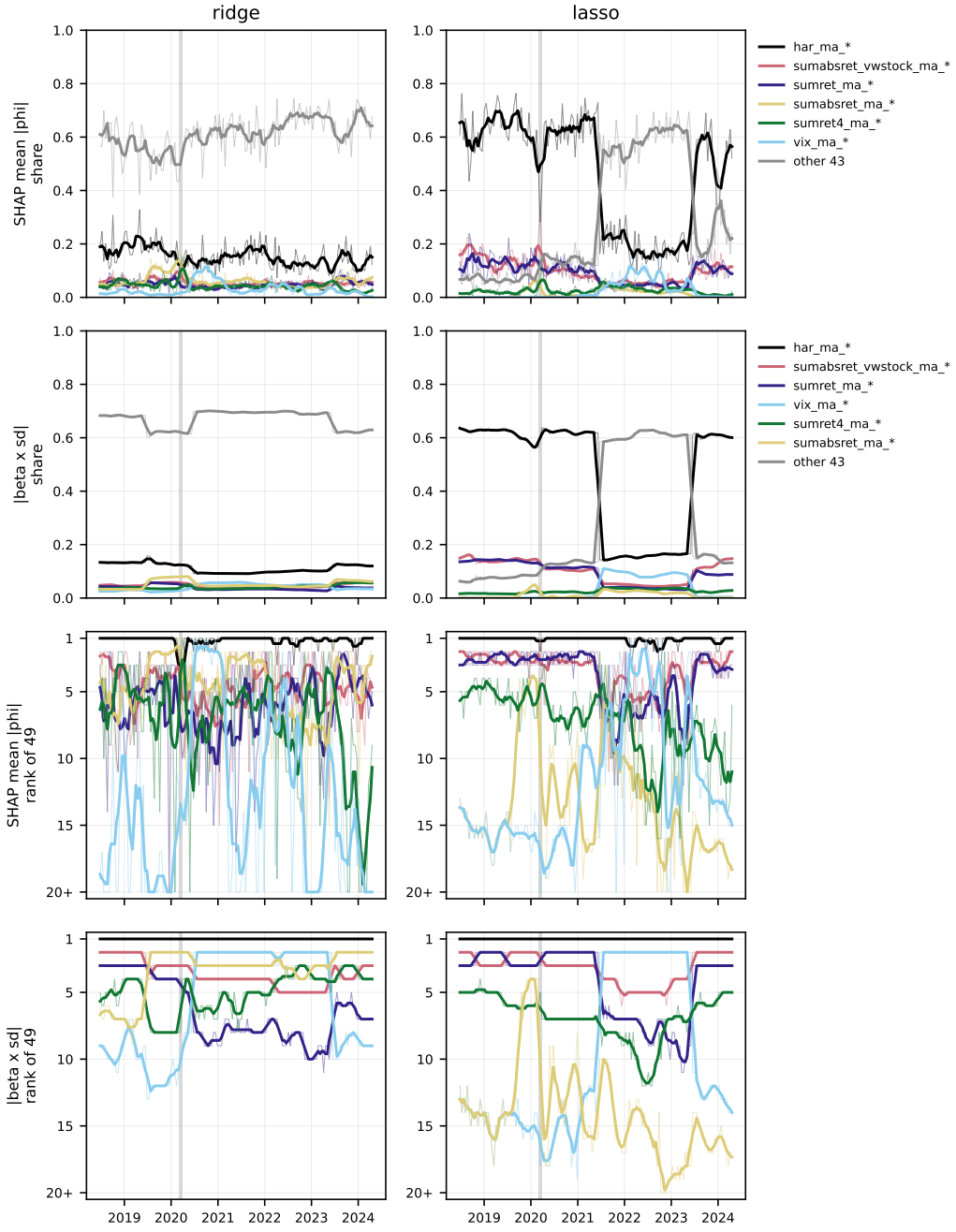


Figure 13: Draft, measures under review. The same for the per-bar ridge and lasso: SHAP mean $|\phi|$, with $\phi_{t,j} = \beta_j(x_{t,j} - \bar{x}_j)$ and the window mean as background, and $|\beta \cdot \mathrm{sd}|$ (the coefficient times the column's standard deviation in the refit's window, summed over a series' columns), as shares and as ranks. A linear model spreads or concentrates credit across correlated columns differently from a tree, so the shares compare over time within a model, not across the two figures.

Table 17: Feature importance by the coefficient and SHAP measures: the leading units per model, all features.

Model	Measure	Leading units	W	Above noise
Lasso	N_A	Calendar 147; HAR ladder 147; `sumret` 147	0.74	23
Ridge	S_A (%)	Calendar 31.0; HAR ladder 22.8; `sumret` 3.6	0.85	38
LightGBM	$\bar{r}_A$	HAR ladder 1.0; `sumret` 2.9; `sumabsret` 4.8	0.57	27
XGBoost	$\bar{r}_A$	HAR ladder 1.0; `sumret` 3.0; `sumabsret` 3.9	0.60	35
Random forest	$\bar{r}_A$	HAR ladder 1.0; `sumabsret` 2.6; `sumret` 4.0	0.62	29

Notes: Per-bar 16:00 models of record, all features ($p = 628$ columns in 49 units: 47 exogenous series, the HAR ladder and the calendar block), 147 refits, one every 10 sessions. Lasso and ridge: pre-tuned penalties $\alpha = 10^{-2}$ and 10^{3}, backbone penalised at 1/100. N_A: refits in which a column of unit A has $\beta_j \neq 0$; S_A: share of $\sum_k \sum_j |\beta_j^{(k)}|$ on A; $\bar{r}_A$: mean over refits of A's rank by mean $|\phi_{t,A}|$ on the refit's held-out rows (1 = largest). W: Kendall's coefficient of concordance of the per-refit rankings (by $\sum_{j \in A} |\beta_j|$ for the linear models). Noise bar, exogenous series clearing it: lasso, a column selected in a larger share of the refits than the continuous probe in its 49 probe fits (1); ridge, a column with $|\beta_j|$ above the continuous probe's in at least half of the probe fits; trees, a column with mean $|\phi_j|$ above the continuous probe's in at least half of the probe fits.

Table 18: Lasso and ridge: how often each unit is selected and its share of the coefficient mass, 147 refits.

			Lasso			Ridge		
Unit	Bucket	Cols	N_A	Active	$\sum \lvert\beta\rvert$ (%)	$\sum \lvert\beta\rvert$ (%)	$\sum \lvert\beta\rvert$ sd (%)	Mean rank
Calendar	BB	16	147	10.0	53.2	31.0	9.3	1.0
HAR ladder	BB	6	147	5.9	34.6	22.8	30.6	2.0
`sumret`	Mom	12	147	3.0	5.4	3.6	4.5	3.0
`sumabsret_vwstock`	VW	12	147	1.0	2.6	2.2	3.7	4.6
`sumautocov`	Mom	12	147	1.3	1.2	2.1	2.2	5.0
`sumpret2_ewstock`	EW	12	126	1.2	0.6	1.1	1.9	17.8
`effspread_vwstock`	Liq	12	119	0.8	0.2	1.3	2.5	13.1
`sellturnover_vwstock`	Liq	12	109	0.7	1.0	1.2	1.4	16.5
`sumret4`	Mom	12	107	0.8	0.2	1.2	2.2	15.8
`sumret4_vwstock`	VW	12	106	0.7	0.1	0.7	1.2	34.2
`sumpret2_vwstock`	VW	12	105	0.8	0.2	1.0	1.7	21.2
`ofi_ewstock`	Liq	12	83	0.8	0.1	1.5	1.5	10.4
`vix`	IV	12	83	0.6	0.1	1.7	2.2	8.1
`sumret4_ewstock`	EW	12	65	0.5	0.1	0.7	1.1	33.6
`effspread_ewstock`	Liq	12	56	0.5	0.0	1.0	1.6	23.6
`numobs`	Liq	18	45	0.3	0.0	0.4	0.4	44.7
`fomc_day`	FOMC	18	42	0.3	0.0	0.4	0.6	44.4
`spread_vwstock`	Liq	12	35	0.2	0.1	0.8	0.7	29.7
`sumbipow_ewstock`	EW	12	26	0.2	0.1	0.6	1.1	37.2
`sumbipow`	Mom	12	25	0.2	0.0	0.5	1.0	41.2

Notes: Units ordered by the lasso's N_A (ties by its $\sum |\beta|$); the first 20 of 49 shown, 20 units are never selected. Cols: columns of the unit. Lasso: N_A = refits (of 147) with a nonzero coefficient on the unit; active = mean number of its nonzero columns; $\sum |\beta|$ = its share of $\sum_k \sum_j |\beta_j^{(k)}|$. Ridge: the same share; $\sum |\beta|$ sd = the share with each $|\beta_j|$ scaled by the column's standard deviation on the refit's 2,000-session window; mean rank = mean over refits of the unit's rank by $\sum_{j \in A} |\beta_j^{(k)}|$. Value ladders are robust-scaled (median window standard deviation 0.84), but the 324 availability and activity ladders are not (0.022), nor the calendar dummies (0.39). Backbone (BB) columns carry 1/100 of the penalty. Buckets: Mom return moments, Liq liquidity, EW / VW equal- / value-weighted market, Sent sentiment, IV implied volatility, VD volatility demand.

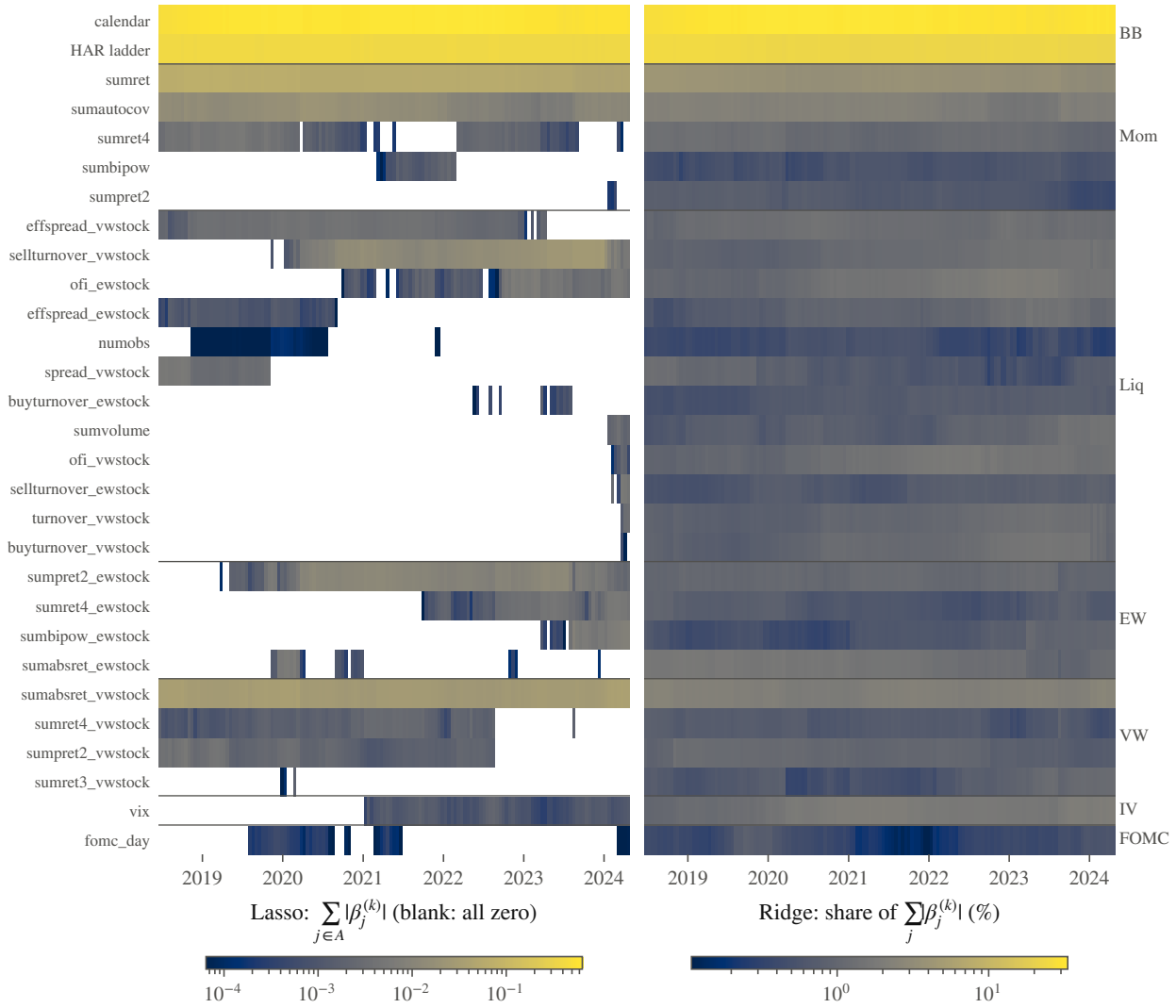


Figure 14: The linear coefficients over time, by unit, at the 147 refits (every 10 sessions, 25 June 2018 to 30 April 2024). Left: the pre-tuned lasso's $\sum_{j \in A} |\beta_j^{(k)}|$, log colour scale, blank where every coefficient of the unit is zero (the on/off pattern). Right: the pre-tuned ridge's share of the refit's $\sum_j |\beta_j^{(k)}|$ on the unit, %. Rows: the 29 units the lasso selects at least once, grouped by bucket (backbone first) and ordered by N_A; the other 20 are never selected. Coefficients are per unit of the robust-scaled input (see Table 18).

Table 19: Trees: the per-refit SHAP rankings of the units, aggregated over 147 refits.

		LightGBM			XGBoost			Random forest		
Unit	Bucket	$\bar{r}$	Top 5	Borda	$\bar{r}$	Top 5	Borda	$\bar{r}$	Top 5	Borda
HAR ladder	BB	1.0	100	1	1.0	100	1	1.0	100	1
`sumret`	Mom	2.9	97	2	3.0	97	2	4.0	82	3
`sumabsret`	Mom	4.8	71	4	3.9	87	3	2.6	97	2
`sumvolume`	Liq	4.9	71	3	4.6	75	4	5.1	69	4
`sumabsret_ewstock`	EW	6.7	39	5	6.7	38	5	5.7	46	5
`sumabsret_vwstock`	VW	10.0	15	7	8.6	13	7	6.1	46	6
Calendar	BB	7.9	37	6	8.1	31	6	15.1	7	9
`stocktwits_sentiment`	Sent	11.9	16	8	11.6	17	8	11.3	19	7
`sumautocov`	Mom	11.9	3	10	11.6	5	11	16.1	2	13
`ofi_ewstock`	Liq	14.6	14	9	14.8	11	9	12.9	9	8
`sumret3`	Mom	17.9	3	13	16.5	1	13	15.4	2	12
`stocktwits_sentcount`	Sent	17.1	7	11	16.2	4	12	17.6	3	11
`sellturnover_vwstock`	Liq	20.0	7	12	16.4	8	10	18.3	7	10
`ofi_vwstock`	Liq	25.8	1	24	26.0	2	19	18.4	1	15
`stocktwits_attention`	Sent	22.7	1	20	24.4	1	18	23.7	0	19
Kendall's W			0.57			0.60			0.62	
Exog. series above the probe			27			35			29	

Notes: Per refit k, unit A's importance is $I_{k,A} = |T_k|^{-1} \sum_{t \in T_k} |\phi_{t,A}|$, $\phi_{t,A} = \sum_{j \in A} \phi_{t,j}$ (path-dependent TreeSHAP of the refit's model on its held-out rows); $r_{k,A}$ is its rank among the 49 units (1 = largest, ties averaged). $\bar{r}$: mean of $r_{k,A}$ over the 147 refits; top 5: share of refits with $r_{k,A} \leq 5$, %; Borda: rank of $\sum_k \max(0, 11 - r_{k,A})$ (a full Borda count is affine in $\bar{r}$). Units: the 15 with the lowest $\bar{r}$ averaged over the three models. Kendall's W of the per-refit rankings ($W = 1$: identical rankings at every refit). Probe: an exogenous series counts if one of its columns has mean $|\phi_j|$ above the continuous noise column's in at least half of the 49 probe fits (every third refit). SHAP values are the de-dup study's stored values of the cluster refits (LightGBM lgbm_opt_00, XGBoost xgb_opt_00, random forest trial 255, window-masked).

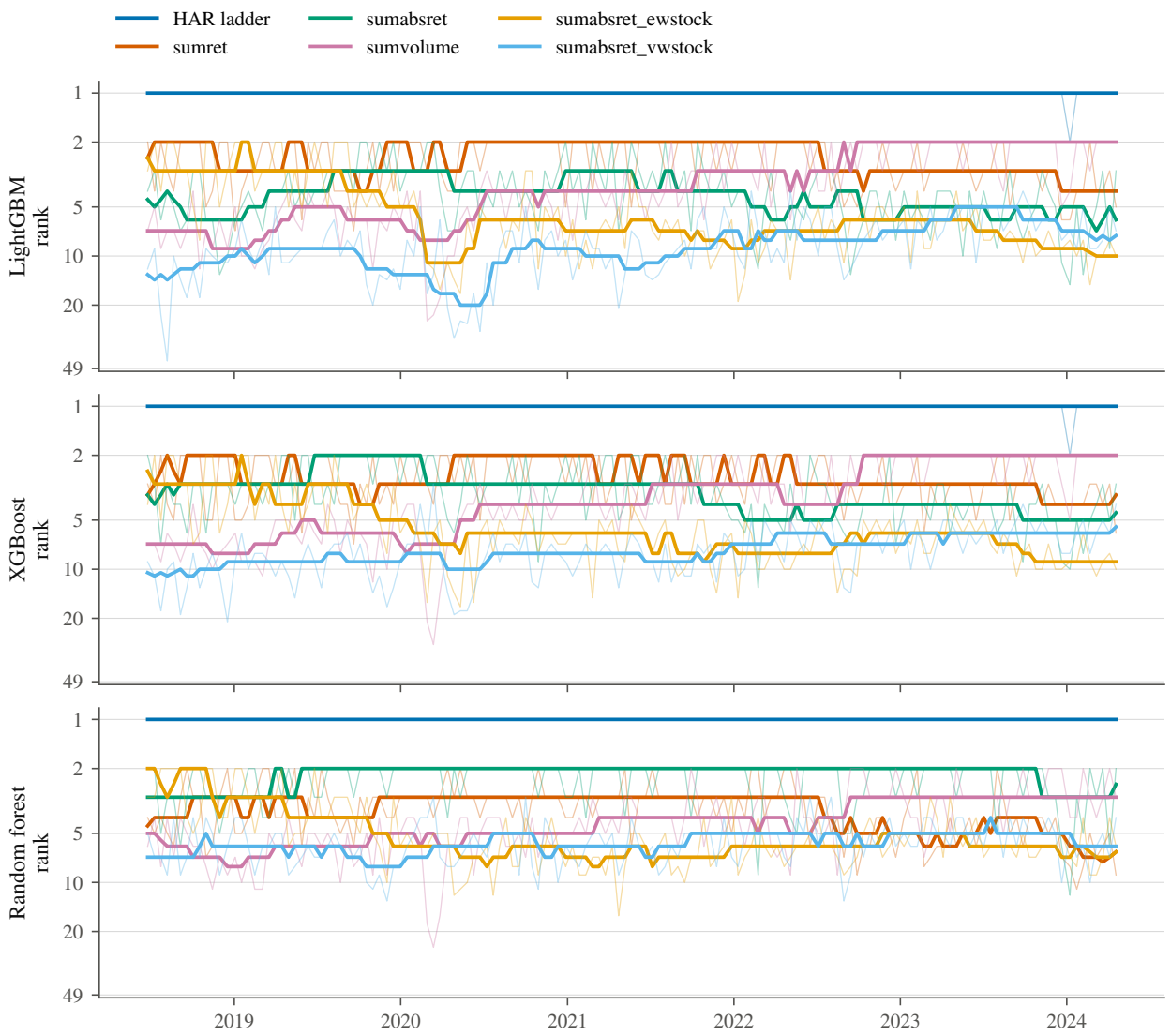


Figure 15: The trees' per-refit SHAP ranks over time, the six units with the lowest mean rank averaged over the three models. Rank $r_{k,A}$ of unit A among the 49 by mean $|\phi_{t,A}|$ on refit k's held-out rows (1 = largest; log scale), at the start of each refit's tail. Thin: one point per refit; thick: centred rolling median over 11 refits (about 110 sessions). Kendall's W over the 147 refits: LightGBM 0.57, XGBoost 0.60, random forest 0.62.

Table 20: Trees: the strongest SHAP interactions, global and per refit, with the H-statistic cross-check.

Pair	$\bar{I}$ ($\times 10^3$)	$\bar{r}$	Top 10 (%)	H^2 (%)	$H^2_{\min}$
LightGBM: series pairs					
HAR ladder × `sumabsret`	12.64	1.7	100	0.33	0.19
HAR ladder × `sumret`	10.95	2.4	99	0.29	0.09
HAR ladder × `sumvolume`	6.62	6.3	88	0.05	0.05
HAR ladder × `sumabsret_ewstock`	5.46	6.3	88	0.05	0.12
HAR ladder × `stocktwits_sentiment`	4.70	14.4	67	0.06	0.17
LightGBM: column pairs					
`har_ma_1` × `har_ma_5`	10.25	–	–	0.36	0.02
`har_ma_1` × `har_ma_25`	4.63	–	–	0.10	0.02
`har_ma_5` × `adj_sumabsret_ma_1`	4.53	–	–	0.67	0.42
Probe floor	0.49	$W = 0.45$			
XGBoost: series pairs					
HAR ladder × `sumabsret`	15.23	1.6	100	0.37	0.39
HAR ladder × `sumret`	11.03	2.5	100	0.68	0.21
HAR ladder × `sumvolume`	7.40	6.0	90	0.08	0.08
HAR ladder × `sumabsret_ewstock`	5.92	5.3	93	0.06	0.17
HAR ladder × `stocktwits_sentiment`	4.87	11.8	69	0.09	0.25
XGBoost: column pairs					
`har_ma_1` × `har_ma_5`	9.95	–	–	0.39	0.02
`har_ma_1` × `har_ma_25`	4.99	–	–	0.14	0.02
`har_ma_1` × `adj_sumabsret_ma_5`	4.94	–	–	0.28	0.43
Probe floor	0.41	$W = 0.47$			
Random forest: series pairs					
HAR ladder × `sumabsret`	23.85	1.1	100	0.85	1.50
HAR ladder × `sumret`	9.13	3.5	99	0.20	0.48
HAR ladder × `sumvolume`	8.52	3.6	99	0.05	0.17
HAR ladder × `sumabsret_ewstock`	6.87	4.6	99	0.06	0.33
HAR ladder × `sumabsret_vwstock`	6.59	4.7	98	0.05	0.26
Random forest: column pairs					
`har_ma_1` × `har_ma_5`	25.31	–	–	1.16	0.06
`har_ma_1` × `adj_sumabsret_ma_1`	8.67	–	–	0.89	1.48
`har_ma_1` × `adj_sumabsret_ma_5`	6.74	–	–	0.32	1.04
Probe floor	0.50	$W = 0.57$			

Notes: SHAP interaction values $\phi_{t,jl}$ (path-dependent TreeSHAP, Lundberg et al. 2020; $\sum_l \phi_{t,jl} = \phi_{t,j}$) of the local refits on each refit's held-out rows, all 1,469 scored sessions. Series pair (A, B): $\Phi_{t,AB} = \sum_{j\in A}\sum_{l\in B}(\phi_{t,jl} + \phi_{t,lj})$; $\bar{I}_{AB}$ = mean of $|\Phi_{t,AB}|$ over the scored sessions, $\times 10^3$, in the fit space of $\hat{y}$. Column pairs: $|\phi_{t,jl} + \phi_{t,lj}|$. $\bar{r}$ and top 10: the pair's mean rank and share of refits in the top 10 when the 1176 series pairs are ranked per refit by their mean $|\Phi_{t,AB}|$ on the refit's rows. H^2 (Friedman–Popescu): the variance of $\mathrm{PD}_{AB} - \mathrm{PD}_A - \mathrm{PD}_B$ as a share of the variance of PD_{AB}, %; $H^2_{\min}$: the same numerator over $\min(\mathrm{var}\,\mathrm{PD}_A, \mathrm{var}\,\mathrm{PD}_B)$, a ratio. Both pooled over 15 refits (every 10th), each on the 100 held-out sessions from its anchor as evaluation and background points; computed for each model's top 10 series and top 5 column pairs. Probe floor: median over the 49 probe fits of the largest $\bar{I}$ between the continuous noise column and a unit. W: Kendall's W of the per-refit rankings of each model's top 50 pairs. XGBoost and the forest are refitted locally with other library versions than the stored refits (forecast gaps up to 0.17).

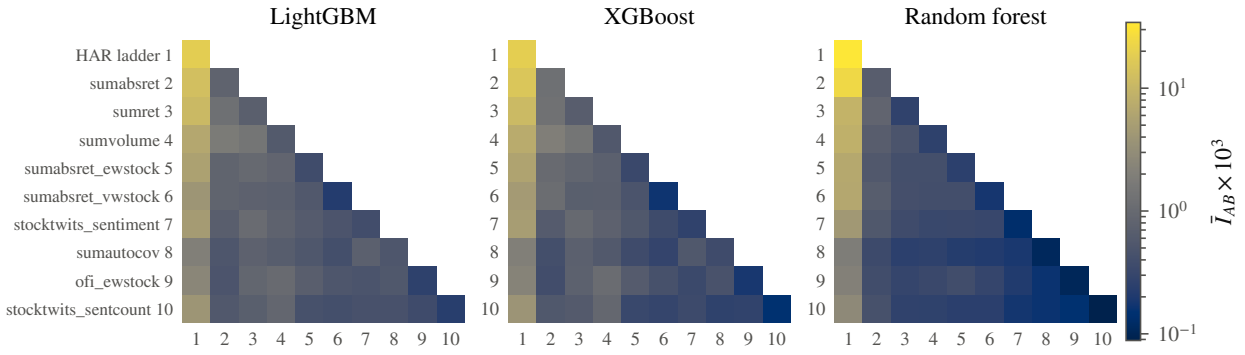


Figure 16: The trees' series-level SHAP interactions: $\bar{I}_{AB}$, the mean over the 1,469 scored sessions of $|\Phi_{t,AB}|$, $\Phi_{t,AB} = \sum_{j \in A} \sum_{l \in B} (\phi_{t,jl} + \phi_{t,lj})$, $\times 10^3$, log colour scale shared by the panels; lower triangle, the diagonal is the interaction among a unit's own columns, $\sum_{j \neq l \in A} \phi_{t,jl}$. Units: the 10 with the largest share of their model's off-diagonal interaction mass, summed over the three models; numbered as in the left panel. Local refits on the held-out rows of each of the 147 refits.

9 Conclusion

- **Exogenous inputs add to HAR + calendar, consistently but weakly.**
  - QLIKE improves in most pairs; Sharpe in nearly all.
  - Few gains individually significant on 866 days.
- **Per bar beats pooled on accuracy,** significantly on HAR + calendar.
  - The trade does not separate the two.
- **A better QLIKE is not a better trade.** The trade sees $\hat{\sigma}_t^2$ only through

$$q_t = \text{sign}(\hat{\sigma}_t^2 - (IV_t^{30})^2).$$

- **The trade is a demonstration, not a trading claim.**
  - Every forecast beats always-short.
  - It is a tail trade.
  - Crossing the spread costs Sharpe.
- **A recurrent network matches the linear models** once regularised like them:
  - ridge-initialised;
  - exogenous block shrunk;
  - trained on the scored loss, $\min \sum_t \text{QLIKE}(RV_t, \hat{y}_t^2 B_t)$.
- **Limits:**
  - 866 days;
  - one asset and one bar;
  - no hedge or market impact;
  - tree forecasts depend on CPU instruction class.
- **Future work:**
  - score the post-2024 expirations;
  - distributional forecasts;
  - a second market;
  - intraday entry timing;
  - calibrated trading costs.

References

Kjersti Aas, Martin Jullum, and Anders Løland. Explaining individual predictions when features are dependent: More accurate approximations to shapley values. *Artificial Intelligence*, 298: 103502, 2021.

Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631, 2019.

Torben G. Andersen and Tim Bollerslev. Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2-3):115–158, 1997.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003.

Torben G. Andersen, Tim Bollerslev, Peter F. Christoffersen, and Francis X. Diebold. Volatility and correlation forecasting. In *Handbook of Economic Forecasting*, volume 1, pages 777–878. Elsevier, 2006.

Torben G. Andersen, Tim Bollerslev, and Francis X. Diebold. Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility. *The Review of Economics and Statistics*, 89(4):701–720, 2007.

Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004.

Jeremias Bekierman and Hans Manner. Forecasting realized variance measures using time-varying coefficient models. *International Journal of Forecasting*, 34(2):276–287, 2018.

Mikkel Bennedsen, Asger Lunde, and Mikko S. Pakkanen. Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5):961–1006, 2022.

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

Tim Bollerslev, Andrew J. Patton, and Rogier Quaedvlieg. Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18, 2016.

Tim Bollerslev, Benjamin Hood, John Huss, and Lasse Heje Pedersen. Risk everywhere: modeling and managing volatility. *The Review of Financial Studies*, 31(7):2729–2773, 2018.

Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009.

Fulvio Corsi, Stefan Mittnik, Christian Pigorsch, and Uta Pigorsch. The volatility of realized volatility. *Econometric Reviews*, 27(1-3):46–78, 2008.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.

Naihua Duan. Smearing estimate: A nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383):605–610, 1983.

Werner Ehm, Tilmann Gneiting, Alexander Jordan, and Fabian Krüger. Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(3):505–562, 2016.

Robert F. Engle and Giampiero M. Gallo. A multiple indicators model for volatility using intra-daily data. *Journal of Econometrics*, 131(1-2):3–27, 2006.

Pierre Garrigues and Laurent El Ghaoui. An homotopy algorithm for the lasso with online observations. In *Advances in Neural Information Processing Systems*, volume 21, 2008.

Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri. Economic predictions with big data: The illusion of sparsity. *Econometrica*, 89(5):2409–2437, 2021.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273, 2020.

Peter R. Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79 (2):453–497, 2011.

Peter R. Hansen, Zhuo Huang, and Howard Howan Shek. Realized garch: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6):877–906, 2012.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291, 1997.

Rodrigo Hizmeri and Mattia Bevilacqua. Early birds get the vol: Morning volatility uncertainty and variance risk premium. `https://ssrn.com/abstract=6212458`, 2025. Working paper, SSRN 6212458.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Serhiy Kozak, Stefan Nagel, and Shrihari Santosh. Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271–292, 2020.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Scott M. Lundberg, Gabriel G. Erion, and Su-In Lee. Consistent individualized feature attribution for tree ensembles. *arXiv preprint arXiv:1802.03888*, 2018.

Willard G. Manning and John Mullahy. Estimating log models: to transform or not to transform? *Journal of Health Economics*, 20(4):461–494, 2001.

Robert C. Merton. Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

Jacob A. Mincer and Victor Zarnowitz. The evaluation of economic forecasts. In *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, pages 3–46. NBER, 1969.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011.

Zhouyu Shen and Dacheng Xiu. Can machines learn weak signals? Technical report, NBER Working Paper 33421, 2025.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1):267–288, 1996.

Mark A. van de Wiel, Mirrelijn M. van Nee, and Armin Rauschenberger. Fast cross-validation for multi-penalty high-dimensional ridge regression. *Journal of Computational and Graphical Statistics*, 30(4):835–847, 2021. doi: 10.1080/10618600.2021.1904962.

Wessel N. van Wieringen. Lecture notes on ridge regression, 2023. Version 0.60 (v8), 27 June 2023.

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B*, 67(2):301–320, 2005.

A Appendix structure

- **A. Data and features:**
  - sources and spans;
  - feature dictionary: 47 series → 628 columns;
  - missing and stale data;
  - column de-duplication; window mask;
  - SPXW chain; implied-volatility inversion;
  - FOMC calendar.
- **B. Tuning per model:**
  - linear penalties; block ridge;
  - trees: settings, Optuna budgets, picks;
  - tuning frequency; refit cadence;
  - network grid and selection.
- **C. Robustness, statistical:**
  - full model table, trees included;
  - DM and HAC conventions;
  - MCS, SPA, StepM;
  - regime and break tests;
  - per bar vs pooled;
  - remaining variance through the session.
- **D. Robustness, trade:**
  - date-shift test;
  - Murphy diagrams; sample splits;
  - P&L decomposition;
  - compounding;
  - sizing;
  - strategy variations; other entry times; month-end.
- **E. Reproducibility:**
  - lock files;
  - design hashes;
  - environment;
  - CPU classes;
  - compute;
  - generator script → every reported number.

- **F. Network implementation:**
  - native PufferLib 5.0 MinGRU trainer;
  - float64 reference checks; byte-identity gates;
  - walk scripts.

Table 21: The 16:00 ladder: one change at a time, from the paper's forecast to the per-bar specification and through the tree rungs.

	QLIKE		Sharpe ratio			Same
Step (to minus from)	Δ%	DM	Δ, mid	HAC t	Δ, crossed	side (%)
From the paper's forecast to the per-bar specification						
block-diag. ridge → pooled ridge, all features	−4.3	−1.75	+0.16	+0.34	+0.16	88
Fit the 16:00 bar alone: pooled twin → per-bar ridge						
HAR + calendar	−11.3	−3.42	+0.46	+1.05	+0.46	88
all features	−6.2	−1.56	−0.03	−0.07	−0.02	86
Inputs of the per-bar ridge						
HAR + calendar → all features	+1.2	+0.26	+0.05	+0.09	+0.04	86
Estimator on all features: per-bar ridge → alternative						
lasso	−4.5	−1.60	+0.46	+1.28	+0.47	92
elastic net	−1.6	−1.04	+0.50	+1.59	+0.50	93
LightGBM, shipped, refit every 10	+0.5	+0.14	−0.05	−0.13	−0.05	85
XGBoost, shipped, refit every 10	+1.3	+0.37	+0.13	+0.35	+0.13	85
random forest, shipped, refit every 10	+2.5	+0.56	−0.15	−0.34	−0.13	85
LightGBM, random search, refit every 10	+2.5	+0.56	+0.39	+0.93	+0.40	85
XGBoost, random search, refit every 10	+5.1	+1.14	+0.26	+0.46	+0.26	84
random forest, random search, refit every 10	+10.6	+1.85	−0.30	−0.81	−0.29	86
Campaign, design: the 12 session-edge duplicates dropped						
per-bar ridge, one penalty (campaign's)	+0.0	—	+0.00	—	+0.00	100
Campaign, per-window mask: masked minus unmasked						
LightGBM, shipped, refit every 10	−1.0	−1.01	−0.21	−0.69	−0.21	96
LightGBM, shipped, refit daily	+1.4	+1.39	−0.23	−0.55	−0.23	94
LightGBM, random search, refit every 10	+2.6	+1.34	−0.08	−0.32	−0.08	94
LightGBM, random search, refit daily	+0.1	+0.03	−0.05	−0.17	−0.05	94
Campaign, refit cadence: every 10 sessions → every session						
LightGBM, shipped settings	−2.4	−1.23	−0.04	−0.08	−0.04	93
Campaign, tuning (LightGBM, refit daily)						
shipped settings → random search	+2.9	+1.05	+0.28	+0.58	+0.28	91
shipped settings → Optuna, tuned daily	+7.5	+2.87	−0.10	−0.31	−0.10	91
Optuna: tuned every 250 sessions → daily	+2.8	+0.81	−0.32	−0.65	−0.33	90

Notes: each row is a paired step, to minus from, over 866 trade days; point estimates. QLIKE Δ%: $100\,(\text{QLIKE}_{\text{to}}/\text{QLIKE}_{\text{from}} - 1)$. DM: Diebold–Mariano statistic of the daily QLIKE difference (—: not stored for that pair). Sharpe Δ: change in the annualised sign(s) Sharpe ratio at the mid and crossed fills. HAC t: Newey–West t of the daily mid-return difference. Same side: share of days on which the two forecasts take the same position. Shipped settings: LightGBM `lgbm_opt_00`, XGBoost `xgb_opt_00`, random forest trial 255, the top entries of the 1998–2002 development searches.

Table 22: Refit cadence and random search, by tree and input set.

	T1 − T10		RS1 − RS10		RS1 − T1	
Model	QLIKE	ΔSR	QLIKE	ΔSR	QLIKE	ΔSR
HAR + calendar						
LightGBM	−0.8	+0.06	+0.2	−0.31	+5.9	−0.15
XGBoost	+0.1	−0.25	−0.2	−0.42	+9.3	−0.38
Random forest	−1.5	−0.31	+0.1	+0.03	+9.4	+0.01
All features						
LightGBM	−2.4	−0.04	−1.6	−0.20	+2.9	+0.28
XGBoost	−3.6	−0.43	−2.2	+0.09	+5.3	+0.65
Random forest	−1.6	−0.43	−3.7	+0.06	+5.6	+0.33

Notes: T1 − T10: the shipped settings refit every session minus every 10th session; RS1 − RS10: the random search (a fixed candidate set, re-chosen every 250 sessions on validation MSE), likewise; RS1 − T1: the random search minus the shipped settings, both refit every session. QLIKE: the difference in % of the second forecast's QLIKE. ΔSR: the difference in the annualised sign(s) Sharpe ratio at the mid fill. Point estimates over 866 trade days; every fit window-masked. HAR + calendar: 22 columns; all features: 628.

Table 23: The per-window mask, masked minus unmasked, per rung.

	T10		T1		RS10		RS1	
Model	QLIKE	ΔSR	QLIKE	ΔSR	QLIKE	ΔSR	QLIKE	ΔSR
HAR + calendar								
LightGBM	+0.4	−0.17	−0.1	−0.12	0.0	−0.13	−0.1	+0.06
XGBoost	−0.6	−0.04	+0.7	−0.08	—		—	
Random forest	+0.3	+0.05	0.0	−0.10	—		—	
All features								
LightGBM	−1.0	−0.21	+1.4	−0.23	+2.6	−0.08	+0.1	−0.05
XGBoost	+1.0	−0.22	+0.8	−0.30	—		—	
Random forest	+0.1	−0.18	−0.4	−0.34	—		—	

Notes: the mask drops, at every fit, the columns that are constant on the training window and exact copies of an earlier column. Rungs as in Table 22; cells as there, the unmasked forecast being the comparator. —: no unmasked run of XGBoost or the random forest on their current settings. The campaign's unmasked rungs were not pinned to a CPU class.

Table 24: How often to tune: Optuna re-tuned every 1, 5 or 25 sessions minus every 250 sessions.

		1 vs 250		5 vs 250		25 vs 250	
Model	QLIKE, 250	QLIKE	ΔSR	QLIKE	ΔSR	QLIKE	ΔSR
HAR + calendar							
LightGBM	0.1068	−3.8	−0.13	−4.9	−0.62	−3.9	−0.26
XGBoost	0.1062	−4.8	−0.42	−5.9	−0.42	−5.2	−0.55
Random forest	0.1036	−4.4	0.00	−4.5	−0.24	−4.2	−0.20
All features							
LightGBM	0.0983	+2.8	−0.32	+1.2	−0.59	+6.1	+0.23
XGBoost	0.0991	+0.2	−0.03	−0.9	−0.08	+2.4	−0.01
Random forest	0.1001	−1.6	−0.35	−1.6	−0.08	+0.8	+0.08

Notes: at each tuning point a fresh TPE study runs 50 sequential trials, trial 0 being the shipped settings, and keeps the best on validation MSE; every path is refit every session. QLIKE, 250: the QLIKE of the path tuned every 250 sessions. Other cells as in Table 22, the 250-session path being the comparator.

Table 25: Is the Optuna budget enough? Where the best of 50 trials falls, and what the last trials add, within the tuning points.

	Best of 50 trials			Val. MSE gain to 50 trials (%)			Pick
	median trial	shipped (%)	in 41–50 (%)	from 10 mean / med.	from 25 mean / med.	from 40 mean / med.	changes (%)
HAR + calendar							
LightGBM	35	0.1	36.8	5.22 / 4.42	1.66 / 0.99	0.50 / 0.00	72
XGBoost	34	0.3	36.5	5.22 / 4.36	1.95 / 1.04	0.66 / 0.00	70
Random forest	32	1.0	26.8	2.69 / 1.95	0.62 / 0.28	0.16 / 0.00	63
All features							
LightGBM	33	0.7	31.9	5.96 / 4.86	2.15 / 1.13	0.69 / 0.00	66
XGBoost	33	0.2	33.6	5.91 / 4.90	2.15 / 1.18	0.71 / 0.00	68
Random forest	33	1.0	31.2	3.22 / 2.32	0.81 / 0.44	0.23 / 0.00	68

Notes: over all 1,469 tuning points of each arm. Median trial: the median index of the trial that holds the best of 50. Shipped: the share of points where that trial is the shipped settings (trial 0). In 41–50: the share where it falls in the last ten trials (20 % if its position were exchangeable). Val. MSE gain: the relative gain in the best validation MSE from the first 10, 25 or 40 trials to all 50, mean / median over points. Pick changes: the share of points whose best of 25 differs from the best of 50.

Table 26: The Optuna budget out of sample: best of 50 trials minus best of the first 10 or 25.

	Tuned every session				Tuned every 25 sessions			
	50 vs 10		50 vs 25		50 vs 10		50 vs 25	
Model	QLIKE	ΔSR	QLIKE	ΔSR	QLIKE	ΔSR	QLIKE	ΔSR
HAR + calendar								
LightGBM	+0.3	+0.21	−0.1	+0.49	−0.6	+0.12	+1.1	+0.30
XGBoost	+1.3	+0.21	+0.1	−0.06	+0.4	+0.19	−0.2	−0.05
Random forest	+0.1	−0.15	+0.2	+0.10	−0.3	+0.16	+0.2	+0.12
All features								
LightGBM	+0.4	−0.06	+0.6	−0.02	+1.1	+0.28	+3.9	+0.70
XGBoost	−1.4	−0.30	+1.7	+0.07	−3.6	−0.25	+0.6	0.00
Random forest	+0.4	−0.80	+0.6	+0.02	+3.1	−0.56	+1.4	+0.22

Notes: Optuna re-tuned every session (left) or every 25 sessions (right), every path refit every session. Cells as in Table 22, the smaller budget being the comparator.

Table 27: Optuna against the simpler rungs and the per-bar ridge: arms below and above zero.

		QLIKE		Sharpe, mid	
TUNE_PER	Arms	below 0	above 0	below 0	above 0
Optuna minus the shipped settings, refit every session (T1)					
1	6	0 (0)	6 (3)	4 (0)	2 (0)
5	6	0 (0)	6 (2)	4 (0)	2 (0)
25	6	0 (0)	6 (4)	2 (0)	4 (0)
250	6	0 (0)	6 (3)	1 (0)	5 (0)
all	24	0 (0)	24 (12)	11 (0)	13 (0)
Optuna minus the random search, refit every session (RS1)					
1	6	4 (0)	2 (0)	4 (0)	2 (0)
5	6	5 (0)	1 (0)	5 (0)	1 (0)
25	6	4 (0)	2 (1)	3 (0)	3 (0)
250	6	3 (0)	3 (0)	4 (0)	2 (1)
all	24	16 (0)	8 (1)	16 (0)	8 (1)
Optuna minus the per-bar ridge					
1	6	0 (0)	6 (2)	6 (1)	0 (0)
5	6	0 (0)	6 (3)	6 (0)	0 (0)
25	6	0 (0)	6 (3)	5 (0)	1 (0)
250	6	0 (0)	6 (3)	5 (0)	1 (0)
all	24	0 (0)	24 (11)	22 (1)	2 (0)

Notes: Optuna, best of 50 trials, re-tuned every TUNE_PER sessions, minus the comparator. Arms: tree × input set. Each cell counts the arms whose difference, a point estimate, lies below or above zero; in parentheses, how many of them are significant at 5 %: Diebold–Mariano for QLIKE, and for the Sharpe ratio the Newey–West t of the daily mid-return difference. The *all* rows pool the four frequencies.

Table 28: Where the Optuna picks sit in each tree's search space.

		At the low end (%)			At the high end (%)			Median pick	
Axis	Search range	H	A	prior	H	A	prior	H	A
LightGBM									
`num_leaves`	[2, 128], log, int	20.6	5.7	11.1	2.0	3.3	4.8	6	15
`min_child_samples`	[1, 256], log, int	19.1	14.2	18.3	0.7	0.8	4.2	5	12
`feature_fraction`	[0.05, 1]	1.4	5.0	5.2	7.9	5.3	5.3	0.646	0.553
`bagging_fraction`	[0.3, 1]	5.2	4.1	4.7	5.0	4.6	4.9	0.638	0.666
`lambda_l2`	$[10^{-3}, 10^{4}]$, log	5.4	5.6	5.5	0.0	0.0	5.1	0.633	0.637
`learning_rate`	[0.005, 0.1], log	3.1	3.7	4.9	6.3	4.7	5.2	0.0256	0.0203
`rounds_cap`†	rounds $\leq$ cap(lr)	—	—	—	1.1	0.6	16.9–17.5	272	294
XGBoost									
`max_depth`	[1, 8], int	18.4	6.1	12.2	5.3	10.6	12.5	3	5
`min_child_weight`	[0.1, 256], log	6.3	4.4	5.3	0.7	0.8	4.8	2.24	5.88
`subsample`	[0.15, 1]	7.5	3.6	5.2	3.8	5.1	5.3	0.516	0.598
`colsample_bytree`	[0.05, 1]	0.8	4.3	4.7	8.3	5.6	4.9	0.691	0.569
`reg_lambda`	$[10^{-5}, 10^{4}]$, log	4.6	4.2	5.5	0.0	0.0	5.1	0.122	0.0899
`learning_rate`	[0.005, 0.1], log	3.9	5.6	4.9	4.2	3.9	5.2	0.024	0.0189
`rounds_cap`†	rounds $\leq$ cap(lr)	—	—	—	1.6	1.1	14.5–15.6	298	316
Random forest									
`max_features`	[0.02, 1]	0.1	1.2	5.1	6.4	10.3	4.8	0.612	0.452
`min_samples_leaf`	[1, 256], log, int	31.7	19.1	17.7	0.0	0.0	4.4	2	4
`max_depth`	{3,5,10,20,40,none}	4.2	3.9	17.0	19.5	22.6	16.5	20	20

Notes: the share of best-of-50 picks in the outermost twentieth of each axis, at its low and high end, per input set (H: HAR + calendar; A: all features), beside the share of the random start-up draws there (prior), and the median pick. Each tree's axes are its own run's search space. †A pseudo-axis: the early-stopped number of boosting rounds, capped by a function of the learning rate.

Table 29: MinGRU development at 16:00: each walk of record in order, beside the per-bar linear references.

	HAR + calendar				All features			
Walk	QLIKE	Δ%	p	Sharpe	QLIKE	Δ%	p	Sharpe
Per-bar walk (the 16:00 bar)								
(1) Walk of record (softplus output)	0.1002	+5.9	0.004	1.29	0.1082	+14.8	0.005	1.20
(2) Exogenous decay (fix 1, B)	0.0990	+4.7	0.034	1.49	0.1054	+11.9	0.014	1.47
(3) Linear shortcut (g8, B)	0.0972	+2.7	0.236	1.34	0.1174	+24.6	0.017	1.30
(4) Ridge-initialised shortcut (b), MSE	0.0987	+4.4	0.053	1.47	0.1021	+8.4	0.017	0.78
(5) (4), QLIKE selection	0.0966	+2.1	0.235	1.68	0.0980	+4.0	0.292	1.67
(6) (4), QLIKE training and selection	0.0985	+4.2	0.059	1.04	0.0948	+0.6	0.853	1.53
Pooled walk (every bar), read at 16:00								
(7) Weight decay (B)	0.1134	+19.9	<0.001	1.04	0.1174	+24.6	0.004	1.83
(8) Ridge-initialised shortcut, QLIKE	0.1151	+21.7	<0.001	0.78	0.1121	+19.0	<0.001	0.34
Per-bar linear references								
Elastic net (the comparator)	0.0946	—	—	1.37	0.0942	—	—	1.92
Ridge	0.0946	+0.0	—	1.37	0.0958	+1.7	—	1.42

Notes: every row is a full walk-forward over 866 trade days, scored on the plain forecast $\hat{f}^2 B$. Δ%: $100\,(\text{QLIKE}/\text{QLIKE}_{\text{enet}} - 1)$ against the per-bar elastic net on the same input set; p: Diebold–Mariano p-value of the daily QLIKE difference. Sharpe: annualised sign(s) Sharpe ratio at the mid fill. (1) softplus output, trained, early-stopped and selected on MSE; (2) adds weight decay on the encoder's input columns in proportion to the linear stack's penalty factors; (3) adds a linear shortcut beside the recurrent branch, started at zero; (4) starts the shortcut at each window's closed-form ridge; (5) selects the configuration on validation QLIKE; (6) also trains and early-stops on QLIKE. B: the new penalty is chosen at each of the 5 block starts, jointly with the base grid. (7) the pooled network, fit on every half-hour bar of the trailing window, its weight decay chosen per block, read at the 16:00 bar. (8) the pooled analogue of (6). Over every bar of the pooled walks, the fit-space QLIKE is 0.1764 (all features) and 0.1842 (HAR + calendar) for (8), against 0.1997 and 0.1892 for (7). Ridge: p not computed. Every variant on disk is in Table 30.

Table 30: Every MinGRU walk on disk (35 variants), grouped by the change it tests.

	HAR + calendar				All features			
Walk	QLIKE	Δ%	p	Sharpe	QLIKE	Δ%	p	Sharpe
Output link								
identity	0.1013	+7.1	0.006	0.89	0.1272	+35.0	0.072	0.84
softplus = (1)	0.1002	+5.9	0.004	1.29	0.1082	+14.8	0.005	1.20
Uniform weight decay w on every weight								
$w = 0.01$	0.1010	+6.8	0.004	1.33	0.1220	+29.5	0.019	0.85
$w = 0.03$	0.0991	+4.7	0.043	1.29	0.1103	+17.1	0.007	1.43
$w = 0.1$	0.1020	+7.8	0.001	1.54	0.1066	+13.2	0.007	1.20
$w = 0.3$	0.0991	+4.7	0.026	1.12	0.1110	+17.9	0.003	0.99
$w = 1$	0.0980	+3.6	0.091	1.48	0.1077	+14.3	0.009	1.33
w per block (B)	0.1002	+5.9	0.007	1.29	0.1223	+29.8	0.017	0.76
Exogenous decay w_{enc} (fix 1)								
$w_{\text{enc}} = 0.03$		*not run*			0.1119	+18.8	<0.001	1.47
$w_{\text{enc}} = 0.3$		*not run*			0.1072	+13.8	<0.001	0.80
$w_{\text{enc}} = 1$		*not run*			0.1099	+16.7	0.029	0.90
$w_{\text{enc}} = 3$		*not run*			0.1063	+12.8	0.050	1.05
$w_{\text{enc}} = 6$		*not run*			0.1051	+11.6	0.005	1.28
$w_{\text{enc}} = 15$		*not run*			0.1015	+7.7	0.054	0.60
w_{enc} per block (B) = (2)	0.0990	+4.7	0.034	1.49	0.1054	+11.9	0.014	1.47
Linear shortcut, zero start (g8)								
shortcut alone, branch frozen		*not run*			0.1039	+10.3	0.005	1.53
grid at $w_{\text{enc}} = 0$		*not run*			0.1312	+39.3	0.002	0.82
grid at $w_{\text{enc}} = 15$		*not run*			0.1149	+22.0	0.018	1.51
per block (B) = (3)	0.0972	+2.7	0.236	1.34	0.1174	+24.6	0.017	1.30
Shortcut started at the window's ridge (b)								
check, identity link		*not run*			0.0957	+1.6	0.308	1.42
check, softplus link		*not run*			0.0966	+2.5	0.205	1.78
grid at $w_{\text{enc}} = 15$		*not run*			0.1078	+14.4	0.059	1.45
per block (B) = (4)	0.0987	+4.4	0.053	1.47	0.1021	+8.4	0.017	0.78
QLIKE in selection and training								
(1), QLIKE selection	0.0998	+5.5	0.010	1.34	0.1082	+14.8	0.005	1.20
(2), QLIKE selection	0.0993	+5.0	0.011	1.27	0.1025	+8.8	0.042	1.28
(4), QLIKE selection = (5)	0.0966	+2.1	0.235	1.68	0.0980	+4.0	0.292	1.67
(4), QLIKE training = (6)	0.0985	+4.2	0.059	1.04	0.0948	+0.6	0.853	1.53
Pooled walk (every bar), read at 16:00								
$w = 0$	0.1164	+23.1	<0.001	0.98	0.1244	+32.0	0.003	1.90
$w = 0.01$	0.1167	+23.4	<0.001	1.01	0.1191	+26.5	0.016	2.23
$w = 0.03$	0.1141	+20.6	<0.001	0.92	0.1177	+25.0	0.019	1.84
$w = 0.1$	0.1140	+20.6	<0.001	1.33	0.1173	+24.5	0.020	2.30
$w = 0.3$	0.1135	+20.0	<0.001	0.88	0.1148	+21.9	0.009	1.73
$w = 1$	0.1140	+20.5	<0.001	1.15	0.1099	+16.7	0.003	1.58
w per block (B) = (7)	0.1134	+19.9	<0.001	1.04	0.1174	+24.6	0.004	1.83
ridge start, QLIKE = (8)	0.1151	+21.7	<0.001	0.78	0.1121	+19.0	<0.001	0.34

Notes: columns and the elastic-net comparator as in Table 29; (1)–(8) are its rows. w: uniform weight decay on every weight; w_{enc}: decay on the encoder's input columns only. A fixed value is held in every block; B chooses it per block. Branch frozen: the recurrent branch is held at zero, so the forecast is the shortcut alone. Checks: the shortcut held at the linear stack's own ridge and the branch frozen, under the identity and the softplus output link. QLIKE training: trained, early-stopped and selected on QLIKE. Not run: the variant was tried on all features only. All-bars QLIKE of (7), (8): notes to Table 29.

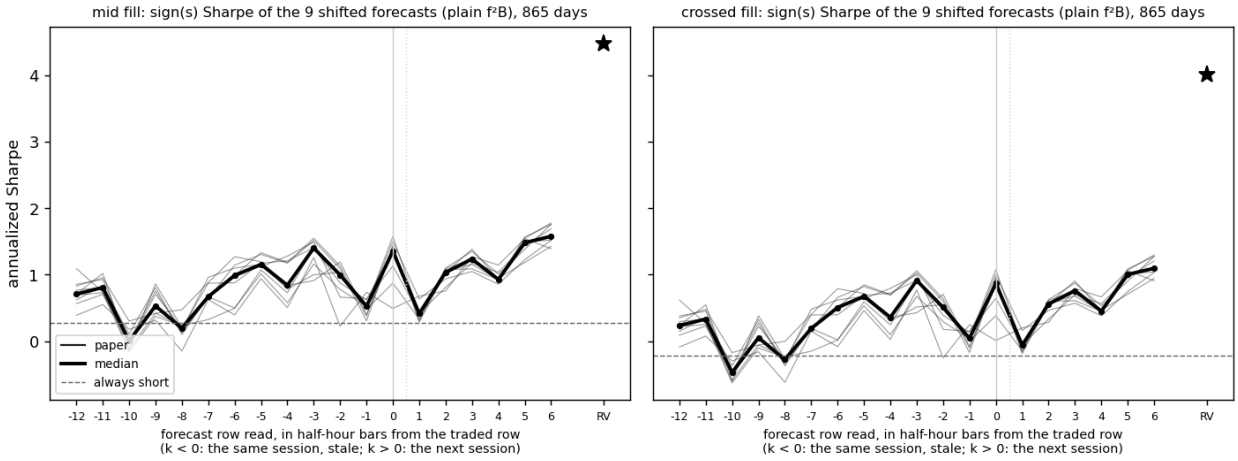


Figure 17: Date-shift test. The annualised sign(s) Sharpe ratio when each forecast is read k half-hour bars away from the traded 16:00 row ($k < 0$: earlier in the same session, stale; $k > 0$: the next session), mid fill (left) and crossed fill (right), k from -12 to 6. Thin lines: the 9 forecasts whose tables carry every session row (the paper's own forecasts; the other 103 forecasts of the model set hold only the 16:00 row); thick: their median; star: the bar's realized variance used as the forecast; dashed: always short. 865 days on which every shifted row exists.

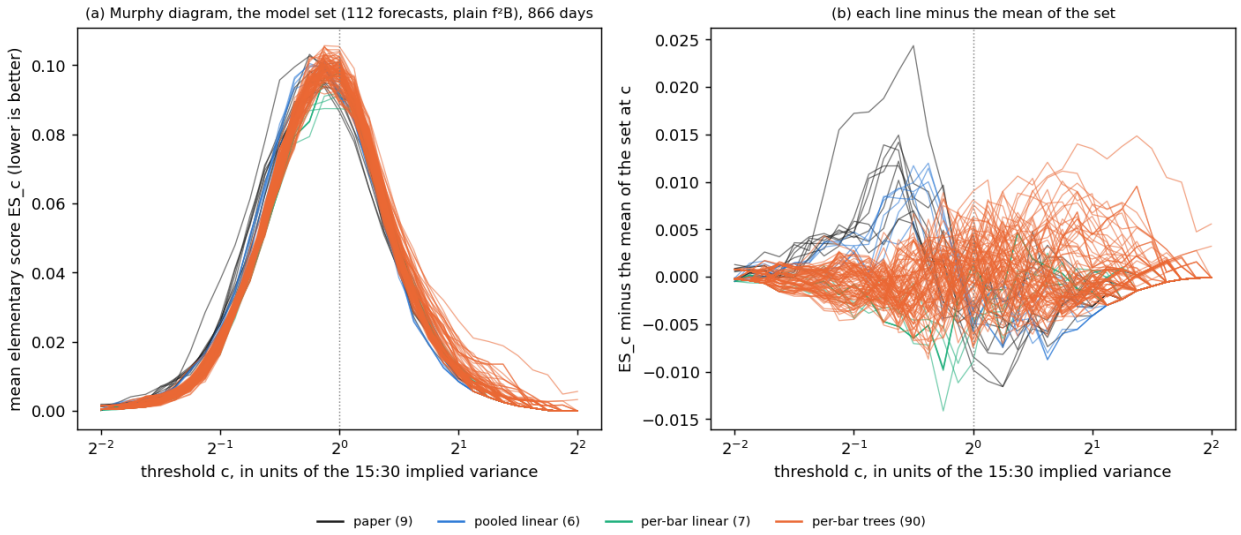


Figure 18: Murphy diagram of the model set (112 forecasts: paper 9, pooled linear 6, per-bar linear 7, per-bar trees 90), 866 days. (a) The mean elementary score ES_c at each of 33 thresholds c, in units of the 15:30 implied variance, log scale (lower is better); $c = 1$ is the threshold the sign(s) rule uses, and QLIKE is the area under each line weighted by c^{-2}. (b) Each line minus the mean of the set at c.

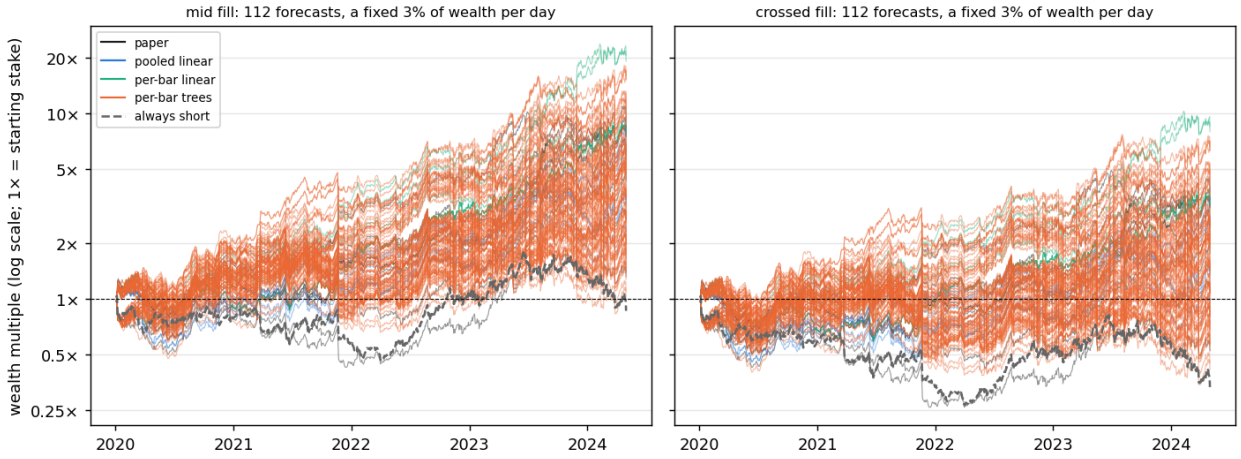


Figure 19: Compounding at a fixed fraction: wealth $W_t = \prod_{s \le t}(1 + f q_s R_s)$ with $f = 3\,\%$ of wealth staked as straddle premium every day, $q_s = \pm 1$ the forecast's sign(s) position and R_s the straddle's return, for each of the 112 forecasts of the model set, mid fill (left) and crossed fill (right), log scale; dashed: always short.

Table 31: Where the all-features MinGRU loses: QLIKE by day class, per walk.

Walk	Month-ends ($n = 43$)	Extreme inputs ($n = 60$)	Other days ($n = 763$)	All days ($n = 866$)
QLIKE, HAR + calendar				
(1) Walk of record	0.1478	0.1328	0.0949	0.1002
(2) Exogenous decay (B)	0.1534	0.1091	0.0952	0.0990
(3) Linear shortcut (B)	0.1444	0.0902	0.0951	0.0972
(4) Ridge start (b), MSE	0.1420	0.1136	0.0951	0.0987
(5) (4), QLIKE selection	0.1350	0.1062	0.0936	0.0966
(6) (4), QLIKE training	0.1271	0.1190	0.0953	0.0985
Elastic net	0.1542	0.1163	0.0895	0.0946
Ridge	0.1542	0.1163	0.0896	0.0946
QLIKE, all features				
(1) Walk of record	0.2323	0.2165	0.0927	0.1082
(2) Exogenous decay (B)	0.1948	0.1735	0.0950	0.1054
(3) Linear shortcut (B)	0.1399	0.3316	0.0992	0.1174
(4) Ridge start (b), MSE	0.1586	0.1818	0.0927	0.1021
(5) (4), QLIKE selection	0.1274	0.1362	0.0934	0.0980
(6) (4), QLIKE training	0.1198	0.1527	0.0888	0.0948
Elastic net	0.1376	0.1764	0.0853	0.0942
Ridge	0.1361	0.1852	0.0865	0.0958
All features minus HAR + calendar				
(1) Walk of record	0.0845	0.0837	−0.0023	0.0080
(2) Exogenous decay (B)	0.0413	0.0644	−0.0002	0.0063
(3) Linear shortcut (B)	−0.0045	0.2414	0.0042	0.0202
(4) Ridge start (b), MSE	0.0166	0.0682	−0.0024	0.0034
(5) (4), QLIKE selection	−0.0076	0.0299	−0.0003	0.0015
(6) (4), QLIKE training	−0.0074	0.0337	−0.0065	−0.0037
Elastic net	−0.0166	0.0601	−0.0042	−0.0004
Ridge	−0.0181	0.0689	−0.0031	0.0011

Notes: walks (1)–(6) of Table 29. Month-ends: the sessions flagged `is_month_end` (5 of them also have extreme inputs). Extreme inputs: at least 10 kept columns beyond their training-window maximum, month-ends excluded. The classes partition the 866 days, so the day-weighted mean of the three class columns is the overall QLIKE. Of the walk of record's summed daily gap between its all-features and HAR + calendar forecasts, extreme-input days carry 72 %, month-ends 52 % and the other days −25 %.

Table 32: Feature dictionary: the 47 exogenous series.

Series	Raw statistic	Source file	Transform	Diurnal	Active
ES return moments (7)					
`sumret`	bar return $\sum r$	`core_stats`	x	s.d.	no
`sumabsret`	$\sum \lvert r \rvert$	`core_stats`	x	mean	no
`sumret3`	$\sum r^3$	`core_stats`	$x^{1/3}$	s.d.	no
`sumret4`	$\sum r^4$	`core_stats`	$\pm\lvert x \rvert^{1/4}$	mean	no
`sumpret2`	upside semivariance $\sum r^2 \mathbf{1}\{r > 0\}$	`core_stats`	$\sqrt{x}$	mean	no
`sumbipow`	bipower variation	`core_stats`	$\sqrt{x}$	mean	no
`sumautocov`	first-order autocovariance $\sum r_i r_{i-1}$	`core_stats`	$\pm\sqrt{\lvert x \rvert}$	s.d.	no
Liquidity (14)					
`sumvolume`	contract volume	`core_stats`	$\log x$	mean	no
`numobs`	1-min returns in the bar (≤ 30)	`core_stats`	$\log x$	mean	yes
`turnover_ewstock`	turnover	`ewstock_stats`	$\sqrt{x}$	mean	no
`buyturnover_ewstock`	buyer-initiated turnover	`ewstock_stats`	$\sqrt{x}$	mean	no
`sellturnover_ewstock`	seller-initiated turnover	`ewstock_stats`	$\sqrt{x}$	mean	no
`effspread_ewstock`	effective spread	`ewstock_stats`	$\sqrt{x}$	mean	no
`spread_ewstock`	quoted spread	`ewstock_stats`	$\log x$	mean	no
`turnover_vwstock`	turnover	`vwstock_stats`	$\sqrt{x}$	mean	no
`buyturnover_vwstock`	buyer-initiated turnover	`vwstock_stats`	$\sqrt{x}$	mean	no
`sellturnover_vwstock`	seller-initiated turnover	`vwstock_stats`	$\sqrt{x}$	mean	no
`effspread_vwstock`	effective spread	`vwstock_stats`	$\sqrt{x}$	mean	no
`spread_vwstock`	quoted spread	`vwstock_stats`	x	s.d.	no
`ofi_ewstock`	order-flow imbalance $(b-s)/(b+s)$ of turnover	`ewstock_stats` (derived)	x	s.d.	no
`ofi_vwstock`	order-flow imbalance $(b-s)/(b+s)$ of turnover	`vwstock_stats` (derived)	x	s.d.	no
Market, equal-weighted (6)					
`sumret2_ewstock`	realized variance $\sum r^2$	`ewstock_stats`	$\sqrt{x}$	mean	no
`sumret3_ewstock`	$\sum r^3$	`ewstock_stats`	$x^{1/3}$	s.d.	no
`sumret4_ewstock`	$\sum r^4$	`ewstock_stats`	$\pm\lvert x \rvert^{1/4}$	mean	no
`sumabsret_ewstock`	$\sum \lvert r \rvert$	`ewstock_stats`	x	mean	no
`sumbipow_ewstock`	bipower variation	`ewstock_stats`	$\sqrt{x}$	mean	no
`sumpret2_ewstock`	upside semivariance $\sum r^2 \mathbf{1}\{r > 0\}$	`ewstock_stats`	$\sqrt{x}$	mean	no
Market, value-weighted (6)					
`sumret2_vwstock`	realized variance $\sum r^2$	`vwstock_stats`	$\sqrt{x}$	mean	no
`sumret3_vwstock`	$\sum r^3$	`vwstock_stats`	$x^{1/3}$	s.d.	no
`sumret4_vwstock`	$\sum r^4$	`vwstock_stats`	$\pm\lvert x \rvert^{1/4}$	mean	no
`sumabsret_vwstock`	$\sum \lvert r \rvert$	`vwstock_stats`	x	mean	no
`sumbipow_vwstock`	bipower variation	`vwstock_stats`	$\sqrt{x}$	mean	no
`sumpret2_vwstock`	upside semivariance $\sum r^2 \mathbf{1}\{r > 0\}$	`vwstock_stats`	$\sqrt{x}$	mean	no
Sentiment (3)					
`stocktwits_attention`	message volume (attention)	`spy_and_sentiment`	$\log x$	mean	no
`stocktwits_sentiment`	net sentiment score	`spy_and_sentiment`	x	none	no
`stocktwits_sentcount`	messages with a sentiment tag	`spy_and_sentiment`	$\log x$	mean	no
Implied volatility (3)					
`vix`	VIX level	`vix_and_voldemand`	$\log x$	none	no
`vvix`	VVIX level	`vix_and_voldemand`	$\log x$	none	no
`vix3m`	VIX3M level	`vix_and_voldemand`	$\log x$	none	no

continued on the next page

Table 32 (continued)

Series	Raw statistic	Source file	Transform	Diurnal	Active
Volatility demand (4)					
`voldemand_spx_open_and_close`	SPX options, opening and closing	`vix_and_voldemand`	$\operatorname{asinh}(x/s_t)$	none	yes
`voldemand_spx_open_only`	SPX options, opening only	`vix_and_voldemand`	$\operatorname{asinh}(x/s_t)$	none	yes
`voldemand_all_open_and_close`	all options, opening and closing	`vix_and_voldemand`	$\operatorname{asinh}(x/s_t)$	none	yes
`voldemand_all_open_only`	all options, opening only	`vix_and_voldemand`	$\operatorname{asinh}(x/s_t)$	none	yes
FOMC calendar (4)					
`fomc_release`	1 on the 14:30 statement bar	`releases` (derived)	$\log x$	none	yes
`fomc_day`	1 on statement days	`releases` (derived)	$\log x$	none	yes
`fomc_until_inv`	$1/(1 + \text{bars to next statement})$	`releases` (derived)	$\log x$	none	no
`fomc_since_inv`	$1/(1 + \text{bars since last})$	`releases` (derived)	$\log x$	none	no

Notes: One row per exogenous series, by bucket. Derived programmatically: the bucket from `src/data/loading.py`; the transform from the branch `apply_semantic_transform` takes for the column name and sign (`src/features/transforms/target.py`); the diurnal adjustment and the asinh routing from `robust_transform` run on the forward-filled panel; the activity flag from the executor's mode-inflation rule (modal value on at least 20 % of rows), checked against the design's `_active` columns. Diurnal: mean, divided by the trailing same-slot mean (20 observations, at least 5, lagged one bar); s.d., divided by the trailing same-slot standard deviation (floored); both banded to $[0.5, 2]$ times the slot's trailing 250-observation median; none: the VIX family, sentiment and the FOMC flags. asinh: a signed series whose s.d. divisor sits on its floor on at least 50 % of rows skips the division and takes $\operatorname{asinh}(x/s_t)$, s_t a causal EWMA of $|x|$. x: no transform (signed series and `sumabsret`); a series counts as signed if it has any negative value (`spread_vwstock` is signed by 1 negative print, 2018-11-12 11:00, and so takes x and s.d. where its EW twin takes $\log x$ and mean). After the transform every series is winsorised at rolling 240-bar 1 / 99 % bounds, set to its median where unobserved, laddered and scaled. Modes of the active series (share of rows): `numobs` 30 (72%); `voldemand_spx_open_and_close` 0 (59%); `voldemand_spx_open_only` 0 (60%); `voldemand_all_open_and_close` 0 (59%); `voldemand_all_open_only` 0 (60%); `fomc_release` 0 (98%); `fomc_day` 0 (95%).

B Notes to the authors (not part of the paper)

- **The current title, abstract, introduction and conclusion describe the old 24-hour panel** and need replacing.
- **The results text anchors on the pooled block ridge;** re-base it on the per-bar OLS. Against that OLS, 7 of 111 forecasts have lower QLIKE, none significantly.
- **Data issue to fix before final numbers:**
  - the activity flags are computed after the forward fill, so the FOMC flags read 1 on all 122 trade days after November 2023;
  - VIX, volume-demand and FOMC feeds end early and are carried forward without limit into 2024.
- **Missing outputs:** paired-comparison tables (axis 1, axis 2), DM against the OLS in the master table, and the model confidence set over models.
- **Pending:** the pooled random forest (full table) and the pooled ridge-init network.
- **The full outline,** with source and macro pointers, is `writeup/PAPER_REHAUL_OUTLINE_2026-10-15.md`.